



SINGLE ANSWER TYPE

- The value of $\int_0^1 \sqrt[5]{2x^3 - 3x^2 + 3x - 1} dx$ equals
- If $f(\theta) = \frac{d}{d\theta} \left\{ \int_0^\theta \frac{dx}{1 - \cos \theta \cdot \cos x} \right\}$ then the value of $\frac{4}{\pi} \left\{ \frac{df}{d\theta} \sin \theta + 2f(\theta) \cdot \cos \theta \right\}$ equals
- Let $I_1 = \int_0^{\pi/2} \cos \left(\frac{2\pi}{3} \sin^2 x \right) dx$ and $I_2 = \int_0^{\pi/2} \cos \left(\frac{\pi}{3} \sin x \right) dx$ then the value of $\frac{I_2}{I_1}$ equals
- If $\int_{-4}^{26} \operatorname{sgn} \left(\left\{ \frac{x-8}{2} \right\} \right) dx = N$ where $\{ \}$ is fractional part function then the sum of digits of N equals (sgn is signum function)
- $\int_{\frac{1}{2}}^2 \frac{1}{x} \operatorname{cosec}^{101} \left(x - \frac{1}{x} \right) dx = \underline{\hspace{2cm}}$
- $\int_0^{\pi/2} \frac{\sin^2 nx}{\sin x} dx = \sum_{r=1}^{\lambda n} \frac{1}{2r-1}$ then $\lambda^2 - \lambda + 6 = \dots\dots\dots$
- Let $g(x) = x^C e^{2x}$ where C is a constant and $f(x) = \int_0^x e^{2t} \sqrt{3t^2 + 1} dt$. $\lim_{x \rightarrow +\infty} \frac{f'(x)}{g'(x)}$ exists finitely and non-zero then $C =$
- $f: R^+ \rightarrow R$ is continuous and satisfies the condition $\int_1^{xy} f(t) dt = y \int_1^x f(t) dt + x \int_1^y f(t) dt \quad \forall x, y \in R^+$ and $f(1) = 3$ then $f(e) =$
- $f: R^+ \rightarrow R$ is continuous and satisfies the property that $\forall x, y \in R^+$ the integral $\int_x^{xy} f(t) dt$ is independent of x (and therefore depends only on y). If $f(2) = 2$, the value of $\int_1^e f(t) dt =$
- f is a differentiable function satisfying $f(x) = x^2 + \int_0^x e^{-t} f(x-t) dt \quad \forall x \in R$. $f(-3) =$
- If $f(x) = \int_0^x (1 + \sin(\sin t)) dt$ then $(f^{-1})'(0) =$

12. If the value of definite integral $\int_1^a x \cdot a^{\lfloor \log_a x \rfloor} dx$ where $a > 1$, and $\lfloor \cdot \rfloor$ denotes the greatest integer, is $\frac{e-1}{2}$ then the value of $5\lfloor a \rfloor$ is _____

MULTIPLE ANSWER TYPE

13. If $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$ then
- A) $\int_0^\infty \frac{\sin 2x}{x} dx = \pi$ B) $\int_0^\infty \frac{\sin^3 x}{x} dx = \frac{\pi}{4}$
 C) $\int_0^\infty \frac{\sin^5 x}{x} dx = \frac{3\pi}{16}$ D) $\int_0^\infty \frac{\sin 3x}{x} dx = \frac{\pi}{2}$
14. Let $f(x)$ is a real valued function defined by $f(x) = x^2 + x^2 \int_{-1}^{+1} t f(t) dt + x^3 \int_{-1}^{+1} f(t) dt$
 Then which of the following options are correct.
- A) $\int_{-1}^{+1} t f(t) dt = \frac{4}{11}$ B) $\int_{-1}^{+1} f(t) dt = \frac{10}{11}$
 C) $\int_{-1}^{+1} t f(t) dt < \int_{-1}^{+1} f(t) dt$ D) $f(1) - f(-1) = \frac{20}{11}$
15. Let $f(x)$ be a continuous function and "a" is constant satisfying $\int_0^x f(t) dt = e^x - ae^{2x} \int_0^1 f(t) e^{-t} dt$ then
- A) $f(x) = \frac{e^x + 2e^{2x}}{3}$ B) $f(x) = e^x - 2e^{2x}$ C) $a = \frac{3}{1-2e}$ D) $a = \frac{1}{3-2e}$
16. Let $f: R \rightarrow R$ be defined by $f(x) = \int_1^3 \frac{1}{|t-x|+1} dt$ then
- A) $f(x) = \log_e(4x - x^2)$ B) $\int_1^3 f(x) dx = (6 \log_e 3) - 4$
 C) Maximum of $f(x)$ is $\log_e 4$ D) f attains max. at $x = 2$
17. $f(x) = x + \int_0^1 (xy^2 + x^2y) f(y) dy$ then $f(x)$ satisfies
- A) $119f(x) - 80x^2 = 180x$ B) $129f(x) - 119x^2 = 120x$
 C) $140f(x) + f'(x) = x + 2$ D) $f(x)$ has local minimum value at $x = -\frac{9}{8}$
18. If $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1+e^x) \sin x} dx$ for $n = 0, 1, 2, \dots$ then
- A) $I_n = I_{n+1}$ B) $I_n = I_{n+2}$ C) $\sum_{K=1}^{20} I_K = 10\pi$ D) $\sum_{K=1}^{10} I_{2K+1} = 10\pi$

19. $f : [0, 1] \rightarrow [0, 1]$ is differentiable and f' is continuous with $f(0) = 0$ satisfies the relation

$$\int_0^x \sqrt{1 - (f'(t))^2} dt = \int_0^x f(t) dt. \text{ Then } \dots\dots\dots$$

- A) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ B) $\lim_{x \rightarrow 0^+} x f\left(\frac{1}{x}\right) = 0$ C) $f\left(\frac{1}{3}\right) > \frac{1}{3}$ D) Maximum value of $f(x)$ is $f(1)$

20. Let $a \in R, a \neq -1$, $\lim_{n \rightarrow \infty} \frac{1^a + 2^a + \dots + n^a}{(n+1)^{a-1} [(na+1) + (na+2) + \dots + (na+n)]}$ is equal to $\frac{1}{60}$. Then $a =$

- a) 5 b) 7 c) $-\frac{15}{2}$ d) $-\frac{17}{2}$

21. Let $\lambda = \int_{\pi/4}^{\pi/3} \frac{\sin x}{x} dx$. Then which of the following statements are true?

- a) $\frac{\sqrt{3}}{8} \leq \lambda \leq \frac{\sqrt{2}}{6}$ b) $\lambda < \frac{1}{4\sqrt{2}}$ c) $\lambda > \frac{1}{2\sqrt{3}}$ d) $\frac{1}{6} < \lambda < \frac{\pi}{12}$

22. Let g be a continuous function such that $f(x) = \int_0^x g(y) dy$ and $h(x) = \int_0^x f(t) dt$. Then

$$\int_0^b x^2 f(x) dx =$$

a) $\int_0^b 2x [f(b) - f(x)] dx$

b) $\int_0^b 2x [h(b) - h(x)] dx$

c) $\int_0^b \left(\frac{b^3 - x^3}{3} \right) g(x) dx$

d) $\int_0^b \left(\frac{b^3}{3} - x^2 \right) g(x) dx$

MATCHING TYPE

23. Column – I

Column – II

A) If $f(\pi) = 2$ and $\int_0^\pi (f(x) + f''(x)) \sin x dx = 5$, then

p) 2

$f(0)$ is equal to (It is given that $f(x)$ is continuous in $[0, \pi]$)

B) The value of $\int_{1/e}^{\tan x} \frac{t}{1+t^2} dt + \int_{1/e}^{\cot x} \frac{1}{t(1+t^2)} dt$, where $x \in \left(\frac{\pi}{6}, \frac{\pi}{3}\right)$

q) $1/2$

is equal to

C) $\lim_{n \rightarrow \infty} \left(\sin \frac{\pi}{2n} \cdot \sin \frac{2\pi}{2n} \cdot \sin \frac{3\pi}{2n} \dots \sin \frac{(n-1)\pi}{n} \right)^{1/n}$ is equal to

r) 3

D) If $f(x) = \begin{cases} 0 & , \text{ where } x = \frac{n}{n+1}, n = 1, 2, 3, \dots \\ 1 & , \text{ elsewhere} \end{cases}$, then the value of $\int_0^2 f(x) dx$ is

24. If $2^{2010} \frac{\int_0^1 x^{1004} (1-x)^{1004} dx}{\int_0^1 x^{1004} (1-x^{2010})^{1004} dx} = abcd$, where $abcd$ is a four digit number and

a, b, c, d are the digits lying from 0 to 9. Then Match the following from Column – I to Column – II

	Column – I		Column – II
(A)	$a + b + d =$	(p)	2
(B)	$a^2 - b^2 + d^2 =$	(q)	4
(C)	$b + c - d =$	(r)	8
(D)	$b^2 + c^2 + d^2 =$	(s)	16

KEY

1	0	2	2	3	3	4	3	5	0	6	3	7	1	8	3
9	4	10	0	11	1	12	5	13	BCD	14	ABCD	15	BD	16	BCD
17	AD	18	BCD	19	ABD	20	BD	21	AD	22	BC				

23. A-Q,B-S,C-P,D-Q

24. A-R,B-S,C-Q,D-P



Sec: : SC-R

Definite Integration

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SINGLE ANSWER TYPE

1. $\int_{-2\pi}^{2\pi} \frac{\sin^6 x}{(\sin^6 x + \cos^6 x)(1 + e^{-x})} dx =$

- A) 2π B) π C) $\frac{\pi}{2}$ D) 4π

2. If $f(x)$ is a differentiable function and $\int_0^{x^3} t^2 f(t) dt = \frac{3}{13} x^{13} + 5$ then $f\left(\frac{8}{27}\right) =$

- A) $\frac{8}{27}$ B) $\frac{16}{27}$ C) $\frac{16}{81}$ D) $\frac{8}{9}$

3. Let $f(x), g(x), h(x)$ be continuous in $[0, 2a]$ and satisfies

$f(2a-x) = f(x), g(2a-x) = g(x), h(x) + h(2a-x) = 3,$

$f(2a-x) \cdot g(2a-x) = f(x) \cdot g(x)$ then $\int_0^{2a} f(x) \cdot g(x) \cdot h(x) dx =$

- A) $\int_0^{2a} f(x) \cdot g(x) dx$ B) $3 \int_0^a f(x) \cdot g(x) dx$ C) $2 \int_0^a f(x) \cdot g(x) dx$ D) $\int_0^a f(x) \cdot g(x) dx$

4. If $\int_a^b \frac{x^n}{x^n + (16-x)^n} dx = 6, (n \in R)$ then

- A) $a = 4, b = 12$ B) $a = 2, b = 14$ C) $a = -4, b = 20$ D) $a = -2, b = 18$

5. The value of $\int_0^{n^2} [\sqrt{x}] dx$, (where $[.]$ denotes G.I.F), $n \in N$, is

- A) $\frac{n^2(n+1)^2}{4}$ B) $\frac{n(n-1)(4n+1)}{6}$ C) $\frac{n(n+1)}{2}$ D) $\frac{n(n+1)(n+2)}{6}$

6. $\int_0^1 |x - x^2| dx + \int_1^2 |x^2 - 3x + 2| dx =$

- A) $\frac{1}{2}$ B) $\frac{1}{3}$ C) $\frac{1}{4}$ D) $\frac{1}{8}$

7. $\int_1^4 \{x - 0.4\} dx$ equals, (where $\{.\}$ denotes fractional part of x)

- A) 1.3 B) 6.3 C) 1.5 D) 7.5

8. The value of $\int_0^{[x]/3} \frac{8^x}{2^{[3x]}} dx$, (where $[.]$ denotes G.I.F) is equal to

- A) $\frac{[x]}{\ln 2}$ B) $\frac{[x]}{\ln 4}$ C) $\frac{2[x]}{\ln 2}$ D) $\frac{[x]}{\ln 8}$

MULTIPLE ANSWER TYPE

9. Let $f(x)$ be a continuous function for all x , which is not identically zero such that

$$\{f(x)\}^2 = \int_0^x f(t) \cdot \frac{2 \sec^2 t}{4 + \tan t} dt \text{ and } f(0) = \ln 4, \text{ then}$$

- A) $f\left(\frac{\pi}{4}\right) = \ln 5$ B) $f\left(\frac{\pi}{2}\right) = \ln 4$ C) $f\left(\frac{3\pi}{4}\right) = \ln 3$ D) $f(x) = 0$

10. The value(s) of ' α ' which satisfy $\int_{\pi/2}^{\alpha} \sin x dx = \sin 2\alpha$, ($\alpha \in [0, 2\pi]$) is (are)

- A) $\frac{\pi}{2}$ B) $\frac{3\pi}{2}$ C) $\frac{7\pi}{6}$ D) $\frac{11\pi}{6}$

11. The value of $\int_0^{\pi/4} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x}$ is

- A) $\frac{1}{ab} \tan^{-1}\left(\frac{b}{a}\right)$, ($a > 0, b > 0$) B) $\frac{1}{ab} \tan^{-1}\left(\frac{b}{a}\right)$, ($a < 0, b < 0$)
C) $\frac{\pi}{4}$, ($a = 1, b = 1$) D) $\frac{1}{ab} \tan^{-1}\left(\frac{a}{b}\right) + \frac{1}{ab}$

12. Let $I = \int_0^1 \frac{dx}{1+x^e}$ then

- A) $I > \frac{\pi}{4}$ B) $I < 1$ C) $I > \frac{e}{1+e}$ D) $I < \frac{e}{1+e}$

PASSAGE-1

$\phi(x)$ defined $[a, b]$ has discontinuities $c_1, c_2, c_3, \dots, c_n$ such that $a < c_1 < c_2 < c_3 < \dots < c_n < b$ then

$$\int_a^b \phi(x) dx = \int_a^{c_1} \phi(x) dx + \int_{c_1}^{c_2} \phi(x) dx + \dots + \int_{c_{n-1}}^{c_n} \phi(x) dx + \int_{c_n}^b \phi(x) dx$$

13. $\int_{-1}^1 [2x-3] dx =$ (where $[.]$ denotes G.I.F)

- A) -7 B) -9 C) 5 D) $\frac{11}{2}$

14. $\int_0^{50\pi} [\tan^{-1} x] dx =$ (where $[.]$ denotes G.I.F)

- A) $\tan 1 + 50\pi$ B) $-\tan 1$ C) $50\pi - \tan 1$ D) $20\pi - 2 \tan 1$

15. $\int_{\pi/2}^{3\pi/2} [\sin x] dx =$ (where $[\cdot]$ denotes G.I.F)

- A) $-\pi$ B) $\frac{\pi}{2}$ C) $\frac{-\pi}{2}$ D) π

PASSAGE-2

We know that for a continuous function f in $[a, b]$,

$$\lim_{n \rightarrow \infty} h f(a + r h) = \int_a^b f(x) dx, h = \frac{b-a}{n} \quad \dots\dots\dots(1)$$

On putting $a = 0, b = 1$, (1) becomes $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} f\left(\frac{r}{n}\right) = \int_0^1 f(x) dx$

This formula enables us to evaluate limits of the form $\lim_{n \rightarrow \infty} [\phi_1(n) + \phi_2(n) + \dots\dots + \phi_n(n)]$

To evaluate this limit we express the r^{th} term as $\phi_r(n) = \frac{1}{n} f\left(\frac{r}{n}\right)$ and then replace $\frac{r}{n}$ by x , $\frac{1}{n}$ by dx and

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \text{ by } \int_0^1 \text{ also } \lim_{n \rightarrow \infty} \sum_{r=1}^{kn} \frac{1}{n} f\left(\frac{r}{n}\right) = \int_0^k f(x) dx, k \in [a, b]$$

16. $\lim_{n \rightarrow \infty} \left[\frac{1^2}{1^3 + n^3} + \frac{2^2}{2^3 + n^3} + \dots\dots + \frac{1}{2n} \right] =$

- A) $\frac{1}{3} \log 2$ B) $\frac{1}{2} \log 2$ C) $\log 2$ D) $\frac{1}{2}$

17. $\lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right) \left(1 + \frac{3}{n}\right) \dots\dots \left(1 + \frac{n}{n}\right) \right]^{1/n} =$

- A) $\frac{2}{e}$ B) $\frac{e}{2}$ C) $\frac{e}{4}$ D) $\frac{4}{e}$

18. $\lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \cdot \sec^2\left(\frac{1}{n^2}\right) + \frac{2}{n^2} \cdot \sec^2\left(\frac{2^2}{n^2}\right) + \dots\dots + \frac{1}{n} \sec^2(1) \right] =$

- A) $\tan 1$ B) $\frac{1}{2} \tan 1$ C) $\frac{1}{2} \sec 1$ D) $\frac{1}{2} \cos ec 1$

MATRIX MATCHING

19. Match the following:

Column I

(A) $\int_0^{\pi/2} \frac{1 + 2 \cos x}{(2 + \cos x)^2} dx =$

Column -II

(p) $\frac{5}{2}$

(B) $\int_0^{16\pi/3} |\sin x| dx =$ (q) $\frac{7}{3}$

(C) $\int_{e^{-1}}^{e^2} \left| \frac{\log_e x}{x} \right| dx =$ (r) $\frac{1}{2}$

(D) $\int_0^4 \{\sqrt{x}\} dx =$ (where $\{\cdot\}$ denotes the fractional part function) (s) $\frac{21}{2}$

20. Match the following:

Column I

Column -II

(A) $\int_0^{\pi} x (\sin^2(\sin x) + \cos^2(\cos x)) dx =$ (p) $\frac{\pi^2}{16}$

(B) $\int_0^{\pi^2/4} (2 \sin \sqrt{x} + \sqrt{x} \cos \sqrt{x}) dx =$ (q) $\frac{\pi^2}{2}$

(C) $\left| \frac{\pi}{\ln 2} \int_{-\pi/4}^{\pi/4} \ln(\sqrt{1 + \sin 2x}) dx \right| =$ (r) $\frac{\pi^2}{4}$

(D) $\int_0^{\pi} \frac{8x^3 \cos^4 x \sin^2 x}{\pi^2 - 3\pi x + 3x^2} dx =$ (s) $\frac{\pi^2}{8}$

KEY

1 B	2 C	3 B	4 B	5 B	6 B
7 C	8 D	9 A,C	10 A,B,C,D	11 A,B,C	12 A,B,C
13 A	14 C	15 C	16 A	17 D	18 B
19 (A) $\rightarrow$ (r)	20 (A) $\rightarrow$ (q)				
(B) $\rightarrow$ (s)	(B) $\rightarrow$ (q)				
(C) $\rightarrow$ (p)	(C) $\rightarrow$ (r)				
(D) $\rightarrow$ (q)	(D) $\rightarrow$ (r)				



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Definite Integration

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SINGLE ANSWER TYPE

- If $I_1 = \int_0^{\frac{\pi}{2}} \frac{\sin(\sin x)}{\sin x} dx$, $I_2 = \int_0^{\frac{\pi}{2}} \frac{\sin x}{x} dx$ and $I_3 = \int_0^{\frac{\pi}{2}} \frac{\sin(\tan x)}{\tan x} dx$
then which of the following is true
(A) $I_1 > I_3$ (B) $I_2 > I_3$ (C) $I_1 > I_2$ (D) $I_1 < I_2$
- If $f(x)$ is monotonic and differentiable function, then
 $\int_{f(a)}^{f(b)} 2x(b - f^{-1}(x)) dx = \underline{\hspace{2cm}}$
(A) $\int_a^b (f^2(x) - f^2(a)) dx$ (B) $\int_a^b (f^2(x) - f^2(b)) dx$
(C) $\int_a^b f^2(x) dx + (a - b)f^2(b)$ (D) $\int_a^b f^2(x) dx + (a - b)f^2(a)$
- If $u = \int_0^{\infty} \frac{dx}{x^4 + 7x^2 + 1}$ and $v = \int_0^{\infty} \frac{x^2 dx}{x^4 + 7x^2 + 1}$ then
(A) $u > v$ (B) $u < v$ (C) $u = v$ (D) $u = \frac{\pi}{6}$
- If $I = \int_0^{\frac{\pi}{2}} \log(\sin 2x) dx$ then
A) $I = \int_0^{\frac{\pi}{2}} \log(\sin x) dx$ B) $I = \int_0^{\frac{\pi}{2}} \log(\cos x) dx$
C) $I = \int_0^{\frac{\pi}{2}} \log(\cos 2x) dx$ D) $I = -\frac{\pi}{2} \log 2$
- If $I_m = \int_0^{\pi/2} \cos^m x \sin^m x dx$ where $m \in \mathbb{N} \cup \{0\}$ then
A) $I_m - \frac{1}{2} I_{m-1} = \frac{1}{2m}$ B) $I_m + \frac{1}{2} I_{m-1} = \frac{1}{2m}$
C) $I_m = \frac{1}{2^m} \left[2 + \frac{2^2}{2} + \frac{2^3}{3} + \frac{2^4}{4} + \dots + \frac{2^m}{m} \right]$ D) $I_m = \frac{1}{2^{m+1}} \left[2 + \frac{2^2}{2} + \frac{2^3}{3} + \dots + \frac{2^m}{m} \right]$
- If $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1 + \pi^x) \sin x} dx$; $n=0, 1, 2, \dots$

A) $I_n = I_{n+2}$ B) $\sum_{m=0}^{10} I_{2m+1} = 10\pi$ C) $\sum_{m=0}^{10} I_{2m} = 0$ D) $I_n = I_{n+1}$

7. $I = \int_0^{\frac{\pi}{2}} \sqrt{\sin x} dx$ and $J = \int_0^{\frac{\pi}{4}} \sqrt{\tan x} dx$ Then _____

A) $I > \frac{\pi}{3}$ B) $J > \frac{\pi}{6}$ C) $I < \frac{\pi\sqrt{\pi}}{3\sqrt{2}}$ D) $J < \frac{\pi^{\frac{3}{2}}}{12}$

8. Which of the following statements is/are false?

A) $\int_1^4 e^{x^2} d(x^2) = e^{16} - e^1$

B) If $0 < x < \frac{\pi}{2}$, $\int_{1/2}^{\sqrt{3}/2} \ln(e^{\cos x}) d(\sin x) = \frac{\pi}{12}$

C) Let $f(x) = \int_0^x |\cos t| dt$. Then $f'(x)$ is continuous but not differentiable at $x = \frac{\pi}{2}$

D) Let $f(x) = \int_{-2}^x |t+1| dt$, $x \in \left[-\frac{3}{2}, 1\right]$. Then, f is continuous but not differentiable at $x = -1$

9. Which of the following statements is/are true?

A) Let f be strictly increasing continuous function defined on the set of real numbers. If $f(1) = 2, f(2) = 4$ then $\int_1^2 f(x) dx$ can be equal to $\frac{25}{8}$

B) Let $f: \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is such that $f''(x) \geq 0, \forall x \in [a, b]$. Then the value of $\int_a^b f(x) dx$ can not exceed $\frac{(f(a) + f(b))(b-a)}{2}$

C) The value of $\int_0^1 \frac{dx}{1+x^{\pi/2}} < \frac{\pi}{4}$

D) the value of $\int_0^1 \frac{dx}{1+x^{\pi/2}} > \ln 2$

10. Let $I_1 = \int_0^{\pi/4} e^{x^2} dx$, $I_2 = \int_0^{\pi/4} e^x dx$, $I_3 = \int_0^{\pi/4} e^{x^2} \cos x dx$, $I_4 = \int_0^{\pi/4} e^{x^2} \sin x dx$ then which of the following is/are not TRUE?

A) $I_2 > I_4 > I_3$

B) $I_2 \geq I_1 \geq I_3$

C) $I_1 > I_3 > I_4$

D) $I_1 \geq I_3 \geq I_4$

11. Let 'f' be a real valued function defined on the interval $(-1, 1)$ such that

$f(x) = e^{2x} \int_0^x \sqrt{t^3 + 1} dt$, for all $x \in (-1, 1)$ and let f^{-1} be the inverse function of f . Then

$(f^{-1})'(0)$ is equal to

a) 1 b) $\int_{-1/2}^0 [x^2 + x + 1] + [\pi] - [e]$ where $[]$ is G.I.F. c) $\ln e^e$ d) e

12. The value(s) of $\int_{-\infty}^0 \frac{e^{3x}(1-e^x)^4}{1+e^{-2x}} dx$ is/are

a) $\frac{22}{7} - \pi$ b) $\frac{2}{105}$ c) 0 d) $\log_e e - \int_1^2 [x] dx$ where $[]$ G.I.F.

13. Let $f(n) = \int_0^{\pi/4} \tan^n x dx$, where n is a +ve integer exceeding 2 then

- A) $f(n+1) > f(n)$ B) $f(n) < \frac{1}{n-1}$ C) $f(n) > \frac{1}{2}$ D) $f(n) + f(n-2) = \frac{1}{n-1}$

14. The point(s) of extremum of $\int_0^{x^2} \frac{t^2 - 5t + 4}{2 + e^t} dt$ is (are)

- (A) $x = -2$ (B) $x = 1$ (C) $x = 0$ (D) $x = -1$

15. If x satisfies the equation $x^2 \left(\int_0^1 \frac{dt}{t^2 + 2t \cos \alpha + 1} \right) - x \left(\int_{-3}^3 \frac{t^2 \sin 2t}{t^2 + 1} dt \right) - 2 = 0$ ($0 < \alpha < \pi$),

then the value of x is

- A) $2\sqrt{\left(\frac{\sin \alpha}{\alpha}\right)}$ B) $-2\sqrt{\left(\frac{\sin \alpha}{\alpha}\right)}$ C) $4\sqrt{\left(\frac{\sin \alpha}{\alpha}\right)}$ D) $-4\sqrt{\left(\frac{\sin \alpha}{\alpha}\right)}$

16. If $f(x) = 2x^3 - 15x^2 + 24x$ and $g(a) = \int_0^a f(x) dx + \int_0^{5-a} f(x) dx$ ($0 < a < 5$), then

- A) g is increasing in $\left(1, \frac{5}{2}\right)$ B) g is decreasing in $\left(\frac{5}{2}, 4\right)$
C) g is increasing in $(0, 5)$ D) None of these

17. Given $\frac{\int_y^{f(x)} e^t dt}{\int_y^x \frac{1}{t} dt} = 1 \quad \forall x, y \in \left(\frac{1}{e^2}, \infty\right)$ where $f(x)$ is continuous and differentiable function

and $f\left(\frac{1}{e}\right) = 0$. If $g(x) = \begin{cases} e^x, & x \geq k \\ e^{x^2}, & 0 < x < k \end{cases}$, then

- A) $f(g(x))$ is continuous for $k = 1$ B) $f(g(x))$ is differentiable for $k = 1$,
C) $f(g(x))$ is non-differentiable for $k = 1$ D) $f(g(x))$ is continuous for $k = 2$

18. If $f(x) = \frac{\sin x}{x}$, $I_1 = \int_0^{\pi} f(x) dx$; $I_2 = \int_0^{\pi/2} f(x) \cdot f\left(\frac{\pi}{2} - x\right) dx$, then

- A) $I_1 > \pi$ B) $I_1 < \pi$ C) $\frac{I_1}{I_2} = \pi$ D) $\frac{I_1}{I_2} = \frac{\pi}{2}$

19. If $\tan \theta + \sec \theta = e^x$, then

- A) $\sec \theta = \frac{e^x + e^{-x}}{2}$ B) $\tan \theta = \frac{e^x - e^{-x}}{2}$ C) $\int_{-\log 3}^{\log 2} \cos \theta dx = \frac{\pi}{2}$ D) $\int_{\log 4}^{\log 16} \sin \theta dx = \log\left(\frac{68}{257}\right)$

PASSAGE-1

Let f be a function defined by $f(k) = \int_0^{\pi/2} |k - 2 \sin t| \cos t dt$ where k is real parameter.

Then

20. Which of the following statements is correct?

- A) $f(k)$ is continuous $\forall k$ but not derivable at two values of k .
B) $f(k)$ is differentiable $\forall k$
C) $f(k) = 1 - k \quad \forall k \geq 2$
D) $f(k) = k - 1 \quad \forall k \geq 0$

21. The minimum value of $f(k)$ is

A) $\frac{1}{2}$

B) $\frac{3}{2}$

C) 0

D) 1

PASSAGE-2

Let $f(x)$ and $g(x)$ be two differentiable functions on $\mathbb{R}$ satisfying

$$f(x) = \frac{x^3}{2} + 1 - x \int_0^x g(t) dt \text{ and } g(x) = x - \int_0^1 f(t) dt. \text{ Then}$$

22. The value of definite integral $\int_0^1 f(t) dt$ lies in the interval

A) $\left(0, \frac{1}{2}\right)$

B) $\left(\frac{1}{2}, 1\right)$

C) $\left(1, \frac{4}{3}\right)$

D) $\left(\frac{4}{3}, \frac{5}{3}\right)$

23. Minimum vertical distance between two curves $y = f(x), y = g(x)$ is

A) $\frac{7}{3}$

B) $\frac{1}{6}$

C) $\frac{8}{3}$

D) $\frac{7}{6}$

PASSAGE-3

If $f(x)$ is discontinuous and not having same definition between a and b , then we must break the interval such that $f(x)$ becomes continuous and having same definition in the breaking intervals. Now, if $f(x)$ is discontinuous at $x = c$ ($a < c < b$),

then $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$ and also if $f(x)$ discontinuous $x = a$ in $(0, 2a)$,

then we can write $\int_0^{2a} f(x) dx = \int_0^a \{f(a-x) + f(a+x)\} dx$

24. $\int_0^{10} \left[\frac{x^2 + 2}{x^2 + 1} \right] dx$ (where $[.]$ denotes the greatest integer function) is equal to

A) 0

B) 2

C) 5

D) 10

25. $\int_{-1}^1 [x] d\left(\frac{1}{1+e^{-1/x}}\right)$ (where $[.]$ denotes the greatest integer function) is equal to

A) -3

B) -2

C) -1

D) 0

INTEGER TYPE

26. If $I = \int_0^\pi x (\sin^2(\sin x) + \cos^2(\cos x)) dx$, then $[I] = \underline{\hspace{1cm}}$, where $[.]$ denotes the greatest integer function

27. If f, g, h are continuous functions on $[0, a]$ such that $f(a-x) = f(x)$, $g(a-x) = -g(x)$, $3h(x) - 4h(a-x) = 5$ then $\int_0^a f(x) \cdot g(x) h(x) dx$

28. If $\int_0^x \left(\int_0^u f(t) dt \right) du = \lambda \int_0^x f(u) \cdot (x-u) du$ then find the value of λ

29. If $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$, then $\frac{8}{\pi} \int_0^\infty \frac{\sin^3 x}{x} dx =$

30. If $\int_0^{2\pi} \frac{dx}{2 + \sin 2x} = k \int_0^{\pi/2} \frac{dt}{7 + \cos 2t}$, then the value of $\frac{k}{4}$ is $\underline{\hspace{1cm}}$

31. Let 'f' be twice differentiable function satisfying $\frac{e^x f''(x) - e^x f'(x)}{e^{2x}} = 1$ and $f(0) = 0$, $f'(0) = 1$, then the value of $\left[\int_0^1 \frac{f(x)}{x} dx \right]$ (where [.] denotes G.I.F) is
32. Let $a \in \left(0, \frac{\pi}{2}\right)$, then the value of $\lim_{a \rightarrow 0} \frac{1}{a^3} \int_0^a \ln(1 + \tan a \tan x) dx$ is equal to $\frac{1}{k}$ where k is ____
33. Let $f(x)$ be a periodic function with fundamental period 12. If $\int_{-6}^{12} f(x) dx = 9$, $\int_0^{12} f(x) dx = 12$ then the value of $\frac{1}{3} \left| \int_0^6 f(x) dx \right|$ is _____
34. Using $\int_0^1 x^n dx = \frac{1}{n+1}$, $n \neq -1$, the sum of the series $1 - \frac{1}{5} + \frac{1}{7} - \frac{1}{11} + \frac{1}{13} - \frac{1}{17} + \frac{1}{19} - + \dots \infty = \frac{\pi}{k\sqrt{3}}$ where k is equal to _____
35. If $\lim_{n \rightarrow \infty} n^2 \int_{-1/n}^{1/n} (2015 \sin x + 2016 \cos x) |x| dx = \lambda$, then the sum of digits of λ is ____
36. Let $f(x) = \begin{cases} 1 - |x|, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases}$ and $g(x) = f(x-1) + f(x+1) \forall x \in \mathbb{R}$, then the value of $\int_{-2011}^{2011} g(x) dx$ is equal to

MATCHING TYPE

37. For the function $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$; $-\infty < x < \infty$

(Here μ, σ are constants : $-\infty < \mu < \infty, \sigma > 0$)

Using i) $\int_0^\infty e^{-x} x^{n-1} dx = G(n)$; $n \in \mathbb{R}^+$

ii) $G(n+1) = nG(n)$, iii) $G\left(\frac{1}{2}\right) = \sqrt{\pi}$

Matching the following

	Column – I		Column – II
(A)	$\int_{-\infty}^\infty f(x) dx$	(P)	μ
(B)	$\int_{-\infty}^\infty x f(x) dx$	(Q)	1
(C)	$\int_{-\infty}^\infty (x - \mu)^2 f(x) dx$	(R)	$\sigma\sqrt{\frac{2}{\pi}}$
(D)	$\int_{-\infty}^\infty x - \mu f(x) dx$	(S)	σ^2

38.

	Column – I		Column – II
(A)	For $x > 0$, let $f(x) = \int_1^x \frac{\log_e t}{1+t} dt$	(P)	3

	Then $f(e) + f\left(\frac{1}{e}\right) = \underline{\hspace{2cm}}$		
(B)	$\int_0^2 [x + [x + [x]]] dx = \underline{\hspace{2cm}}$ Where all the symbols $[.]$ are representing G.I.F	(Q)	2
(C)	$\text{Lt}_{n \rightarrow \infty} \left(\frac{1}{1 + \sqrt{n}} + \frac{1}{2 + \sqrt{2n}} + \frac{1}{3 + \sqrt{3n}} + \dots + \frac{1}{2n} \right)$ $= 2 \log_e k$ then $k = \underline{\hspace{2cm}}$	(R)	8
(D)	$\text{Lt}_{n \rightarrow \infty} n \sum_{k=0}^{n-1} \int_{\frac{k}{n}}^{\frac{k+1}{n}} \sqrt{\left(x - \frac{k}{n}\right) \left(\frac{k+1}{n} - x\right)} dx$ $= \frac{\pi}{k}$ then $k = \underline{\hspace{2cm}}$	(S)	$\frac{1}{2}$

KEY

1	ABC	2	AD	3	CD	4	ABD	5	AD	6	AC	7	AC	8	AD
9	ABCD	10	A	11	AB	12	A	13	BD	14	ABCD	15	AB	16	AB
17	AC	18	BD	19	ABC	20	B	21	A	22	D	23	A	24	D
25	D	26	4	27	0	28	1	29	2	30	4	31	1	32	2
33	5	34	2	35	9	36	2								

37. A-Q;B-P;C-S;D-R

38. A-S;B-P;C-Q;D-R



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Sec: Star SC

Test on Definite Integration

Date: 11-09-2020

SINGLE ANSWER TYPE

1. If $I_1 = \int_0^{\pi/4} \frac{1}{(1 + \tan x)^2} dx$ and $I_2 = \int_0^{\pi/4} (1 + \tan x)^2 dx$ then $\frac{I_2}{I_1} =$
- A) 1 B) 2 C) 3 D) 4
2. The value of the definite integral $\int_0^{\pi} ((1 - x \sin 2x)e^{\cos^2 x} + (1 + x \sin 2x)e^{\sin^2 x}) dx$ is equal to
- A) π B) πe C) $\pi(e-1)$ D) $\pi(e+1)$
3. $\int_{1/2}^2 \frac{1}{x} \sin\left(x - \frac{1}{x}\right) dx =$
- A) 0 B) 1 C) 2 D) $\sqrt{2}$
4. If $I = \int_0^{\pi/2} (\cos 2nx) \log \cos x dx$. Then I is same as ($n \in N$)
- A) $\int_0^{\pi/2} (\cot x) \cos 2nx dx$ B) $\int_0^{\pi/2} (\cot x) \sin 2nx dx$
- C) $\int_0^{\pi/2} (\tan x) \cos 2nx dx$ D) $\frac{1}{2n} \int_0^{\pi/2} (\tan x) \sin 2nx dx$
5. The value of the expression $\frac{\int_0^a x^4 \sqrt{a^2 - x^2} dx}{\int_0^a x^2 \sqrt{a^2 - x^2} dx}$ is equal to ($a \neq 0$)
- A) $\frac{a^2}{6}$ B) $\frac{3a^2}{2}$ C) $\frac{3a^2}{4}$ D) $\frac{a^2}{2}$
6. Let $\lambda = \int_0^1 \frac{1}{1+x^3} dx$; $p = \lim_{n \rightarrow \infty} \left(\frac{\prod_{r=1}^n (n^3 + r^3)}{n^{3n}} \right)^{1/n}$ then $\ln p =$
- A) $\ln 2 - 1 + \lambda$ B) $\ln 2 - 3 + 3\lambda$

C) $2\ln 2 - \lambda$

D) $\ln 4 - 3 + 3\lambda$

7. $\int_0^{\infty} \frac{\tan^{-1} ax - \tan^{-1} bx}{x} dx = \quad (a > 0, b > 0)$

A) 0

B) $\frac{\pi}{2} \ln ab$

C) $\frac{\pi}{2} \ln \left(\frac{a}{b} \right)$

D) $\left(\frac{\pi}{4} \right) \ln \left(\frac{a}{b} \right)$

8. Let $g(x)$ be inverse of $f(x)$ and $f(x)$ is given by

$f(x) = \int_3^x \frac{1}{\sqrt{t^4 + 3t^2 + 13}} dt$ then $g'(0) =$

A) $\frac{1}{\sqrt{13}}$

B) 11

C) 121

D) $\frac{1}{13}$

9. $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{4n} \frac{\sqrt{n}}{\sqrt{r} (3\sqrt{r} + 4\sqrt{n})^2} =$

A) $\frac{1}{14}$

B) $\frac{1}{10}$

C) $\frac{1}{20}$

D) $\frac{1}{7}$

10. $\int_0^{\frac{\pi}{2}} \sin x \sin 2x \sin 3x \sin 4x dx =$

A) $\frac{\pi}{4}$

B) $\frac{\pi}{8}$

C) $\frac{\pi}{16}$

D) $\frac{\pi}{32}$

MULTIPLE ANSWER TYPE

11. If $f(x), g(x), h(x)$ be twice differentiable function on $[0, 2]$ satisfying

$f''(x) = g''(x) = h''(x)$, $f'(1) = 2g'(1) = h'(1) = 4$, and $f(2) = 3g(2) = h(2) = 9$ then which of the following is/are true?

A) $\int_0^2 (f(x) + h(x) - 2g(x)) dx = 16$

B) $\int_0^2 (f(x) + h(x) - 2g(x)) dx = 8$

C) $f(x) - g(x) = 2x^2 + 2$ has two real roots

D) $f(x)$ and $h(x)$ are identical functions

12. If $L = \lim_{n \rightarrow \infty} \frac{1}{n^3} \frac{e^{1/n} + e^{2/n} + \dots + e}{(n+1)^m (1^m + 4^m + \dots + n^{2m})}$ is non zero finite real number then, which of the following is/are true?

A) $L = 3(e-1)$

B) $L = e-1$

C) $m = 1$

D) $m = \frac{4}{3}$

13. Let $f: R \rightarrow (0, \infty)$ be a real valued function such that $\int_0^x t f(x-t) dt = e^{2x} - 1$, then which of the following is/are correct?

A) $f(0) = 4$ B) $(f^{-1})'(4) = \frac{1}{8}$ C) $\lim_{x \rightarrow 0} \frac{f(x) - 4}{x} = 4$ D) $\left(\frac{df}{d(e^x)} \right)_{at x=0} = 8$

PASSAGE-1

Let f be a function defined by $f(k) = \int_0^{\pi/2} |k - 2 \sin t| \cos t dt$ where k is real parameter.

Then

14. Which of the following statements is correct?

A) $f(k)$ is continuous $\forall k$ but not derivable at two values of k .

B) $f(k)$ is differentiable $\forall k$

C) $f(k) = 1 - k \forall k \geq 2$

D) $f(k) = k - 1 \forall k \geq 0$

15. The minimum value of $f(k)$ is

A) $\frac{1}{2}$

B) $\frac{3}{2}$

C) 0

D) 1

PASSAGE-2

Let $f_n(x) = \sum_{r=1}^n \frac{\sin^2 x}{\cos^2\left(\frac{x}{2}\right) - \cos^2\left(\frac{2r+1}{2}\right)x}$

$g_n(x) = \prod_{r=1}^n f_r(x) \quad \forall n = 1, 2, 3, \dots$

16. If $I_n = \int_0^\pi \frac{f_n(x)}{g_n(x)} dx$ then $\sum_{K=1}^{100} I_K =$

A) 25π

B) 50π

C) 75π

D) 100π

17. The value of $\lim_{x \rightarrow 0} \int_0^x \frac{dt}{x f_9(t) g_9(t)} =$

A) $\frac{50}{9}$

B) $\frac{10}{9}$

C) $\frac{100}{9}$

D) $\frac{25}{9}$

INTEGER ANSWER TYPE

18. Let $y = p(x)$ be a quadratic function with $p'(2) = 1$, then the value of

$\int_{2-\pi}^{2+\pi} p(x) \sin\left(\frac{x-2}{2}\right) dx$, is _____

19. Let 'f' be twice differentiable function satisfying $\frac{e^x f''(x) - e^x f'(x)}{e^{2x}} = 1$ and $f(0) = 0$,

$f'(0) = 1$, then the value of $\left[\int_0^1 \frac{f(x)}{x} dx \right]$ (where [.] denotes G.I.F) is

20. Suppose $f(x)$ is a continuous function such that $f(x) = \int_0^x f(t) dt$. then $e^{-4} f(4) = \underline{\hspace{2cm}}$

21. $\int_0^\pi \sqrt{(\cos x + \cos 2x + \cos 3x)^2 + (\sin x + \sin 2x + \sin 3x)^2} dx$ has the value equal to $\frac{\pi}{k} + \sqrt{w}$, where k

and w are positive integers. The sum of digits of HCF of k^w and w^k is

22. The value of a positive integer $n \leq 5$, such that $\int_0^1 e^x (x-1)^n dx = 16 - 6e$ is

23. $\int_{-2}^2 \min(x - [x], -x - [-x]) dx$, where [.] is the greatest integer function =

24. Let a, b, c be positive constants. When the value of integral $\int_0^1 (acx^{b+1} + a^3bx^{3b+5}) dx$ is independent of b , then $a^p c = \frac{r}{q}$, where p, q, r are expressed in simplest form, then $p + q + r =$

MATCHING TYPE

25. Match the following: $a < b; a, b \in N$

Column I		Column II	
(A)	$\int_a^b \sqrt{\frac{x-a}{b-x}} dx$	(P)	$\frac{\pi}{\sqrt{ab}}$
(B)	$\int_a^b \frac{1}{\sqrt{(x-a)(b-x)}} dx$	(Q)	$(b-a)\pi/2$
(C)	$\int_a^b \frac{1}{x\sqrt{(x-a)(b-x)}} dx$	(R)	$(a+b)\pi/2$
(D)	$\int_a^b \frac{x}{\sqrt{(x-a)(b-x)}} dx$	(S)	π



SINGLE ANSWER TYPE

- Given that $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{\log(n^2 + r^2) - 2 \log n}{n} = \log 2 + \frac{\pi}{2} - 2$, then $\lim_{n \rightarrow \infty} \frac{1}{n^{2m}} \left[(n^2 + 1^2)^m (n^2 + 2^2)^m \dots (2n^2)^m \right]^{1/n}$ is equal to
 A) $2^m e^{m(\pi/2-2)}$ B) $2^m e^{m(2-\pi/2)}$ C) $e^{m(\pi/2-2)}$ D) $e^{2m(\pi/2-2)}$
- If $I_n = \int_0^{\pi} e^x (\sin x)^n dx$, then $\frac{I_3}{I_1}$ is equal to
 A) $\frac{3}{5}$ B) $\frac{1}{5}$ C) 1 D) $\frac{2}{5}$
- Evaluate : $\int_0^a \ln(\cot a + \tan x) dx$; $a \in \left(0, \frac{\pi}{2}\right)$
 A) $a \ln(\sin a)$ B) $-a \ln(\sin a)$ C) $\ln(\cos a)$ D) $-\ln(\cos a)$
- Let $\lambda = \int_0^1 \frac{dx}{1+x^3}$, $p = \lim_{n \rightarrow \infty} \left[\frac{\prod_{r=1}^n (n^3 + r^3)}{n^{3n}} \right]^{1/n}$, then $\ln p$ is equal to
 A) $\ln 2 - 1 + \lambda$ B) $\ln 2 - 3 + 3\lambda$ C) $2 \ln 2 - \lambda$ D) $\ln 4 - 3 + 3\lambda$
- Consider the integrals $I_1 = \int_0^1 e^{-x} \cos^2 x dx$, $I_2 = \int_0^1 e^{-x} \cos^2 x dx$, $I_3 = \int_0^1 e^{-\frac{x^2}{2}} \cos^2 x dx$, $I_4 = \int_0^1 e^{-\frac{x^2}{2}} dx$, then
 A) $I_2 > I_4 > I_1 > I_3$ B) $I_2 < I_4 < I_1 < I_3$ C) $I_1 < I_2 < I_3 < I_4$ D) $I_1 > I_2 > I_3 > I_4$
- If $y = \int_1^x x^2 \sqrt{\ln t} dt$; then the value of $\frac{d^3 y}{dx^3}$ at $x = e$
 A) 33 B) 11 C) $\frac{33}{4}$ D) None of these
- The value of definite integral $\int_0^{\pi^2} \sin \sqrt{x} dx$
 A) $\frac{\pi}{2}$ B) 2π C) π D) 2
- $\lim_{n \rightarrow \infty} \frac{\sqrt{1} + \sqrt{2} + \sqrt{3} + \dots + \sqrt{n}}{8n + 10n^{3/2}}$ is equal to
 A) $\frac{2}{15}$ B) $\frac{1}{15}$ C) $\frac{4}{15}$ D) $\frac{7}{15}$
- If $I = \int_{\sin^{-1}(x^2-4x+5)-2x}^{\cos^{-1}(x^2-4x+5)+x^2} \frac{\sqrt{f(x)} dx}{\sqrt{f\left(\frac{\pi}{2}-x\right)} + \sqrt{f(x)}}$ then $[I]$ is equal to (where $[.]$ denotes greatest integer function)

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- A) 2 B) 3 C) 4 D) 5
10. If $\int_0^{\infty} \left(\frac{\sin x}{x} \right)^3 dx = A$ and $\int_0^{\infty} \left(\frac{x - \sin x}{x^3} \right) dx = \frac{aA}{b}$, where a and b are relative prime then the value of $(a+b)$ equals.
- A) 3 B) 4 C) 5 D) 6
11. If $f(x) = x^3 + 3x + 4$ then the value of $\int_{-1}^1 f(x) dx + \int_0^4 f^{-1}(x) dx$ equals
- A) 4 B) $\frac{17}{4}$ C) $\frac{21}{4}$ D) $\frac{23}{4}$
12. Let $f(x)$ and $g(x)$ be two differentiable functions on $\mathbb{R}$ satisfying $f(x) = \frac{x^3}{2} + 1 - x \int_0^x g(t) dt$ and $g(x) = x - \int_0^1 f(t) dt$. Then If the distance of the point $p(x_1, y_1)$ lying on the curve $y = f(x)$, from the curve $y = g(x)$ is least, then $x_1 =$
- A) $\frac{1}{6}$ B) $\frac{1}{3}$ C) $\frac{2}{3}$ D) $\frac{7}{6}$

INTEGER ANSWER TYPE

13. If $\int_0^{\infty} \frac{x^2}{(1+x^2)^{7/2}} dx = \frac{k}{15}$, then $k =$ _____
14. Let $f(x) = \begin{cases} x - [x], & \text{if } [x] \text{ is odd} \\ 1 + [x] - x, & \text{if } [x] \text{ is even} \end{cases}$. then $\frac{\pi^2}{10} \int_{-10}^{10} f(x) \cos \pi x |dx|$ is _____
15. $\int \frac{dx}{\cos^3 x \sqrt{2 \sin 2x}} = \left(\tan^{1/2} x \right) \left(1 + \frac{\tan^2 x}{\lambda} \right) + C$ then $\lambda =$ _____
16. $I_1 = \int_0^1 (1 - x^{50})^{100} dx$, $I_2 = \int_0^1 (1 - x^{50})^{101} dx$ then $\left[\frac{I_1}{I_2} \right] =$ _____
(Here $[.]$ denotes G.I.F)
17. $\int_{-\pi/2}^{\pi/2} \frac{\cos 6x + 6 \cos 4x + 15 \cos 2x + 10}{\cos 5x + 5 \cos 3x + 10 \cos x} dx =$ _____
18. Let $F(x) = \int_0^{x^2} f(t) dt$. If $F(x) = x^2(1+x)$ then $f(4) =$ _____
19. $\int_0^{\infty} \frac{x^2}{(x^2+1)(x^2+4)(x^2+9)} dx = \frac{\pi}{(n+2)!}$ then $n =$ _____
20. If $f(x) = ae^{2x} + be^x - cx$, $f(0) = -1$, $f'(\ln 2) = 30$ and $\int_0^{\ln 4} (f(x) + cx) dx = \frac{57}{2}$ then $a - b - c =$ _____
21. If $\int_0^1 |x-t| \cdot \cos \pi t dt = \pm \frac{a}{\pi^2}$ where $x \geq 1$. Then $a =$ _____
22. $\int_{\pi/4}^{\pi/2} \frac{dx}{(\sin x + \cos x + 2\sqrt{\sin x \cos x}) \sqrt{\sin x \cos x}}$ is _____

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23. If $I_n = \int_0^1 (1+x+x^2+\dots+x^{n-1})(1+3x+5x^2+\dots+(2n-3)x^{n-2}+(2n-1)x^{n-1})dx$, $n \in N$, then value of $\sqrt{I_9}$ is
24. If $\int_1^\infty \frac{(x^3+3)dx}{x^6(x^2+1)} = \frac{a+b\pi}{c}$, then find the value of $\frac{c}{2}$
25. Let $g: R \rightarrow \{4\}$ be a function given by $g(x) = x^3(f'(t)-2) + x^2 f''(t) + 4x(f(0)+6) + 4$ and $h(x)$ is defined as $h(x) = \begin{cases} \int_0^x |f(t)-2|dt, & 0 \leq x \leq 6 \\ (x-6)^2 + 20, & 6 < x \leq 12 \end{cases}$. If number of integers in the range of $h(x)$ is N then find the value of $\frac{N}{19}$.
26. If $\int_0^\infty \left(\frac{\ln x}{1-x}\right)^2 dx + \int_0^1 k \frac{\ln(1-x)}{x} dx = 0$ then k is ____

KEY

1	A	2	A	3	A	4	B	5	C	6	C	7	B	8	B
9	B	10	C	11	D	12	B	13	2	14	4	15	5	16	1
17	4	18	4	19	3	20	5	21	2	22	1	23	9	24	6
25	3	26	4												

DEFINITE INTEGRATION

Section - A

- Let $\frac{d}{dx}f(x) = \frac{e^{\sin x}}{x}$, $x > 0$. If $\int_1^4 \frac{3}{x} e^{\sin x^3} dx = F(k) - F(1)$ then one of the possible value of k is
A) 63 B) 64 C) 16 D) 32
- If $\int_{-3}^2 f(x) dx = \frac{7}{3}$ and $\int_{-3}^9 f(x) dx = \frac{5}{6}$ then $\int_2^9 f(x) dx =$
A) $\frac{3}{2}$ B) $-\frac{3}{2}$ C) $-\frac{19}{6}$ D) $\frac{19}{6}$
- If $0 < a < c$, $0 < b < c$ then $\int_0^{\infty} \frac{a^x - b^x}{c^x} dx =$
A) $\ln\left(\frac{b}{c}\right) - \ln\left(\frac{a}{c}\right)$ B) $\frac{\ln a - \ln b}{\ln c}$ C) $\frac{1}{\ln\left(\frac{b}{c}\right)} - \frac{1}{\ln\left(\frac{a}{c}\right)}$ D) 0
- Let $F(x) = f(x) + f\left(\frac{1}{x}\right)$, where $f(x) = \int_1^x \frac{\ln t}{1+t} dt$, then $F(e) =$
A) 2 B) $\frac{1}{2}$ C) 0 D) 1
- If $\int_{\ln 2}^t \frac{dx}{\sqrt{e^x - 1}} = \frac{\pi}{6}$ then $t =$
A) $\ln 8$ B) $\ln 6$ C) $\ln 4$ D) 1
- If $\int_0^{b-c} f(x+c) dx = k \int_b^c f(x) dx$ then $k =$
A) 0 B) 1 C) 2 D) -1
- $\int_2^3 \frac{2-x}{\sqrt{5x-6-x^2}} dx =$
A) $\frac{\pi}{2}$ B) $-\frac{\pi}{2}$ C) $-\pi$ D) π
- $\int_{-1}^1 \left(\sqrt{1+x+x^2} - \sqrt{1-x+x^2} \right) dx =$
A) 2 B) 4 C) 6 D) 0
- If $\frac{d}{dx}f(x) = g(x)$ then $\int_a^b f(x)g(x) dx =$
A) $\frac{f(b)-f(a)}{2}$ B) $\frac{f(a)-f(b)}{2}$ C) $\frac{f^2(b)-f^2(a)}{2}$ D) $\frac{f^2(a)-f^2(b)}{2}$

- If $f(x) = (1 + \tan x) \left(1 + \tan\left(\frac{\pi}{4} - x\right) \right)$ and $g(x)$ is a function with domain R , then $\int_0^1 x^3 g(f(x)) dx =$
A) $\frac{1}{2} g\left(\frac{\pi}{4}\right)$ B) $\frac{1}{4} g(2)$ C) $\frac{1}{4} g(1)$ D) 1
- $\int_0^{\frac{\pi}{2}} \frac{1}{1 + \tan^3 x} dx =$
A) π B) $\frac{\pi}{4}$ C) $\frac{\pi}{2}$ D) 2π
- Let $f(x)$ be a function satisfying $f'(x) = f(x)$ with $f(0) = 1$ and $g(x)$ be a function that satisfies $f(x) + g(x) = x^2$. Then $\int_0^1 f(x)g(x) dx =$
A) $e + \frac{e^2}{2} + \frac{5}{2}$ B) $e - \frac{e^2}{2} - \frac{3}{2}$ C) $e - \frac{e^2}{2} - \frac{5}{2}$ D) $e + \frac{e^2}{2} - \frac{3}{2}$
- $\int_{\frac{3\pi}{2}}^{\frac{\pi}{2}} \left((x+\pi)^3 + \cos^2(x+3\pi) \right) dx =$
A) $\frac{\pi}{2}$ B) $\frac{\pi}{4} - 1$ C) $\frac{\pi^4}{32}$ D) $\frac{\pi^4}{32} + \frac{\pi}{2}$
- $\int_0^{\frac{\pi}{2}} \frac{200 \sin x + 100 \cos x}{\sin x + \cos x} dx =$
A) 50π B) 25π C) 75π D) 150π
- If $\int_0^{\pi} f(\tan x) dx = \lambda$ then $\int_0^{2\pi} f(\tan x) dx =$
A) $\frac{\lambda}{2}$ B) -2λ C) 2λ D) $-\frac{\lambda}{2}$
- $\int_0^{2\pi} \ln(1 + \cos x) dx =$
A) $\pi \ln 2$ B) $-\pi \ln 2$ C) $-2\pi \ln 2$ D) $2\pi \ln 2$
- $\int_0^{4014} \frac{2^x}{2^x + 2^{4014-x}} dx =$
A) 2^{2007} B) 2^{4014} C) 4014 D) 2007

18. $\int_{-1}^1 (ax^3 + bx) dx = 0$ for
 A) any values of a and b B) $a > 0, b > 0$ only
 C) $a > 0, b < 0$ only D) $a < 0, b < 0$ only
19. If $\int_{-1}^4 f(x) dx = 4$ and $\int_2^4 (3 - f(x)) dx = 7$ then $\int_{-1}^2 f(x) dx =$
 A) -2 B) 3 C) 5 D) 8
20. $\int_{-\pi/2}^{\pi/2} \ln \left(\frac{2 - \sin \theta}{2 + \sin \theta} \right) d\theta =$
 A) 0 B) 1 C) 2 D) -1
21. If $f(a + b - x) = f(x)$ then $\int_a^b x f(x) dx =$
 A) $\left(\frac{a+b}{2} \right) \int_a^b f(x) dx$ B) $\int_a^b f(x) dx$ C) $(a+b) \int_a^b f(x) dx$ D) 0
22. $\int_2^5 \frac{[x^2 + 49 - 14x]}{[x^2 - 14x + 49] + [x^2]} dx =$ (where [.] indicates G.I.F)
 A) $\frac{1}{2}$ B) $\frac{3}{2}$ C) $\frac{5}{2}$ D) $\frac{7}{2}$
23. $\int_{-\pi}^{\pi} \frac{2x(1 + \sin x)}{1 + \cos^2 x} dx =$
 A) π^2 B) $\frac{\pi^2}{2}$ C) $\frac{\pi^2}{4}$ D) $2\pi^2$
24. $\sum_{n=1}^{10} \int_{2n-1}^{2n} \sin^{27} x \, dx + \sum_{n=1}^{10} \int_{2n}^{2n+1} \sin^{27} x \, dx =$
 A) 0 B) 1 C) 26 D) 27
25. $\int_{-1}^1 \sqrt{\left(\frac{x+2}{x-2} \right)^2 + \left(\frac{x-2}{x+2} \right)^2} - 2 \, dx =$
 A) $8 \ln \frac{4}{3}$ B) $8 \ln \frac{3}{4}$ C) $4 \ln \frac{4}{3}$ D) 0

26. $f(x) = \int_0^x \ln \left(\frac{1-t}{1+t} \right) dt \Rightarrow$
 A) $f(x)$ is an even function
 B) $f(x)$ is an odd function
 C) $f(x)$ is neither even nor odd
 D) $f(x)$ is cannot be a function
27. If $f(x) = x^2$ for $0 \leq x \leq 1 = \sqrt{x}$ for $1 \leq x \leq 2$ then $\int_0^2 f(x) dx =$
 A) $\frac{4\sqrt{2}}{3}$ B) $\frac{4\sqrt{2}-1}{3}$ C) $\frac{\sqrt{2}}{3}$ D) $\frac{1}{3}$
28. $\int_{-2}^3 |1 - x^2| dx =$
 A) $\frac{28}{3}$ B) $\frac{1}{3}$ C) $\frac{7}{3}$ D) $\frac{14}{3}$
29. $\int_{\frac{1}{\sqrt{3}}}^1 \frac{(\tan^{-1} x)^3}{1 + x^2} dx =$
 A) $\frac{1}{4} \left(\frac{\pi}{12} \right)^6$ B) $\frac{1}{4} \left(\frac{\pi}{12} \right)^4$ C) $\frac{1}{4} \left(\left(\frac{\pi}{4} \right)^4 - \left(\frac{\pi}{6} \right)^4 \right)$ D) $\left(\frac{\pi}{6} \right)^4$
30. $\int_0^1 \frac{dx}{x^2 + 2x \sin \alpha + 1} =$
 A) $\left(\frac{\pi}{4} - \frac{\alpha}{2} \right) \operatorname{cosec} \alpha$ B) $\left(\frac{\pi}{4} + \frac{\alpha}{2} \right) \operatorname{cosec} \alpha$ C) $\left(\frac{\pi}{4} + \frac{\alpha}{2} \right) \sec \alpha$ D) $\left(\frac{\pi}{4} - \frac{\alpha}{2} \right) \sec \alpha$

Section - B

1. $\int_0^{\frac{\pi}{4}} \frac{\sin^9 x}{\cos^{11} x} dx =$
A) $\frac{1}{9}$ B) $\frac{1}{10}$ C) $\frac{1}{99}$ D) $\frac{1}{90}$
2. $\int_0^1 \frac{(1-x^2)dx}{x^4+x^2+1} =$
A) $-\frac{1}{2} \ln 3$ B) $\frac{1}{2} \ln 3$ C) $\ln 3$ D) $2 \ln 3$
3. $\int_0^1 \frac{x^5 dx}{1+x^4} =$
A) $\frac{2-\pi}{4}$ B) $\frac{2+\pi}{4}$ C) $\frac{4+\pi}{8}$ D) $\frac{4-\pi}{8}$
4. $\int_0^1 \frac{e^x \cdot x}{(x+1)^2} dx =$
A) $\frac{e}{2}$ B) $1+\frac{e}{2}$ C) $\frac{e}{2}-1$ D) $1-\frac{e}{2}$
5. The solution for x of the equation $\int_{\sqrt{2}}^x \frac{dt}{t\sqrt{t^2-1}} = \frac{\pi}{2}$
A) $2\sqrt{2}$ B) 2 C) π D) $-\sqrt{2}$
6. $\int_0^{\infty} \frac{45a}{(3+a+at)^4} dt =$
A) $\frac{15}{(3+a)^2}$ B) $\frac{15}{(3+a)^3}$ C) $\frac{15}{3(3+a)^3}$ D) $\frac{15}{a(3+a)^2}$
7. $\int_0^{\infty} \frac{dx}{\left(x+\sqrt{x^2+1}\right)^3} =$
A) $\frac{3}{8}$ B) $\frac{3}{4}$ C) $\frac{3}{2}$ D) $\frac{8}{5}$
8. $\int_0^{\frac{\pi}{4}} \frac{\sin x + \cos x}{3 + \sin 2x} dx =$
A) $\frac{1}{4} \ln 3$ B) $\frac{1}{2} \ln 3$ C) $\ln 3$ D) $2 \ln 3$

9. If $\int_0^{\infty} \frac{x^2 dx}{(x^2+a^2)(x^2+b^2)(x^2+c^2)} = \frac{\pi}{2(a+b)(b+c)(c+a)}$ then $\int_0^{\infty} \frac{dx}{(x^2+4)(x^2+9)}$
A) $\frac{\pi}{60}$ B) $\frac{\pi}{20}$ C) $\frac{\pi}{40}$ D) $\frac{\pi}{3}$
10. If $A = \int_0^1 \frac{e^t}{1+t} dt$ then $\int_0^1 e^t \ln(1+t) dt =$
A) $e \ln 2 - A$ B) $e \ln 2 + A$ C) $Ae \ln 2$ D) $A \ln 2$
11. $\int_0^{\frac{\pi}{2}} \sin 2x \cdot \tan^{-1}(\sin x) dx =$
A) $\frac{\pi}{2} - 1$ B) $\frac{\pi}{2} + 1$ C) $\frac{\pi}{2}$ D) $1 - \frac{\pi}{2}$
12. Let $f(x) = \int_0^x \frac{\cos t}{t} dt$ ($x > 0$) then for $x = (2n+1)\frac{\pi}{2}$; $f(x)$ has:
A) maxima when $n = 0, 2, 4, 6, \dots$
B) minima when $n = 0, 2, 4, 6, \dots$
C) neither maxima nor minima when $n = 1, -3, -5, \dots$
D) Information not sufficient
13. If f is continuous function then which of the following is correct:
A) $\int_{-2}^2 f(x) dx = \int_0^2 (f(x) - f(-x)) dx$ B) $\int_{-3}^5 2f(x) dx = \int_{-6}^{10} f(x-1) dx$
C) $\int_{-3}^5 2f(x) dx = \int_{-6}^{10} f\left(\frac{x}{2}\right) dx$ D) $\int_{-3}^5 2f(x) dx = \int_{-2}^6 f(x-1) dx$
14. $\int_0^{\infty} \frac{x \cdot \ln x}{(1+x^2)^2} dx =$
A) 1 B) -1 C) 0 D) $\frac{\pi}{2}$
15. $\int_0^{\frac{\pi}{2}} (\sin^{100} x - \cos^{100} x) dx =$
A) $\frac{1}{100}$ B) $\frac{100!}{100^{100}}$ C) $\frac{\pi}{100}$ D) 0
16. For $n \in \mathbb{N}$, $\int_0^{2\pi} \frac{x \sin^{2n} x}{\sin^2 x + \cos^{2n} x} dx =$
A) π B) π^2 C) $\frac{\pi}{2}$ D) $\frac{\pi^2}{2}$

17. If $f(x) = \frac{e^x}{1+e^x}$, $I_1 = \int_{f(-a)}^{f(a)} xg(x(1-x))dx$ and $I_2 = \int_{f(-a)}^{f(a)} g(x(1-x))dx$ then $\frac{I_2}{I_1}$ is

A) 2 B) 1 C) -1 D) -3

18. $\int_{\frac{1}{e}}^e |\ln x| dx =$

A) $1 - \frac{1}{e}$ B) $1 + \frac{1}{e}$ C) $\left(1 - \frac{1}{e}\right)$ D) $\left(1 + \frac{1}{e}\right)$

19. $\int_{\pi}^{10\pi} |\sin x| dx =$

A) 18 B) 16 C) 14 D) 0

20. $\int_0^{25} [\sqrt{x}] dx =$

A) 50 B) 25 C) 75 D) 70

21. $\int_0^{\pi} [\cot x] dx$, where $[.]$ denotes the greatest integer function, is equal to

A) 1 B) -1 C) $-\frac{\pi}{2}$ D) $\frac{\pi}{2}$

22. $\int_1^4 \ln[x] dx =$

A) $\ln 4$ B) $\ln 6$ C) $\ln 8$ D) $\ln 3$

23. $\int_0^2 [x^2 - 1] dx =$

A) $3 - \sqrt{3} - \sqrt{2}$ B) $3 + \sqrt{3} + \sqrt{2}$ C) $\sqrt{3} - 1$ D) $\sqrt{2} + 1$

24. If $[x]$ represents greatest integer $\leq x$, then $\int_0^{\sqrt{2}} [x^2] dx$

A) $2 - \sqrt{2}$ B) $2 + \sqrt{2}$ C) $\sqrt{2} + 1$ D) $\sqrt{2} - 1$

25. $\lim_{x \rightarrow 0} \frac{\int_0^x \sin^2 t \cos t dt}{x^3}$

A) 1 B) $\frac{1}{2}$ C) $\frac{1}{3}$ D) $\frac{2}{3}$

26. $2 \int_0^1 \frac{\tan^{-1} x}{x} dx =$

A) $\int_0^{\frac{\pi}{2}} \frac{\sin x}{x} dx$ B) $\int_0^{\frac{\pi}{2}} \frac{x}{\sin x} dx$ C) $\int_0^{\frac{\pi}{2}} \frac{1}{\sin x} dx$ D) $\int_0^{\frac{\pi}{2}} \frac{x}{\sin x} dx$

27. $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \sec^2 t dt}{x \sin x} =$

A) 1 B) 0 C) 3 D) 2

28. If $I_1 = \int_0^{3\pi} f(\sin^2 x) dx$ and $I_2 = \int_0^{\pi} f(\sin^2 x) dx$ then

A) $I_1 = I_2$ B) $I_1 = 2I_2$ C) $I_1 = 4I_2$ D) $I_1 = 3I_2$

29. $I = \int_0^1 e^{x^2} dx \Rightarrow$

A) $1 \geq 0$ B) $1 \leq 1 \leq e$ C) $1 \geq e$ D) < 0

30. If $I_1 = \int_0^1 2^{x^2} dx$, $I_2 = \int_0^1 2^{x^3} dx$, $I_3 = \int_1^2 2^{x^2} dx$, $I_4 = \int_1^2 2^{x^3} dx$ then

A) $I_1 > I_2$ B) $I_2 > I_1$ C) $I_3 > I_4$ D) $I_3 = I_4$

Section - C

1. Let $f(x)$ be a continuous function such that $f(a-x)+f(x)=0$ for all $x \in [0, a]$. Then

$$\int_0^a \frac{dx}{1+e^{f(x)}}$$
 is equal to

- A) a B) $\frac{a}{2}$ C) $f(a)$ D) $\frac{1}{2}f(a)$

2. If $f(y)=e^y$, $g(y)=y$ and $y>0$, $F(t)=\int_0^t f(t-y)g(y)dy$ then

- A) $F(t)=te^t$ B) $F(t)=te^{-t}$ C) $F(t)=1-e^{-t}(1+t)$ D) $F(t)=e^t-(1+t)$

3. $\int_0^1 x(1-x)^n dx =$

- A) $\frac{1}{n+1} - \frac{1}{n+2}$ B) $\frac{1}{n+1} + \frac{1}{n+2}$ C) $\frac{1}{n+1}$ D) $\frac{1}{n+2}$

4. $\int_0^1 \tan^{-1}\left(\frac{2x-1}{1+x-x^2}\right) dx =$

- A) 0 B) 1 C) $\frac{\pi}{4}$ D) $\frac{\pi}{8}$

5. $\int_0^1 \frac{\ln(1+x)}{1+x^2} dx =$

- A) $\frac{\pi}{4} \ln 2$ B) $\frac{\pi}{2} \ln 2$ C) $\frac{\pi}{8} \ln 2$ D) $\pi \ln 2$

6. If $\int_0^{\frac{\pi}{4}} \ln(\sin x) dx = -\frac{\pi}{2} \ln 2$ then $\int_0^{\frac{\pi}{4}} \ln(1+\cos x) dx =$

- A) $\pi \ln 2$ B) $-\pi \ln 2$ C) $\frac{\pi}{2} \ln 2$ D) $\ln 2$

7. $\int_0^{\frac{\pi}{2}} \frac{x \tan x}{\sec x + \tan x} dx =$

- A) $\frac{\pi}{2} + 1$ B) $\frac{\pi}{2} - 1$ C) $\pi\left(\frac{\pi}{2} - 1\right)$ D) $\pi\left(\frac{\pi}{2} + 1\right)$

8. $\int_0^{\frac{\pi}{2}} x f(\sin x) dx$ is equal to

- A) $\pi \int_0^{\frac{\pi}{2}} f(\cos x) dx$ B) $\pi \int_0^{\frac{\pi}{2}} f(\sin x) dx$ C) $\frac{\pi}{2} \int_0^{\frac{\pi}{2}} f(\sin x) dx$ D) $\pi \int_0^{\frac{\pi}{2}} f(\cos x) dx$

9. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x}{1+e^x} dx =$

- A) 0 B) -1 C) 1 D) 2π

10. $\int_2^8 \frac{[x^2]}{[x^2 - 20x + 100] + [x^2]} dx =$

- A) 0 B) 10 C) 3 D) 12

11. $\int_{-\pi}^{\pi} \frac{x \cos x}{1+\sin^2 x} dx =$

- A) 1 B) 0 C) -1 D) $\frac{1}{2}$

12. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos \theta (1+\sin \theta)^2 d\theta =$

- A) $\frac{14}{3}$ B) $\frac{8}{3}$ C) $\frac{5}{14}$ D) 0

13. $\int_0^{\pi} \frac{\tan x}{\sec x + \cos x} dx =$

- A) π B) $\frac{\pi}{4}$ C) $\frac{\pi}{2}$ D) 2π

14. If n is an integer then the value of $\int_0^{\pi} \cos^2 x \cos^3((2n+1)x) dx =$

- A) 1 B) 0 C) -1 D) $\frac{1}{2}$

15. $\int_0^{100} e^{x-[x]} dx =$

- A) $100 e$ B) $100(e-1)$ C) $100(e+1)$ D) $100(1-e)$

16. $\int_0^{\frac{\pi}{4}} \sqrt{\frac{1-\sin 2x}{1+\sin 2x}} dx =$

- A) $\frac{1}{2} \ln 2$ B) $-\frac{1}{2} \ln 2$ C) $\ln 2$ D) $-\ln 2$

17. $\int_0^1 |\sin 2\pi x| dx =$

- A) 2π B) $\frac{2}{\pi}$ C) $\frac{\pi}{2}$ D) π

18. $\int_0^{\pi} |\cos \theta - \sin \theta| d\theta =$

- A) 2 B) $\sqrt{2}$ C) $2\sqrt{2}$ D) 4

19. The value of $\int_1^a [x] f'(x) dx$, $a > 1$ where $[x]$ denotes the greatest integer not exceeding x is

- A) $af(a) - \{f(1) + f(2) + \dots + f([a])\}$ B) $[a]f(a) - \{f(1) + f(2) + \dots + f([a])\}$
C) $[a]f([a]) - \{f(1) + f(2) + \dots + f([a])\}$ D) $af([a]) - \{f(1) + f(2) + \dots + f([a])\}$

20. $\int_0^{\pi} \sin^8 \frac{x}{2} dx =$

- A) $\frac{35\pi}{128}$ B) $\frac{5\pi}{128}$ C) $\frac{25\pi}{128}$ D) $\frac{3\pi}{128}$

21. If $I_n = \int_0^{\frac{\pi}{4}} \tan^n x dx$ then $\lim_{n \rightarrow \infty} n(I_n + I_{n+2}) =$

- A) $\frac{1}{2}$ B) 1 C) 7 D) 0

22. $\int_0^{\frac{\pi}{2}} \sin^8 x \cdot \cos^2 x dx =$

- A) $\frac{\pi}{512}$ B) $\frac{3\pi}{512}$ C) $\frac{5\pi}{512}$ D) $\frac{7\pi}{512}$

23. If $\int_0^{\frac{\pi}{2}} \sin^6 x dx = \frac{5\pi}{32}$ then $\int_{-\pi}^{\pi} (\sin^6 x + \cos^6 x) dx =$

- A) $\frac{5\pi}{8}$ B) $\frac{5\pi}{16}$ C) $\frac{5\pi}{32}$ D) $\frac{5\pi}{4}$

24. $\int_0^3 \sqrt{\frac{x^3}{3-x}} dx =$

- A) $\frac{17\pi}{8}$ B) $\frac{27\pi}{8}$ C) $\frac{34\pi}{17}$ D) 1

25. $\int_0^a x^3 (ax - x^2)^{\frac{3}{2}} dx =$

- A) $\frac{9\pi a^7}{2048}$ B) $\frac{7\pi a^7}{2048}$ C) $\frac{\pi a^7}{1024}$ D) $\frac{5\pi a^7}{2048}$

26. $\int_0^{2\pi} x \sin^6 x \cos^5 x dx =$

- A) 0 B) $\frac{5\pi}{32}$ C) $\frac{8\pi}{693}$ D) $\frac{35\pi}{64}$

27. $\int_0^{\infty} \frac{x^3}{(1+x^2)^{\frac{9}{2}}} dx =$

- A) $\frac{1}{35}$ B) $\frac{3}{35}$ C) $\frac{2}{35}$ D) $\frac{4}{35}$

28. $\int_0^{2\pi} x \sin^4 x \cos^6 x dx =$

- A) 0 B) $\frac{3\pi^2}{128}$ C) $\frac{5\pi^2}{128}$ D) $\frac{3\pi^2}{64}$

29. If $I_n = \int_1^e (\log x)^n dx$ then $I_8 + 8I_7 =$

- A) 1 B) e C) 2 D) $2e$

30. $I_{mn} = \int_0^1 x^m (\ln x)^n dx =$

- A) 0 B) $\frac{n}{m+1}$ C) $-\frac{n}{m+1} I_{m-1,n}$ D) $-\frac{n}{m+1} I_{m,n-1}$

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Key**Section - A**

1.	B	2.	C	3.	C	4.	B	5.	C	6.	D	7.	B
8.	D	9.	C	10.	B	11.	B	12.	B	13.	A	14.	C
15.	C	16.	C	17.	D	18.	A	19.	C	20.	A	21.	A
22.	B	23.	A	24.	A	25.	A	26.	A	27.	B	28.	A
29.	C	30.	D										

Section - B

1.	B	2.	A	3.	D	4.	C	5.	D	6.	B	7.	A
8.	A	9.	A	10.	A	11.	A	12.	A	13.	D	14.	C
15.	D	16.	B	17.	A	18.	C	19.	A	20.	C	21.	C
22.	B	23.	A	24.	D	25.	C	26.	B	27.	A	28.	D
29.	B	30.	A										

Section - C

1.	B	2.	D	3.	A	4.	A	5.	C	6.	B	7.	C
8.	D	9.	C	10.	C	11.	B	12.	B	13.	C	14.	B
15.	B	16.	A	17.	B	18.	C	19.	B	20.	A	21.	B
22.	D	23.	D	24.	B	25.	A	26.	A	27.	C	28.	B
29.	B	30.	D										

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SINGLE ANSWER TYPE

- $\int \frac{x^2 - 1}{x^3 \sqrt{2x^4 - 2x^2 + 1}} dx$ is equal to
 (a) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x^2} + C$ (b) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x^3} + C$
 (c) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x} + C$ (d) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{2x^2} + C$
- The value of $\int \frac{x \cot x \operatorname{cosec} x}{(x + \cot x)^2} dx$ is equal to
 (A) $\frac{-1}{(x \tan x + 1)} + C$ (B) $\frac{-1}{(x \sin x + \cos x)} + C$ (C) $\frac{-1}{(x + \cot x)} + C$ (D) $\frac{1}{(x \sin x + \cos x)} + C$
- If $\int \frac{x dx}{\sqrt[2012]{(1+x^2)^{1012} (2+x^2)^{3012}}} = \frac{\alpha}{\beta} (1-f(x))^{\frac{\beta}{2\alpha}} + k$ (where α and β are coprime integers) then which of the following options are true
 (A) $\alpha = 503; \beta = 500, f(\sqrt{2}) = \frac{1}{\beta - \alpha}$ (B) $\alpha = 503; \beta = 250, f(\sqrt{2}) = \frac{1}{\alpha - \beta}$
 (C) $\alpha = 503; \beta = 500, f(1) = \frac{1}{\alpha - \beta}$ (D) $\alpha = 503; \beta = 225, f(\sqrt{3}) = \frac{1}{\alpha - \beta}$
- If $\int \frac{\sec^2 x - 2010}{\sin^{2010} x} dx = \frac{P(x)}{\sin^{2010} x} + C$, then value of $P\left(\frac{\pi}{3}\right)$ is
 (A) 0 (B) $\frac{1}{\sqrt{3}}$ (C) $\sqrt{3}$ (D) $-\sqrt{3}$
- If $xf(x) = 3f^2(x) + 2$ then $\int \frac{2x^2 - 12xf(x) + f(x)}{(6f(x) - x)(x^2 - f(x))^2} dx =$
 A) $\frac{1}{x^2 - f(x)} + c$ B) $\frac{1}{x^2 + f(x)} + c$ C) $\frac{1}{2x^2 - f(x)} + c$ D) $\frac{2}{x^2 - f(x)} + c$
- For $x \in \left(0, \frac{\pi}{2}\right)$, If $\left(\int \frac{\tan x \sqrt{\sec x} (1 + \cos^2 x)}{\sqrt{\cos x + \sin^2 x}} dx = 2\sqrt{f(x)+1} + c\right)$ and $f\left(\frac{\pi}{3}\right) = \frac{3}{2}$ then $\left[f\left(\frac{\pi}{4}\right)\right] =$
 (where $[.]$ denotes G.I.F)
 A) 1/2 B) 1 C) 2 D) 0
- If $f'(x) = 3x^2 \sin \frac{1}{x} - x \cos \frac{1}{x}$, $x \neq 0$, $f(0) = 0$ then the value of $f\left(\frac{1}{\pi}\right)$ is
 A) 0 B) 1 C) 2 D) 3
- If y is implicit differentiable function of x such that $y(x+y)^2 = x$ then $\int \frac{dx}{x+3y} =$

A) $\ln((x+y)^2+1)+c$ B) $\frac{1}{2}\ln((x-y)^2+1)+c$ C) $\ln((x-y)^2+1)+c$ D) $\frac{1}{2}\ln((x+y)^2+1)+c$

9. $\int \frac{e^x(x-1)(x-\ln x)}{x^2} dx$ equals to

A) $e^x\left(\frac{x-\ln x}{x}\right)+c$ B) $e^x\left(\frac{x-\ln x+1}{x}\right)+c$ C) $e^x\left(\frac{x-\ln x}{x^2}\right)+c$ D) $e^x\left(\frac{x-\ln x-1}{x}\right)+c$

10. If $x \in (0,1)$, then $\int \sqrt{x} \tan \left(2 \tan^{-1} \left(\frac{\sqrt{\sqrt{1+\sqrt{x}}+1}-\sqrt{\sqrt{1+\sqrt{x}}-1}}{\sqrt{\sqrt{1+\sqrt{x}}+1}+\sqrt{\sqrt{1+\sqrt{x}}-1}} \right) \right) dx$ equals to

A) $\frac{2}{3}x^{3/4}+c$ B) $\frac{4}{3}x^{3/4}+c$ C) $\frac{4}{5}x^{5/4}+c$ D) $\frac{2}{5}x^{5/4}+c$

11. The value of $\frac{\int_0^1 (1-x^2)^{1/3} dx}{\int_0^1 (1-x^3)^{1/2} dx}$ is equal to

A) 1 B) 2 C) $\frac{1}{2}$ D) $\frac{1}{6}$

12. If $\int (x^{24}+x^{16}+x^8)(2x^{16}+3x^8+6)^{1/8} dx = c + \frac{1}{\alpha} (2x^{24}+3x^{16}+6x^8)^{\beta/\gamma}$ (β, γ are coprime natural numbers) then the value of $\alpha+\beta+\gamma$ is

A) 51 B) 61 C) 71 D) 81

13. The value of $\int x \cdot \frac{\ln(x+\sqrt{1+x^2})}{\sqrt{1+x^2}} dx$ equals

a) $\sqrt{1+x^2} \ln(x+\sqrt{1+x^2}) - x + C$ b) $\frac{x}{2} \cdot \ln^2(x+\sqrt{1+x^2}) - \frac{x}{\sqrt{1+x^2}} + C$

c) $\frac{x}{2} \cdot \ln^2(x+\sqrt{1+x^2}) + \frac{x}{\sqrt{1+x^2}} + C$ d) $\sqrt{1+x^2} \ln(x+\sqrt{1+x^2}) + x + C$

14. The value of $2 \int \sin x \cdot \operatorname{cosec} 4x dx$ is equal to

A) $\frac{1}{2\sqrt{2}} \ln \left| \frac{1+\sqrt{2} \sin x}{1-\sqrt{2} \sin x} \right| - \frac{1}{4} \ln \left| \frac{1+\sin x}{1-\sin x} \right| + C$ b) $\frac{1}{2\sqrt{2}} \ln \left| \frac{1+\sqrt{2} \sin x}{1-\sqrt{2} \sin x} \right| + \frac{1}{4} \ln \left| \frac{1+\sin x}{1-\sin x} \right| + C$

c) $\frac{1}{2\sqrt{2}} \ln \left| \frac{1-\sqrt{2} \sin x}{1+\sqrt{2} \sin x} \right| - \frac{1}{4} \ln \left| \frac{1+\sin x}{1-\sin x} \right| + C$ d) $\frac{1}{2\sqrt{2}} \ln \left| \frac{1-\sqrt{2} \sin x}{1+\sqrt{2} \sin x} \right| - \frac{1}{4} \ln \left| \frac{1+\cos x}{1-\cos x} \right| + C$

15. $\int e^{\sin x} \left(\frac{x \cos^3 x - \sin x}{\cos^2 x} \right) dx$, is equal to

1) $e^{\sin x} (\tan x + x) + C$ 2) $e^{\sin x} (x - \sec x) + C$ 3) $e^{\sin x} (\sec x + \tan x) + C$ 4) none of these

16. $\frac{\sin^3 x}{(\cos^4 x + 3\cos^2 x + 1) \tan^{-1}(\sec x + \cos x)} dx =$

1) $\tan^{-1}(\sec x + \cos x) + c$ 2) $\log |\tan^{-1}(\sec x + \cos x)| + c$

3) $\frac{1}{(\sec x + \cos^2 x)^2} + c$ 4) $\log |\sec x + \cos x| + c$

17. $\int (\sqrt{\sin x} + \sqrt{\cos x})^{-4} dx =$

$$1) \frac{2}{(\sqrt{\tan x} + 1)^2} \left[\frac{1}{3(\sqrt{\tan x} + 1)} - \frac{1}{2} \right] + c \quad 2) \frac{2}{(\sqrt{\tan x} + 1)^2} \left[\frac{1}{3(\sqrt{\tan x} + 1)} + \frac{1}{2} \right] + c$$

$$3) \frac{2}{(\sqrt{\tan x} + 1)^2} \left[\frac{1}{\sqrt{3}(\sqrt{\tan x} + 1)} + \frac{1}{2} \right] + c \quad 4) \frac{2}{(\sqrt{\tan x} + 1)^2} \left[\frac{1}{\sqrt{3}(\sqrt{\tan x} + 1)} - \frac{1}{2} \right] + c$$

18. $\int \sqrt{x + \sqrt{x^2 + 2}} dx =$

$$1) \left(\frac{x + \sqrt{x^2 + 2}}{3} \right)^{\frac{3}{2}} - 2(x + \sqrt{x^2 + 2})^{-\frac{1}{2}} + c \quad 2) \frac{1}{3}(x + \sqrt{x^2 + 2})^{\frac{3}{2}} - 2(x + \sqrt{x^2 + 2})^{-\frac{1}{2}} + c$$

$$3) \frac{2}{7}(x + \sqrt{x^2 + 2})^{\frac{7}{2}} - 2(x + \sqrt{x^2 + 2}) + c \quad 4) \frac{2}{7}(x + \sqrt{x^2 + 2})^{\frac{7}{2}} + 2(x + \sqrt{x^2 + 2}) + c$$

19. $\int \frac{\cos x + x \sin x}{x(x + \cos x)} dx =$

$$1) \log \left| \frac{x}{x + \cos x} \right| + c \quad 2) -\log \left| \frac{x}{x + \cos x} \right| + c \quad 3) \log \left| \frac{x + \cos x}{2x} \right| + c \quad 4) -\log \left| \frac{x + \cos x}{2x} \right| + c$$

20. $\int \frac{\cos^3 x + \cos^5 x}{\sin^2 x + \sin^4 x} dx$ equals

$$1) \sin x - 6 \tan^{-1}(\sin x) + c \quad 2) \sin x - 2 \sin^{-1} x + c$$

$$3) \sin x - 2(\sin x)^{-1} - 6 \tan^{-1}(\sin x) + c \quad 4) \sin x - 2(\sin x)^{-1} + 5 \tan^{-1}(\sin x) + c$$

21. The evaluation of $\int \frac{px^{p+2q-1} - qx^{q-1}}{x^{2p+2q} + 2x^{p+q} + 1} dx$ is

$$1) -\frac{x^p}{x^{p+q} + 1} + C \quad 2) \frac{x^q}{x^{p+q} + 1} + C \quad 3) -\frac{x^q}{x^{p+q} + 1} + C \quad 4) \frac{x^p}{x^{p+q} + 1} + C$$

22. $\int \frac{e^{\tan^{-1} x}}{(1+x^2)} \left[\left(\sec^{-1} \sqrt{1+x^2} \right)^2 + \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) \right] dx \quad (x > 0)$

$$1) e^{\tan^{-1} x} \cdot \tan^{-1} x + C \quad 2) \frac{e^{\tan^{-1} x} \cdot (\tan^{-1} x)^2}{2} + C$$

$$3) e^{\tan^{-1} x} \cdot \left(\sec^{-1} \left(\sqrt{1+x^2} \right) \right)^2 + C \quad 4) e^{\tan^{-1} x} \cdot \left(\cos^{-1} \left(\sqrt{1+x^2} \right) \right)^2 + C$$

23. $\int (x^6 + x^4 + x^2) \sqrt{2x^4 + 3x^2 + 6} dx$ is equal to

$$1) \frac{(3x^6 + 2x^4 + 6x^2)^{3/2}}{18} + C \quad 2) \frac{(2x^6 + 3x^4 + 6x^2)^{3/2}}{24} + C$$

$$3) \frac{(2x^6 + 3x^4 + 6x^2)^{3/2}}{18} + C \quad 4) \frac{(3x^6 + 2x^4 + 6x^2)^{3/2}}{24} + C$$

24. If $\int \frac{dx}{\cos^3 x \sqrt{2 \sin 2x}} = \phi(x) + \frac{1}{5}(\tan x)^{5/2} + c$ then $\phi\left(\frac{\pi}{4}\right)$ equals

$$1) 1 \quad 2) \frac{1}{2} \quad 3) 2 \quad 4) 2\sqrt{2}$$

25. If $\int \frac{\cos^3 x + \cos^5 x}{\sin^2 x + \sin^4 x} dx =$

$$1) \sin x - 2 \operatorname{cosec} x - 6 \tan^{-1}(\sin x) + c \quad 2) \sin x - 2 \operatorname{cosec} x + 5 \tan^{-1}(\sin x) + c$$

$$3) \sin x + 2 \sec x + \tan^{-1}(\sin x) + c \quad 4) \sin x - 6 \tan^{-1}(\sin x) + c$$

26. If $\int \frac{x^2 - 1}{x^3 \sqrt{3x^4 - 2x^2 + 1}} dx = \frac{\sqrt{3x^4 - 2x^2 + 1}}{\phi(x)} + c$ then $\phi'(1), \phi'(2), \phi'(3)$ are in ____

- 1) AP 2) GP 3) HP 4) AGP

27. If $\int \ln(\sqrt{1-x} + \sqrt{1+x}) dx = x \ln(\sqrt{1-x} + \sqrt{1+x}) + \frac{1}{2}(f(x) - x) + c$ then the value of $f'(0)$ equals

- 1) 0 2) 1 3) 2 4) $\frac{1}{2}$

28. If $\int \frac{\cos 9x + \cos 6x}{2 \cos 5x - 1} dx = \lambda \sin x (\cos x \cos 2x + 1) + c$ then λ equals

- 1) 1 2) $\frac{1}{2}$ 3) 2 4) $\frac{1}{4}$

29. If $\int \frac{(1 + \ln x)^2}{1 + \ln x^{x+1} + (\ln(x^{\sqrt{x}}))^2} dx = \underline{\hspace{2cm}}$

- 1) $\ln(x + \ln x) + c$ 2) $\ln(1 + x \ln x) + c$ 3) $\ln(x - \ln x) + c$ 4) $\ln(1 - x \ln x) + c$

30. If $\int e^x \left(\frac{x^3 - x + 2}{(x^2 + 1)^2} \right) dx = e^x f(x) + c$ then $f'(-1)$ equals

- 1) 1 2) 2 3) $\frac{1}{2}$ 4) -2

31. Let $I_1 = \int \frac{e^x}{e^{4x} + e^{2x} + 1} dx$, $I_2 = \int \frac{e^{-x}}{e^{-4x} + e^{-2x} + 1} dx$ and $I_2 - I_1 = \lambda \ln \left(\frac{e^{2x} - e^x + 1}{e^{2x} + e^x + 1} \right) + c$ then

- 1) $\lambda = 1$ 2) $\lambda = \frac{1}{2}$ 3) $\lambda = 2$ 4) $\lambda = \frac{1}{\sqrt{2}}$

32. $\int \frac{x^2 - 1}{x \sqrt{(x^2 + \alpha x + 1)(x^2 + \beta x + 1)}} dx$ is equal to

- 1) $\log \left[\frac{\sqrt{x^2 + \alpha x + 1} + \sqrt{x^2 + \beta x + 1}}{\sqrt{x}} \right] + c$ 2) $2 \log \left[\frac{\sqrt{x^2 + \alpha x + 1} - \sqrt{x^2 + \beta x + 1}}{\sqrt{x}} \right] + c$
 3) $\log \left[\sqrt{x^2 + \alpha x + 1} - \sqrt{x^2 + \beta x + 1} \right] + c$ 4) $2 \log \left[\frac{\sqrt{x^2 + \alpha x + 1} + \sqrt{x^2 + \beta x + 1}}{\sqrt{x}} \right] + c$

33. $\int \frac{x^2}{(x \cos x - \sin x)^2} dx =$

- 1) $\cot x + \frac{x \cos x}{x \cos x - \sin x} + c$ 2) $-\left[\cot x + \frac{x \cos x}{x \cos x - \sin x} \right] + c$
 3) $-\cot x + \frac{x \cos x}{x \cos x - \sin x} + c$ 4) $\cot x - \frac{x \cos x}{x \cos x - \sin x} + c$



SINGLE ANSWER TYPE

- The value of $\int \frac{\sin 2x + 2 \tan x}{(\cos^6 x + 6 \cos^2 x + 4)} dx =$
 - $\sqrt[3]{\frac{1 + \cos^2 x}{\cos^7 x}} + c$
 - $\tan^{-1} \frac{1}{\sqrt{2}} \left(\frac{1 + \cos^2 x}{\cos^7 x} \right) + c$
 - $\frac{1}{12} \ln \left(\frac{1 + \cos^2 x}{\cos^7 x} \right) + c$
 - None of these
- $\int e^{x^x} [1 + x^x (x + \ln x^x)] dx =$
 - $e^{x^x} + c$
 - $xe^{x^x} + c$
 - $x^{x^x} \cdot e^{x^x} + c$
 - $\frac{1}{x} e^{x^x} + c$
- If $\int e^x (2 \sin^2 x + 3 \sin 2x) \sin x \, dx = e^x \cdot (\phi(x)) + c$ then which is false
 - $\phi\left(\frac{\pi}{2}\right) = 2$
 - $\phi(\pi) = 0$
 - $\phi(\pi/6) = \frac{1}{2}$
 - $\phi(0) = 0$
- If $\int e^x \left(\frac{x^2 - 2}{x^3} \right) dx = e^x \left(\frac{ax^2 + bx + c}{x^2} \right) + d$ then the value of $a+b+c$ equals
 - 1
 - 2
 - $\frac{1}{2}$
 - 3
- $\int \cos 2\theta \cdot \ln \left(\frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta} \right) d\theta = K \ln(\cos 2\theta) + \sin 2\theta \cdot \ln \sqrt{\frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta}} + c$ then K equals
 - 1
 - $\frac{1}{2}$
 - 2
 - $\frac{1}{4}$
- If $\int \sin \left(2 \tan^{-1} \sqrt{\frac{1-x}{1+x}} \right) dx = A \sin^{-1} x + Bx\sqrt{1-x^2} + c$ then $A+B=$
 - 1
 - 1
 - 0
 - 2
- $\int e^x \frac{(x^2 + 5x + 7)}{(x+3)^2} dx = e^x \phi(x) + c$ then number of solutions of $\phi(x) = x$ equals
 - 0
 - 1
 - 2
 - 3
- $\int \frac{e^x (2 - x^2)}{(1-x)\sqrt{1-x^2}} dx =$
 - $e^x \left(\frac{\sqrt{2+x}}{\sqrt{2-x}} \right) + c$
 - $e^x \left(\frac{\sqrt{2-x}}{\sqrt{2+x}} \right) + c$
 - $e^x \sqrt{\frac{1-x}{1+x}} + c$
 - $e^x \sqrt{\frac{1+x}{1-x}} + c$
- If $I_n = \int \tan^n x \, dx$, ($n \in \mathbb{N}$) then the value of $(n-1)(I_n + I_{n-2}) + (n-3)(I_{n-2} + I_{n-4})$ at $x = \pi/4$ equals
 - 0
 - 1
 - 2
 - 4
- If $I_{m,n} = \int x^m (\ln x)^n \, dx$, ($m, n \in \mathbb{N}$) then $I_{6,5} + \frac{5}{7} I_{6,4} =$

A) $\frac{1}{5}x^5(\ln x)^7$ B) $\frac{1}{7}x^7(\ln x)^5$ C) $\frac{1}{6}x^7(\ln x)^5$ D) $\frac{1}{7}x^6(\ln x)^7$

11. $\int \frac{2e^{5x} + e^{4x} - 4e^{3x} + 4e^{2x} + 2e^x}{(e^{2x} + 4)(e^{2x} - 1)^2} dx$

(A) $\tan^{-1} \frac{e^x}{2} - \frac{1}{e^{2x} - 1} + C$

(B) $\tan^{-1} e^x - \frac{1}{2(e^{2x} - 1)} + C$

(C) $\tan^{-1} \left(\frac{e^x}{2} \right) - \frac{1}{2(e^{2x} - 1)} + C$

(D) $1 - \tan^{-1} \left(\frac{e^x}{2} \right) + \frac{1}{2(e^{2x} - 1)} + C$

12. The integral $\int \frac{\sin 2x \left(1 - \frac{3}{2} \cos x \right) dx}{e^{\sin^2 x + \cos^3 x}} =$

(A) $e^{\sin^2 x + \cos^3 x} + c$ (B) $-e^{-(\sin^2 x + \cos^3 x)} + c$ (C) $e^{(\sin^2 x + \cos^3 x)^2} + c$ (D) None of these

13. $\int \left(\left(\frac{x}{e} \right)^x + \left(\frac{e}{x} \right)^x \right) \log_e x \, dx$

(A) $\left(\frac{x}{e} \right)^x - \left(\frac{e}{x} \right)^x + C$

(B) $\left(\frac{x}{e} \right)^x + \left(\frac{e}{x} \right)^x + C$

(C) $-\left(\frac{x}{e} \right)^x + \left(\frac{e}{x} \right)^x + C$

(D) None

14. If $\int (2 - 3 \sin^2 x) \sqrt{\sec x} dx = 2f(x) \sqrt{g(x)} + c$ and $f(x)$ is non-constant function then

(A) $f^2(x) + g^2(x) = 1$

(B) $f^2(x) - g^2(x) = 1$

(C) $f(x)g(x) = 1$

(D) $f(x) = g(x)$

15. The value of $\int \frac{x \cos x + 1}{\sqrt{2x^3 e^{\sin x} + x^2}} dx$ is

A) $\ln \left| \frac{\sqrt{2x e^{\sin x} + 1} - 1}{\sqrt{2x e^{\sin x} + 1} + 1} \right| + c$

B) $\ln \left| \frac{\sqrt{2x e^{\sin x} - 1} - 1}{\sqrt{2x e^{\sin x} - 1} + 1} \right| + c$

C) $\ln \left| \frac{\sqrt{2x e^{\sin x} - 1} + 1}{\sqrt{2x e^{\sin x} - 1} - 1} \right| + c$

D) $\ln \left| \frac{\sqrt{2x e^{\sin x} + 1} + 1}{\sqrt{2x e^{\sin x} - 1} + 1} \right| + c$

MULTIPLE ANSWER TYPE

16. Let $I = \int \frac{e^x}{e^{4x} + e^{2x} + 1} dx$, $J = \int \frac{e^{-x}}{e^{-4x} + e^{-2x} + 1} dx$. Then for an arbitrary constant c, the value of $J - I =$

A) $\frac{1}{2} \log \left(\frac{e^{4x} - e^{2x} + 1}{e^{4x} + e^{2x} + 1} \right) + c$

B) $-\frac{1}{2} \log \left(\frac{e^{2x} + e^x + 1}{e^{2x} - e^x + 1} \right) + c$

C) $\frac{1}{2} \log \left(\frac{e^{2x} - e^x + 1}{e^{2x} + e^x + 1} \right) + c$

D) $\frac{1}{2} \log \left(\frac{e^{4x} + e^{2x} + 1}{e^{4x} - e^{2x} + 1} \right) + c$

17. $\int \frac{\sqrt{\cos 2x}}{\sin x} dx = A \log |\cot x + \sqrt{\cot^2 x - 1}| + \frac{1}{B} \log \left| \frac{C + \sqrt{1 - \tan^2 x}}{C - \sqrt{1 - \tan^2 x}} \right| + K$, then

A) $A = 1$

B) $A = -1$

C) $B + C = 0$

D) $B - C = 0$

18. If $I = \int e^{x \sin x + \cos x} \left(\frac{x^4 \cos^3 x - x \sin x + \cos x}{x^2 \cos^2 x} \right) dx$ then I equals

(A) $e^{x \sin x + \cos x} \left(x - \frac{\sec x}{x} \right) + C$ (B) $e^{x \sin x + \cos x} \left(x \sin x - \frac{\cos x}{x} \right) + C$

(C) $e^{x \sin x + \cos x} \left(\frac{x}{\tan x} - \frac{\sec x}{x} \right) + C$ (D) $x e^{x \sin x + \cos x} - \int e^{x \sin x + \cos x} \left(1 - \frac{\cos x - x \sin x}{x^2 \cos^2 x} \right) dx$

INTEGER ANSWER TYPE

19. If $\int \frac{dx}{(\sec x + \tan x + \cos ecx + \cot x)^2} = \frac{x}{a} + \frac{\sqrt{2} \cos(x + \pi/4)}{b} + \frac{\cos 2x}{c} + k$ (k is integrating constant) then find $|a + b + c|$
20. $\int \frac{\sec x (2 + \sec x)}{(1 + 2 \sec x)^2} dx = \frac{\sin x}{p + \cos x} + C$, then $P =$
21. $\int \frac{(\sin^{3/2} x + \cos^{3/2} x)}{(\sin^3 x \cos^3 x \sin(x+a))^{1/2}} dx = 2 \left(\frac{1}{P} \sqrt{\cos a + \sin a \cot x} + \frac{1}{Q} \sqrt{\cos a \tan x + \sin a} + c \right)$, then $P^2 + Q^2 =$
22. $\int \frac{x}{x - \sqrt{x^2 - 1}} dx = Ax^3 + B(x^2 - 1)^{3/2} + C$, then $2A + B =$
23. $\int \frac{\cos x - \sin x + 1 - x}{e^x + \sin x + x} dx = \ln f(x) - x + c$ and $f(x)$ is positive then $\frac{f(x) - x}{e^x + \sin x} =$ ($x > 0$)
24. $\sqrt{1 + 2 \cot x (\cot x + \cos ecx)} dx = A \log_e \left(\sin \frac{x}{2} \right) + C$, then $A =$ [$0 < x < \pi$]
25. For $x \in (0, \pi/2)$, If $\int \frac{\tan x \sqrt{\sec x (1 + \cos^2 x)}}{\sqrt{\cos x + \sin^2 x}} dx = 2\sqrt{f} + c$
Then $\left[f\left(\frac{\pi}{4}\right) \right] =$ (where $[.]$ denotes G.I.F)

PASSAGE-1

Integrals of class of functions following a definite pattern can be found by the method of reduction and recursion. Reduction formulas make it possible to reduce an integral dependent on the index $n > 0$, called the order of the integral, to an integral of the same type with a smaller index. Integration by parts helps us to derive reduction formulas.

26. If $I_n = \int \frac{dx}{(x^2 + a^2)^n}$ then $I_{n+1} + \frac{1-2n}{2n} \cdot \frac{1}{a^2} I_n$ is equal to
(A) $\frac{x}{(x^2 + a^2)^n}$ (B) $\frac{1}{2na^2} \frac{1}{(x^2 + a^2)^{n-1}}$ (C) $\frac{1}{2na^2} \frac{1}{(x^2 + a^2)^n}$ (D) $\frac{1}{2na^2} \frac{x}{(x^2 + a^2)^n}$
27. If $I_{n,-m} = \int \frac{\sin^n x}{\cos^m x} dx$, then $I_{n,-m} + \frac{n-1}{m-1} I_{n-2,-m}$ is equal to
(A) $\frac{\sin^{n-1} x}{\cos^{m-1} x}$ (B) $\frac{1}{(m-1)} \frac{\sin^{n-1} x}{\cos^{m-1} x}$ (C) $\frac{1}{(n-1)} \frac{\sin^{n-1} x}{\cos^{m-1} x}$ (D) $\frac{n-1}{m-1} \frac{\sin^{n-1} x}{\cos^{m-1} x}$
28. If $u_n = \int \frac{x^n}{\sqrt{ax^2 + 2bx + c}} dx$, then $(n+1)au_{n+1} + (2n+1)bu_n + nc u_{n-1}$ is equal to
(A) $x^{n-1} \sqrt{ax^2 + 2bx + c}$ (B) $\frac{x^{n-2}}{\sqrt{ax^2 + 2bx + c}}$
(C) $\frac{x^n}{\sqrt{ax^2 + 2bx + c}}$ (D) $x^n \sqrt{ax^2 + 2bx + c}$

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SINGLE ANSWER TYPE

1. $\int \frac{dx}{(\sec x + \csc x + \tan x + \cot x)^2} = \frac{x}{a} + \frac{\sqrt{2} \cos\left(x + \frac{\pi}{4}\right)}{b} + \frac{\cos 2x}{c} + d$ then $a+b+c =$

1) -2 2) -4 3) 2 4) 0

2. If $f(x) = \int \frac{x^2}{(1+x^2)(1+\sqrt{1+x^2})} dx$ and $f(0) = 0$ then $f(1) =$

1) $\log(1+\sqrt{2})$ 2) $\log(1+\sqrt{2}) - \frac{\pi}{4}$ 3) $\log(1+\sqrt{2}) + \frac{\pi}{4}$ 4) $\log(3+\sqrt{2}) - \frac{\pi}{4}$

3. If $I = \int \frac{x^2 + 20}{(x \sin x + 5 \cos x)^2} dx$, then $I =$

1) $\frac{-x}{\cos x(x \sin x + 5 \cos x)} + \tan x + c$ 2) $\frac{x}{\cos x(x \sin x + 5 \cos x)} + \cot x + c$

3) $\frac{-x}{\cos x(x \sin x + 5 \cos x)} + \cot x + c$ 4) $\frac{x}{\cos x(x \sin x + 5 \cos x)} + \tan x + c$

4. $\int \frac{3+2\cos x}{(2+3\cos x)^2} dx$

1) $\frac{\sin x}{3\cos x + 2} + C$ 2) $\frac{2\cos x}{3\sin x + 2} + C$ 3) $\frac{2\cos x}{3\cos x + 2} + C$ 4) $\frac{2\sin x}{3\sin x + 2} + C$

MULTIPLE ANSWER TYPE

5. The value of $\int \frac{dx}{1+e \cos x}$ must be same as
- (A) $\frac{1}{\sqrt{1-e^2}} \tan^{-1} \left(\sqrt{\frac{1-e}{1+e}} \tan \frac{x}{2} \right) + c$ (e lies between 0 and 1)
- (B) $\frac{2}{\sqrt{1-e^2}} \tan^{-1} \left(\sqrt{\frac{1-e}{1+e}} \tan \frac{x}{2} \right) + c$, (e lies between 0 and 1)
- (C) $\frac{1}{\sqrt{e^2-1}} \log \frac{e + \cos x + \sqrt{e^2-1} \sin x}{1+e \cos x} + c$, (e is greater than 1)
- (D) $\frac{2}{\sqrt{e^2-1}} \log \frac{e + \cos x + \sqrt{e^2-1} \sin x}{1+e \cos x} + c$, (e is greater than 1)
6. If $\int \frac{3 \cot 3x - \cot x}{\tan x - 3 \tan 3x} dx = p f(x) + q g(x) + C$, where, 'C' is a constant of integration and $f(2014) = 2014$, then
- A) $p=1; q=\frac{1}{\sqrt{3}}; f(x)=x; g(x)=\ln \left| \frac{\sqrt{3}-\tan x}{\sqrt{3}+\tan x} \right|$

B) $p=1; q=-\frac{1}{\sqrt{3}}; f(x)=x; g(x)=\ln\left|\frac{\sqrt{3}-\tan x}{\sqrt{3}+\tan x}\right|$

C) $p=1; q=-\frac{2}{\sqrt{3}}; f(x)=x; g(x)=\ln\left|\frac{\sqrt{3}+\tan x}{\sqrt{3}-\tan x}\right|$

D) $p=1; q=-\frac{1}{\sqrt{3}}; f(x)=x; g(x)=\ln\left|\frac{\sqrt{3}+\tan x}{\sqrt{3}-\tan x}\right|$

7. $\int \frac{(x-1)dx}{x^2\sqrt{2x^2-2x+1}} = \frac{\sqrt{f(x)}}{g(x)} + C$, then ($x > 0$ and $f(0) = 1$)

a) $f(x) = 2x^2 - 2x + 1$ b) $g(x) = x + 1$ c) $g(x) = x$ d) $f(x) = \sqrt{2x^2 - 2x}$

8. $\int \frac{(x + \sqrt{1+x^2})^{15}}{\sqrt{1+x^2}} dx$ is equal to

(A) $\frac{(x + \sqrt{1+x^2})^{16}}{10} + c$ (B) $\frac{1}{15(\sqrt{1+x^2} + x)} + c$ (C) $\frac{15}{(\sqrt{1+x^2} - x)} + c$ (D) $\frac{(x + \sqrt{1+x^2})^{15}}{15} + c$

9. Let $f(x) = \frac{4+5\cos x}{5+4\cos x} \left(x \in \left(0, \frac{\pi}{2}\right) \right)$ and $\int \frac{dx}{(5+4\cos x)^2} = \frac{5}{27}g(x) - \frac{4}{27}h(x) + c$

then

A) $g(x) = \tan^{-1}(f(x))$ B) $h(x) = \sqrt{1+f^2(x)}$

C) $g(x) = \cos^{-1}(f(x))$ D) $h(x) = \sqrt{1-f^2(x)}$

INTEGER ANSWER TYPE

10. Let 'f' be twice differentiable function satisfying $\frac{e^x f''(x) - e^x f'(x)}{e^{2x}} = 1$ and $f(0) = 0$, $f'(0) = 1$, then the value of $\left[\int_0^1 \frac{f(x)}{x} dx \right]$ (where [.] denotes G.I.F) is

11. $\int \frac{4e^x + 6e^{-x}}{9e^x - 4e^{-x}} dx = Ax + B \ln(9e^{2x} - 4) + c$ then the least integer greater than or equals to $A+B$ is

12. If $\int \frac{x+2}{\sqrt{4x-x^2}} dx$ equals to $-\sqrt{4x-x^2} + k \sin^{-1} \frac{x-2}{2} + c$ then the value of k is

13. If $\int \frac{3x-5}{(3x-2)(x+1)^2} dx = A \log(3x-2) + B \log(x+1) + \frac{C}{x+1} + 1$ then the value of $A+B$ is

14. For $x \in \left(0, \frac{\pi}{2}\right)$, if $\int \sqrt{\sec x + \tan x} + \sqrt{\sec x - \tan x} dx = \sqrt{2} \sin^{-1}(1 - g(x)) + c$ then $\left[g\left(\frac{\pi}{6}\right) \right] =$ (where [.] denotes G.I.F)

15. For $y < 0$, if y is a differentiable function of x such that $y(x+y) = x$ and $\int \frac{dx}{x+2y} = -\ln(k-y) + c$ where $k \in \mathbb{N}$ then $k =$

MATRIX MATCHING TYPE

16. $I_1 = \int \tan x \tan(ax+b) dx$ and $I_2 = \int \cot x \cot(ax+b) dx$, where a and b are non-zero real numbers.

COLUMN-I	COLUMN-II
(A) value of I_1 for $a=1$ is	(P) $x - \cot b \ln \left(\frac{\cos(x-b)}{\cos x} \right) + C$
(B) value of I_2 for $a=1$ is	(Q) $\cot b \ln \left(\frac{\sin x}{\sin(b+x)} \right) - x + C$
(C) value of I_1 for $a=-1$ is	(R) $\cot b \ln \left(\frac{\cos x}{\cos(x+b)} \right) - x + C$
(D) value of I_2 for $a=-1$ is	(S) $x + \cot b \ln \left(\frac{\sin x}{\sin(b-x)} \right) + C$

PASSAGE-1

Let $I_{n,m} = \int \sin^n x \cos^m x dx$. Then we can relate $I_{n,m}$ with each of the following

- (i) $I_{n-2,m}$ ii) $I_{n+2,m}$ iii) $I_{n,m-2}$ iv) $I_{n,m+2}$ v) $I_{n-2,m+2}$ vi) $I_{n+2,m-2}$

Suppose we want to establish a relation between $I_{n,m}$ and $I_{n,m-2}$, then we set

$$P(x) = \sin^{n+1} x \cos^{m-1} x \quad \dots\dots(1)$$

In $I_{n,m}$ and $I_{n,m-2}$ the exponent of $\cos x$ is m and $m-2$ respectively, the minimum of the two is $m-2$ adding 1 to the minimum we get $m-2+1=m-1$. Now choose the exponent $m-1$ of $\cos x$ in $P(x)$. Similarly choose the exponent of $\sin x$ for $P(x)$

Now differentiating both sides of (1), we get

$$\begin{aligned} P'(x) &= (n+1) \sin^n x \cos^m x - (m-1) \sin^{n+2} x \cos^{m-2} x \\ &= (n+1) \sin^n x \cos^m x - (m-1) \sin^n x (1 - \cos^2 x) \cos^{m-2} x \\ &= (n+1) \sin^n x \cos^m x - (m-1) \sin^n x \cos^{m-2} x + (m-1) \sin^n x \cos^m x \\ &= (n+m) \sin^n x \cos^m x - (m-1) \sin^n x \cos^{m-2} x \end{aligned}$$

Now integrating both sides, we get

$$\sin^{n+1} x \cos^{m-1} x = (n+m) I_{n,m} - (m-1) I_{n,m-2}$$

Similarly we can establish the other relations

17. The relation between $I_{4,2}$ and $I_{2,2}$ is

$$\begin{aligned} \text{a) } I_{4,2} &= \frac{1}{6} (-\sin^3 x \cos^3 x + 3I_{2,2}) & \text{b) } I_{4,2} &= \frac{1}{6} (\sin^3 x \cos^3 x + 3I_{2,2}) \\ \text{c) } I_{4,2} &= \frac{1}{6} (\sin^3 x \cos^3 x - 3I_{2,2}) & \text{d) } I_{4,2} &= \frac{1}{4} (-\sin^3 x \cos^3 x + 2I_{2,2}) \end{aligned}$$

18. The relation between $I_{4,2}$ and $I_{4,4}$ is

$$\begin{aligned} \text{a) } I_{4,2} &= \frac{1}{3} (\sin^5 x \cos^3 x + 8I_{4,4}) & \text{b) } I_{4,2} &= \frac{1}{3} (-\sin^5 x \cos^3 x + 8I_{4,4}) \\ \text{c) } I_{4,2} &= \frac{1}{3} (\sin^5 x \cos^3 x - 8I_{4,4}) & \text{d) } I_{4,2} &= \frac{1}{3} (\sin^5 x \cos^3 x + 6I_{4,4}) \end{aligned}$$

PASSAGE-2

Two real valued differentiable functions $f(x), g(x)$ satisfy the following conditions:

$$\begin{aligned} 1) f'(x) &= \frac{f(x) - g(x)}{3} & 2) g'(x) &= \frac{2(g(x) - f(x))}{3} \\ 3) f(0) &= 5 & 4) g(0) &= -1 \quad \text{then} \end{aligned}$$

19. $\int \frac{1}{f(x)} dx =$

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A) $x - \ln(f(x)) + c$ B) $\frac{1}{2}(x - \ln f(x)) + c$ C) $\frac{1}{3}(x - \ln f(x)) + c$ D) $\frac{1}{x - \ln f(x)} + c$

20. $\int \frac{1}{g(x)} dx =$

A) $x - \ln(g(x)) + c$ B) $\frac{1}{2}(x - \ln|g(x)|) + c$ C) $\frac{1}{3}(x - \ln|g(x)|) + c$ D) $\frac{1}{x - \ln|g(x)|} + c$

21. $\int \frac{g(x)}{f(x)} dx =$

A) $x + 3 \ln f(x) + c$ B) $x - 3 \ln f(x) + c$ C) $3 \ln f(x) - x + c$ D) $3(x - \ln f(x)) + c$

PASSAGE-3

Two real valued differentiable functions $f(x), g(x)$ satisfy the following conditions:

1) $f'(x) = \frac{f(x) - g(x)}{3}$

2) $g'(x) = \frac{2(g(x) - f(x))}{3}$

3) $f(0) = 5$

4) $g(0) = -1$ then

22. $\int \frac{1}{f(x)} dx =$

A) $x - \ln(f(x)) + c$ B) $\frac{1}{2}(x - \ln f(x)) + c$ C) $\frac{1}{3}(x - \ln f(x)) + c$ D) $\frac{1}{x - \ln f(x)} + c$

23. $\int \frac{1}{g(x)} dx =$

A) $x - \ln(g(x)) + c$ B) $\frac{1}{2}(x - \ln|g(x)|) + c$ C) $\frac{1}{3}(x - \ln|g(x)|) + c$ D) $\frac{1}{x - \ln|g(x)|} + c$

24. $\int \frac{g(x)}{f(x)} dx =$

A) $x + 3 \ln f(x) + c$ B) $x - 3 \ln f(x) + c$ C) $3 \ln f(x) - x + c$ D) $3(x - \ln f(x)) + c$

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BASARA GNANA SARASWATHI CAMPUS KAKATIYA HILLS

Sec: Star SC

Indefinite Integration

Date: 28-08-2020

SINGLE ANSWER TYPE

1. If $\int \ln(\sin x)^{\sin x} \cdot \sin 2x \cdot dx = \sin^3 x (A \ln(\sin x) + B) + C$ then
A) $A > B$ B) $A = B$ C) A is an integer D) B is an integer
2. If $\int \frac{\sin x}{\sin 3x} dx = \phi(x)$ and if $\phi(0) = 0$ then the value of $\phi\left(\frac{\pi}{6}\right)$ equals
A) $\frac{1}{3} \ln 2$ B) $\frac{1}{\sqrt{3}} \ln 2$ C) $\frac{1}{3} \ln \sqrt{2}$ D) $\frac{1}{\sqrt{3}} \ln \sqrt{2}$
3. If $x > 0, y > 0, xy(xy+1) = 6$ and if the value of $\int \frac{6-xy}{xy^2(y+1)} dx = Ax^2 + Bx + C \ln|x+2| + D$ then
A) $A = 2$ B) $B = -2$ C) $C = 2$ D) $A = 0$
4. If $\int x^3 \frac{d^2}{dx^2} (e^{x^2}) dx = e^{x^2} \cdot f(x) + C$ then the local maximum value of $f(x)$ equals ($x \in R$)
A) $87/8$ B) 2 C) 3 D) 0
5. If $f(\theta) = \int \frac{\sqrt{\tan \theta}}{\sin 2\theta} d\theta$ is continuous function and $f(0) = 0$ then the value of $\sqrt{2f\left(\frac{\pi}{4}\right) + f\left(\frac{\pi}{3}\right)}$ equals
A) $\sqrt{3} + 1$ B) $\sqrt{2} + 1$ C) $\frac{1}{\sqrt{2}}(\sqrt{3} + 1)$ D) $\frac{1}{2}(\sqrt{2} - 1)$
6. If $\int \frac{ax^2 + 2bx + c}{(Ax^2 + 2Bx + C)^2} dx (B^2 \neq AC)$ is a rational function of x then which is necessarily true
A) $2Bb = Ac + aC$ B) $2Bb = Aa + cC$
C) $2Cc = Ab + Ba$ D) $2aA = bB + cC$
7. If $\int e^x \frac{(x^4 + 2)}{(1 + x^2)^{5/2}} dx = \frac{e^x f(x)}{(1 + x^2)^{3/2}} + C$ then which is correct
A) $f(1) = 3$ B) $f(2) = -6$ C) $f'(0) = 2$ D) $f(x) < 0 \forall x \in R$
8. If $f(x)$ is polynomial function such that $f(x) + f'(x) + f''(x) + f'''(x) = x^3$ and $g(x) = \int \frac{f(x)}{x^3} dx$ and $g(1) = 1$ then $g(e) =$
A) $e + 3$ B) $e - 3$ C) $1 + e$ D) $1 - e$

9. $\int \frac{(x^{3x} + x^x)(\log x + 1)}{x^{4x} + 1} dx =, ('c' \text{ is integration constant})$

A) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x^{2x} - 1}{\sqrt{2}x^x} \right) + c$

B) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x^x + 1}{\sqrt{2}x^x} \right) + c$

C) $\sqrt{2} \tan^{-1} \left(\frac{x^{2x} + 1}{x^x} \right) + c$

D) $\sqrt{2} \tan^{-1} (x^x + 1) + c$

MULTIPLE ANSWER TYPE

10. Let f be a differentiable function satisfying $\frac{e^x f^{11}(x) - e^x f^1(x)}{e^{2x}} = 1$ and $f(0) = 0, f^1(0) = 1$ then which is/are correct

A) $\frac{xf^1(x) - f(x)}{x^2 e^x} = 1 (x \neq 0)$

B) For, $x > 0$, $f(x)$ is increasing function

C) For $x < 0$, $f(x)$ is decreasing function

D) Number of solutions of $3f(x) + 1 = 0$ is 2

11. Let $g(x) = x^3 - \frac{3x^2}{2} + x + \frac{1}{4}$ then which is/are correct

A) The curve $y = g(x)$ is symmetric w.r.t $x = \frac{1}{2}$ line

B) $f(x) = \int \frac{1 - g(x)}{g(1 - x)} dx$ and $f(1) = 1$ then $f(2) = 2$

C) The curve $y = g(x)$ is symmetric w.r.t. $x = 1$ line

D) If $g(g(1)) + g(t) = 1$ then $t < \frac{1}{2}$

12. The value of $\int \sqrt{\cot x} \cdot e^{\sqrt{\sin x}} (1 + e^{\sqrt{\sin x}}) dx$ equals

A) $2e^{\sqrt{\sin x}} + c$

B) $\frac{(1 + e^{\sqrt{\sin x}})^2}{2} + C$

C) $e^{\sqrt{\sin x}} (e^{\sqrt{\sin x}} + 2) + C$

D) $(e^{\sqrt{\sin x}} + 1)^2 + C$

13. If $\int \frac{\cos 2x \sin 4x}{\cos^4 x (1 + \cos^2 2x)} dx = A \sec^2 x + B \ln(1 + \cos 2x) + C \ln(1 + \cos^2 2x) + D$ then which is/are correct

A) $A < B$

B) $A = C$

C) $A > C$

D) $B > 1$

14. If $\int \frac{\cos x}{x} (x^2 \ln x + 1) dx = f(x) + c$ ('c,c' is integration constant)

$f(1) = \cos 1$ and $\int f(x) \sec^2 x dx = g(x) + c'$

$g(1) = 0$ and $L = \lim_{x \rightarrow 0^+} g(x)$ then

A) $f(x) = \cos x (1 + \ln x) + x \sin x \ln x$ B) $g(x) = \frac{x \ln x}{\cos x}$

C) $L = 0$ D) $\lim_{x \rightarrow \infty} \frac{g(x)}{x^2} = 0$

15. If $e^{-x} \int \frac{x^2 - x + 1}{(x^2 + 1)^{3/2}} e^x dx = f(x) + c$ ('c' is constant of integration) and $f(0) = 1$ then

which of the following is/are true?

A) $f(x)$ is an even function

B) $f(x)$ is a bounded function

C) The range of $f(x)$ is $(0, 1]$

D) $f(x)$ has two points of maxima

16. If A is square matrix and $e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots$

And $e^A = \frac{1}{2} \begin{bmatrix} f(x) & g(x) \\ g(x) & f(x) \end{bmatrix}$ and $A = \begin{bmatrix} x & x \\ x & x \end{bmatrix}$,

$0 < x < 1$, I is a 2×2 identity matrix then which of the following statement is/are true?

('c' is integration constant)

A) $\int \frac{g(x)}{f(x)} dx = \log(e^x + e^{-x}) + c$

B) $\int \frac{f(x) - g(x)}{f(x) + g(x)} dx = 2e^{-2x} + c$

C) $\int (g(x) + 1) \sin x dx = \frac{e^{2x}}{5} (2 \sin x - \cos x) + c$

D) $f(x) + g(x)$ is an increasing function $\forall x \in R$

PASSAGE-1

Let $\int \frac{\left(\frac{1}{x} + \frac{1}{x^2}\right)(x-1)}{\left(\frac{1}{x^4} + \frac{1}{x^2}\right)\sqrt{(x^4 - x^3 + x^2)(x^4 + x^3 + x^2)}} dx = \sec^{-1}(f(x)) + c$ and $f(1) = 2$. Then for $x > 0$

17. The minimum value of $f(x)$ equals

- A) 2 B) 4 C) $2\sqrt{2}$ D) $\sqrt{2}$

18. Which is correct

- A) $f(\pi) < f(e)$ B) $f(\pi) = f(e)$ C) $f\left(\frac{3}{2}\right) = \frac{13}{6}$ D) $f\left(\frac{3}{2}\right) = \frac{2}{3}$

PASSAGE-2

If $f(x) = \int \frac{x^7 - x^5 + x^3 - x}{x^{10} + 1} dx = A \ln|x^8 - x^6 + x^4 - x^2 + 1| + B \ln|x^2 + 1| + C$, $f(0) = 0$, then answer the following

19. $f(x)$ is

- A) Even function B) Odd function
C) Neither even nor odd function D) Both even and odd functions

20. The value of A equals

- A) $\frac{1}{2}$ B) $\frac{1}{5}$ C) $\frac{4}{5}$ D) $\frac{1}{10}$

21. The value of B equals

- A) $\frac{-1}{5}$ B) $\frac{-2}{5}$ C) $\frac{-3}{5}$ D) $\frac{-4}{5}$

INTEGER ANSWER TYPE

22. If $f(x) = \int \frac{\sin 2x}{(3 + 4\cos x)^3} dx$ and if $f\left(\frac{\pi}{2}\right) = \frac{1}{48}$ then the value of sum of digits of $\left[\frac{1}{f(0)}\right]$ is

([.] is GIF)

23. If $\int \frac{1}{\sin x + \sec x} dx = A \ln \left| \frac{\sqrt{3} + (\sin x - \cos x)}{\sqrt{3} - (\sin x - \cos x)} \right| + B \tan^{-1}(\sin x + \cos x) + C$ then $\left[\frac{B}{A}\right]$ equals

([.] is GIF)

24. If $\int \frac{\tan^4 x}{1 - \tan^2 x} dx = A \tan x + B \ln|\sec 2x + \tan 2x| + C.x + D$ then the value of $\left|\frac{A}{BC}\right|$ equals

25. If $f(x) = \int \frac{x^2}{(1+x^2)(1+\sqrt{1+x^2})} dx$ and $f(0) = 0$ then the value of $[f(1)]$ equals ([.] is GIF)

26. If the integral $I = \int \sqrt{x + \sqrt{x^2 + 3x}} dx$ on substitution $t = \sqrt{x + \sqrt{x^2 + 3x}}$ takes the form

$$\lambda \int \frac{t^4 (t^2 + 3)}{(2t^2 + 3)^2} dt \text{ then the value of } \lambda \text{ equals}$$

27. If $\int \frac{\sqrt{\cot x} - \sqrt{\tan x}}{\sqrt{2}(\cos x + \sin x)} dx = \phi(x)$, $x \in \left(0, \frac{\pi}{2}\right)$ and $\phi\left(\frac{\pi}{4}\right) = \frac{\pi}{2}$ and maximum value of $\phi(x)$ is λ then $[5\lambda]$ equals ([.] GIF)

28. If $\int \left\{ \ln \left(\frac{1 + \sin 2x}{1 - \sin 2x} \right)^{\cos^2 x} + \ln \left(\frac{\cos 2x}{1 + \sin 2x} \right) \right\} dx = A \sin 2x \ln \left(\frac{\sin x + \cos x}{\cos x - \sin x} \right) + B \ln(\cos 2x)$ then the value of $A + B$ equals

25-08-20

KEY			
1. D	2. B	3. C	4. C
5. A	6. D	7. A	8. D
9. D	10. C	11. A	12. C
13. A	14. B	15. B	16. B
17. A	18. B	19. D	20. C
21. C	22. C	23. C	24. A
25. A	26. A	27. B	28. A
29. B	30.	31. C	32. D
33. C			

26-08-20

KEY			
1. D	2. B	3. C	4. B
5. B	6. A	7. A	8. D
9. C	10. B	11. C	12. B
13. A	14. A	15. A	16. BC
17. BD	18. AD	19. 4	20. 2
21. 1	22. 1	23. 1	24. 2
25. 1	26. C	27. B	28. D

27-08-20

KEY			
1. B	2. B	3.	4.
5. BC	6. AC	7. AC	8. D
9. AD	10. 7	11. 0	12. 4
13. 0	14. 1	15. 1	16. A-R; B-Q; C-P; D-S
17. A	18. B	19. C	20. C
21. B	22. C	23. C	24. B



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INDIA



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Sec: Star 5C

Test on Areas

Date: 15-09-2020

SINGLE ANSWER TYPE

1. If $A\left(\frac{3}{\sqrt{2}}, \sqrt{2}\right)$, $B\left(-\frac{3}{\sqrt{2}}, \sqrt{2}\right)$, $C\left(-\frac{3}{\sqrt{2}}, -\sqrt{2}\right)$ and $D(3 \cos \theta, 2 \sin \theta)$ are four points, then the value of θ for which the area of quadrilateral ABCD is maximum, $\left(\frac{3\pi}{2} \leq \theta \leq 2\pi\right)$ is (in sq. units)
(A) $2\pi - \sin^{-1} \frac{1}{3}$ (B) $\frac{7\pi}{4}$ (C) $2\pi - \cos^{-1} \frac{3}{\sqrt{85}}$ (D) $\frac{\pi}{4}$
2. ABCD is a quadrilateral with side lengths $AB = 4$, $BC = 10$, $CD = 6$ and $AD = 6$, and diagonal $BD = 8$ units. If the incircles of triangles ABD and BCD touch BD at P and Q respectively, then area of quadrilateral C_1PC_2Q (where C_1 and C_2 are incentres of triangle ABD and BCD respectively), is
(A) $3 + \frac{\sqrt{15}}{2}$ sq. units (B) 3 sq. units (C) $\frac{\sqrt{15}}{6}$ sq. units (D) 4 sq. units
3. Let $f : [0, \infty) \rightarrow R$ be a continuous and strictly increasing function such that $f^3(x) = \int_0^x tf^2(t)dt$, $\forall x > 0$ then, find the area enclosed by $y = f(x)$, the x axis and the ordinate at $x = 3$, is.....
A) $3/2$ B) $5/2$ C) $1/2$ D) 2
4. Let $f(x) = x^2 + 6x + 1$ and S denotes the set of points (x, y) in the coordinate plane such that $f(x) + f(y) \leq 0$ and $f(x) - f(y) \leq 0$. The area of S is $k\pi$ then $k = \dots$
A) 8 B) 6 C) 7 D) 9
5. Area bounded by $y = \left[\sin^2 \frac{x}{2} + \sqrt{\cos \frac{x}{2}}\right]$ and $y = \left|\sin x - \frac{1}{2}\right|$, $x = 0, x = \pi$ where $[.]$ represents the greatest integer function is.
A) $\frac{1}{6}(7\pi + 12(1 - \sqrt{3}))$ B) $7\pi + 12(1 + \sqrt{3})$
C) $\frac{\pi}{2} + 1 - \sqrt{3}$ D) $\frac{\pi}{2} + 1 + \sqrt{3}$
6. Find the area enclosed between the curves $y = \ln(x + e)$, $x = \ln\left(\frac{1}{y}\right)$ and x-axis
a) 1 sq units b) 2 sq units c) 3 sq units d) 4 sq units
7. The area of the region bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and its pair of latus recta, e is eccentricity of ellipse and $e < 1$, is given by

- a) $2ab(e\sqrt{1-e^2} + \sin^{-1} e)$ square units b) $2ab(e\sqrt{1-e^2} - \sin^{-1} e)$ square units
- c) $ab(e\sqrt{1-e^2} - \sin^{-1} e)$ square units d) $\frac{ab}{2}(e\sqrt{1-e^2} - \sin^{-1} e)$ square units
8. Area bounded by $y = f^{-1}(x)$ and tangent and normal drawn to it at the points with abscissa π and 2π respectively, where $f(x) = \sin x - x$ is
- a) $\frac{\pi^2}{2} - 1$ b) $\frac{3\pi^2}{2} - 2$ c) $\frac{\pi^2}{2} - 4$ d) $\frac{\pi^2}{2} - 2$
9. If A_n is the area bounded by $y=x$ and $y = x^n; n \in N, A_2 \times A_3 \times A_4 \times \dots \times A_n =$
- A) $\frac{1}{(n)(n+1)}$ B) $\frac{1}{2^n(n)(n+1)}$ C) $\frac{1}{2^{n-1}(n)(n+1)}$ D) $\frac{1}{2^{n-2}(n)(n+1)}$
10. For the curve $f(x) = \frac{1}{1+x^2}$, let two points on it be $A(\alpha, f(\alpha)), B\left(\frac{-1}{\alpha}, f\left(\frac{-1}{\alpha}\right)\right), (\alpha > 0)$. The minimum area bounded by the line segments OA, OB and $f(x)$, where 'O' is the origin, is
- A) $\frac{\pi-1}{2}$ B) $\frac{\pi}{2}$ C) $\frac{\pi}{2} - 1$ D) $\frac{\pi+1}{2}$
11. Minimum area of the region bounded by the curves $C_1: y = |x-4|$ and $C_2: y = -|x-a|+b$, provided C_2 always passes through (3,2) as a, b varies, is
- A) $\frac{3}{2}$ B) 0 C) $\frac{3}{\sqrt{2}}$ D) $\frac{1}{\sqrt{2}}$
12. If $|z - (4 + 4i)| = 4$, then the area of the region bounded by the loci of $z, iz, -z$ and $-iz$ in complex plane, is
- A) $4(4 - \pi)$ B) $16(4 - \pi)$ C) $16(\pi - 1)$ D) $4(\pi - 1)$
13. The area bounded by the curves $y = \sin^{-1}|\sin x|$ and $y = (\sin^{-1}|\sin x|)^2$, where $0 \leq x \leq 2\pi$, is
- A) $\frac{1}{3} + \frac{\pi^2}{4}$ sq.units B) $\frac{1}{6} + \frac{\pi^3}{8}$ sq.units C) 2 sq.units D) none of these
14. Let $p(x)$ be a polynomial of least degree whose graph has three points of inflection (where the concavity of the graph changes) at $(-1, -1), (1, 1)$ and a point with abscissa 0 at which the curve is inclined to the axis of abscissa at an angle of 60° . Then $\int_0^1 p(x) dx$ equals to
- A) $\frac{3\sqrt{3}+4}{14}$ B) $\frac{3\sqrt{3}}{7}$ C) $\frac{\sqrt{3}+\sqrt{7}}{14}$ D) $\frac{\sqrt{3}+2}{7}$
15. Let λ be a positive real constant. Consider the curves $S_1: y = e^x$ and $S_2: y = e^{\lambda-x}$. Let A be the area of region surrounded by S_1, S_2 and y-axis. Then $\lim_{\lambda \rightarrow 0} \frac{A}{\lambda^2} = \dots$
- A) 4 B) $\frac{1}{2}$ C) 0 D) $\frac{1}{4}$
16. Let $f(x) = x^2 - 3x + 2; x \in R$ then area bounded by $y = |f(|x|)|$ and $1 \leq |x| \leq 2$ and x-axis is equal to sq units

A) $\frac{1}{5}$

B) $\frac{1}{4}$

C) $\frac{1}{3}$

D) $\frac{1}{2}$

MULTIPLE ANSWER TYPE

17. Area of the region bounded by the curve $y = e^x$ and lines $x=0$ and $y=e$ is

A) $e-1$

B) $\int_1^e \ln(e+1-y)dy$

C) $e - \int_0^1 e^x dx$

D) $\int_1^e \ln y dy$

18. Let $f(x)$ and $g(x)$ be two continuous functions on $a \leq x \leq b$ intersecting at only one point in (a, b) and let $p(x) = \max. \{f(x), g(x)\}$ and $q(x) = \min \{f(x), g(x)\}$ then the area bounded by the curves $y = p(x)$, $y = q(x)$ and ordinates $x = a$ and $x = b$ is given by..

A) $\int_a^b (f(x) - g(x))dx$

B) $\int_a^b (p(x) - q(x))dx$

C) $\int_a^b |f(x) - g(x)|dx$

D) $\int_a^b |p(x) - q(x)|dx$

19. Five regions are defined by

$$C_1 : |x + y| \leq 1, C_2 : |x - y| \leq 1, C_3 : |x| \leq \frac{1}{2}, C_4 : |y| \leq \frac{1}{2}, C_5 : 3x^2 + 3y^2 = 1 : \text{ then}$$

A) The area bounded by C_1, C_2 which doesn't contain area of C_5 is $2 - \frac{\pi}{3}$

B) The part of area of C_5 which doesn't contain area of C_3 and C_4 is $\frac{2\pi}{9} - \frac{1}{\sqrt{3}}$

C) The part of area bounded by C_1, C_2 but not bounded by C_3 and C_4 is 1

D) Area bounded by C_1, C_2, C_3, C_5 but not by C_4 is $\frac{\pi}{2} - \frac{1}{\sqrt{3}}$

INTEGER ANSWER TYPE

20. The ratio in which $\frac{x}{4} + \frac{y}{3} = 1$ divides the region enclosed by $\frac{x^2}{16} + \frac{y^2}{9} = 1$ is $k:1$ ($k > 1$). Then $[k-2] = \dots\dots\dots$ (where $[.]$ denotes G.I.F & take $\pi = 3.14$)

PASSAGE

Consider a square with vertices at $A(1,1)$, $B(-1,1)$, $C(-1,-1)$ and $D(1,-1)$. Let 'S' be the region consisting of all points inside the square which are nearer to the origin than to any edge.

21. The region 'S' is surmounted by

A) A circle

B) A square

C) Four parabolas

D) Two ellipses

22. The point where the function, defined by the boundary (in the first quadrant) of the region 'S' is not differentiable, is

A) $\left(1 - \frac{1}{\sqrt{2}}, 1 - \frac{1}{\sqrt{2}}\right)$

B) $(\sqrt{2}-1, \sqrt{2}-1)$

C) $\left(\frac{1}{2}, \frac{1}{2}\right)$

D) $\left(1 - \frac{1}{\sqrt{3}}, 1 - \frac{1}{\sqrt{3}}\right)$

23. The area of the region 'S' is

A) $\frac{2}{3}(\sqrt{3}+2)(\sqrt{2}+1)$ sq.units

B) $\frac{4}{3}(\sqrt{2}-1)(3-\sqrt{2})$ sq.units

C) $\frac{1}{3}(2-\sqrt{3})(\sqrt{2}+1)$ sq.units

D) $\frac{2}{3}(\sqrt{2}+1)(4-\sqrt{2})$ sq.units

MATCHING TYPE

24. Match the following:

Column -I

Column -II

@bohring_bot

A) The parabolas $y^2 = 4x$ and $x^2 = 4y$ — (P) S_1, S_2, S_3 are in A.P.

divide the square region bounded the lines $x = 4, y = 4$ and the coordinate axes.

If S_1, S_2, S_3 are respectively the areas of these parts numbered from top to bottom, then

B) If S_1, S_2, S_3 are the areas bounded by the x-axis and any three consecutive half waves of the curves $y = e^{ax} \sin \beta x, x \geq 0$, from left to right, then $(a, \beta \neq 0)$

(Q) S_1, S_2, S_3 are in G.P

C) Let $f(x) = \max.\{x^2, (1-x)^2, 2x(1-x)\}$,

(R) $S_1^3 + S_3^3 > 2S_2^3$

where $0 \leq x \leq 1$ if the areas of the region bounded by the curve $y = f(x)$ between

$x \in \left[0, \frac{1}{3}\right], \left[\frac{1}{3}, \frac{2}{3}\right], \left[\frac{2}{3}, 1\right]$ and x-axis

represented by S_1, S_2, S_3 then

D) If the area bounded by $y^2 = x, y = x$,

(S) $S_1 = S_2 = S_3$

$x = \sqrt{y}, y = x$ and $y = x^2, y = x$ are denoted

By S_1, S_2 and S_3 respectively then

D) If the area bounded by $y^2 = x, y = x$,
 $x = \sqrt{y}, y = x$ and $y = x^2, y = x$ are denoted
By S_1, S_2 and S_3 respectively then

(S) $S_1 = S_2 = S_3$

KEY

1	B	2	B	3	A	4	A	5	A	6	B	7	A	8	D
9	D	10	A	11	A	12	B	13	D	14	A	15	D	16	C
17	BCD	18	BCD	19	ABC	20	8	21	C	22	B	23	B		

24. A-PQS; B-OR; C-R; D-PQS

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SINGLE ANSWER TYPE

- Area of the region bounded by $\{(x, y): 0 \leq y \leq x^2 + 1, 0 \leq y \leq x + 1, 0 \leq x \leq 2\}$
A) 23 B) $\frac{23}{3}$ C) $\frac{23}{2}$ D) $\frac{23}{6}$
- Let $A = \left(\frac{1}{2}, 0\right), B = \left(\frac{3}{2}, 0\right), C = \left(\frac{5}{2}, 0\right)$ be 3-points on xy-plane. P is another point on the same plane satisfying $\max\{PA + PB, PB + PC\} < 2$. If the area of the region of P is $\sqrt{3} \left(\frac{\pi}{a} - \frac{\sqrt{3}}{b} \right)$ then $a + b =$ ____
A) 7 B) 5 C) 9 D) 11
- Let $f(x) = x^3 - x^2 + 2x - 8$ and $g(x)$ is inverse of $f(x)$. Then area bounded by $y = g(x)$, x-axis between $x = -12$, and $x = 16$ is
A) $\frac{325}{6}$ sq. units B) $\frac{325}{4}$ sq. units C) $\frac{325}{7}$ sq. units D) $\frac{325}{9}$ sq. units
- The area enclosed by the curve $|x - 60| + |y| = \left| \frac{x}{4} \right|$
A) 240 sq. units B) 360 sq. units C) 480 sq. units D) 600 sq. units

MULTIPLE ANSWER TYPE

- A function $y = f(x)$ satisfies the following conditions
1) $f(1) = 1$
2) The tangent at any point of the graph of $f(x)$ cuts off on the y-axis an intercept equal to the square of the abscissa of point of tangency.
Which of the following are correct.
A) The graph of $y = f(x)$ is passing through $(0, 0)$
B) $y = f(x)$ represents a conic whose focus is $\left(1, \frac{3}{4}\right)$
C) The area enclosed by $y = f(x)$ and the line $y + 3 = 0$ is $\frac{32}{3}$
D) $y = f(x)$ has no maxima
- Let $g(x) = \begin{cases} x^2 & \text{if } 0 \leq x < \frac{1}{2} \\ (x-1)^2 & \text{if } \frac{1}{2} \leq x \leq 1 \end{cases}$

and $f(x)$ is another function such that the area enclosed by curves $g(x)$ and x -axis between the ordinates $x = \beta$ and $x = \beta + f(\beta)$ is half of the area enclosed by $g(x)$ and X -axis (where $0 < \beta < \frac{1}{2}$, $\frac{1}{2} < \beta + f(\beta) < 1$) then which of the following option are correct.

A) $f(x) = 1 - x + \sqrt[3]{x^3 - \frac{1}{8}}$

B) $\lim_{x \rightarrow \frac{1}{2}} \frac{(f(x) + x - 1)^3}{x - \frac{1}{2}} = \frac{3}{4}$

C) If $f(x)$ has local extreme at $x = \alpha$ then $\alpha^3 = \frac{1}{16}$

D) Area enclosed by $y = g(x)$ between $x = 0, x = 1$ and X -axis is $\frac{1}{12}$

7. Which of the following statements are correct

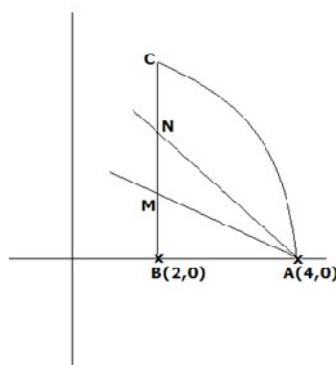
A) Area bounded by $(y - \sin^{-1} x)^2 = x - x^2$ is $\frac{\pi}{4}$

B) Area bounded by $x^2 y^2 = a^2 (y^2 - x^2)$ and $x = \pm a$ is $4a^2$

C) Area bounded by $y^2 = x^2 (a^2 - x^2)$ is $\frac{4}{3}a^2$

D) Area bounded between $x = y^2$ and $x = 3 - 2y^2$ is 4

8. A figure is bounded by the curves $y = \left| \sqrt{2} \sin \frac{\pi x}{4} \right|$, $y = 0$, $x = 2$, $x = 4$. Two straight lines passing through $(4, 0)$ divide the above bounded figure in to 3-equal parts then inclination of the lines



A) 30° B) $\tan^{-1} \left(\frac{4\sqrt{2}}{3\pi} \right)$ C) $\pi - \tan^{-1} \left(\frac{4\sqrt{2}}{3\pi} \right)$ D) $\pi - \tan^{-1} \left(\frac{2\sqrt{2}}{3\pi} \right)$

9. If is the area bounded by $|x - a_i| + |y| = b_i$, $i \in N$ and $a_{i+1} = a_i + \frac{3}{2}b_i$ and $b_{i+1} = \frac{b_i}{2}$, $a_1 = 0, b_1 = 32$ then

A) $A_3 = 128$ B) $\lim_{n \rightarrow \infty} \sum_{i=1}^n A_i = \frac{4}{3}(16)^2$ C) $\lim_{n \rightarrow \infty} \sum_{i=1}^n A_i = \frac{8}{3}(32)^2$ D) $A_3 = 256$

10. Which of the following statement(s) is/are true
- A) The area bounded by the curves $y = -\sqrt{-x}$ and $x = -\sqrt{-y}$, $x, y \leq 0$ is $1/3$
- B) The area bounded by the curves $y = \sqrt{-x}$ and $x = \sqrt{-y}$, $x, y \leq 0$ is $1/3$
- C) The area bounded by the curves $y = \sqrt{-x}$ and $x = \sqrt{-y}$, $x, y \leq 0$ and $y = 1$ is $2/3$
- D) The area bounded by the curves $y = x^2$ and $y = \sqrt{|x|}$, is $2/3$
11. If the area enclosed between the curves $y = \sin^4 x \cos^3 x$, $y = \sin^2 x \cos^3 x$ between $x = 0$ and $x = \frac{\pi}{2}$ is 'A', then which of the following is/are **incorrect**?
- A) $A = \int_0^1 t^2(1-t^2)dt$ B) $A = \int_0^1 t^2(1-t^2)^2 dt$
- C) $A = \int_0^1 t(1-t^2)^2 dt$ D) $A = \frac{\pi}{2} \int_0^1 t(1-t^2)dt$
12. Let $g^1(x) = \frac{192x^3}{2 + \sin^4 \pi x}$ for all $x \in R$ with $g\left(\frac{1}{2}\right) = 0$. If A be the area of the region bounded by $y = g(x)$, with x - axis and between the ordinates $x = \frac{1}{2}$ and $x = 1$ such that $\alpha \leq A \leq \beta$, then the possible values of α and β are
- A) $\alpha = 14, \beta = 26$ B) $\alpha = \frac{1}{12}, \beta = \frac{1}{2}$
- C) $\alpha = -12, \beta = 0$ D) $\alpha = 1, \beta = 12$
13. Let $y = f(x)$ be the continuous function in $[0, \infty)$ satisfy $\int_0^x f(x-t)dt = xf(x) + x^2 \int_0^1 f'(tx)dt \forall x > 0$ & $f(0) = 3$. Then which of the following is/are correct?
- A) Area bounded by $y = f(x)$ & x axis in between $x = 0$ & $x = 3$ is $\frac{9}{2}$
- B) Area bounded by $y = f(x)$ & x axis in between $x = 0$ & $x = 3$ is 9
- C) Number of points of intersection of $y = |f(x)|$ and $x + y = 4$ be 1.
- D) Number of points of intersection of $y = |f(x)|$ and $y - x = 4$ be 1
14. The area bounded by $y = \frac{\sin x}{x}$, x-axis and the ordinates $x = 0, x = \frac{\pi}{4}$ is
- A) ≤ 1 B) $< \frac{\pi}{4}$ C) $\leq \frac{\pi}{2}$ D) $\leq \frac{\pi}{3}$
15. Let $f(x) = \frac{x}{1+x^2}$ and $g(x) = \frac{1}{x(1+x^2)}$. If A_1 be the area of region bounded between $y = f(x)$, x-axis, $x = \frac{1}{e}$ and $x = \pi$, and A_2 is the area of region bounded between $y = g(x)$, x-axis and between $x = \frac{1}{\pi}$ and $x = \frac{1}{e}$. Then $|A_1 - A_2|$ is greater than or equal to

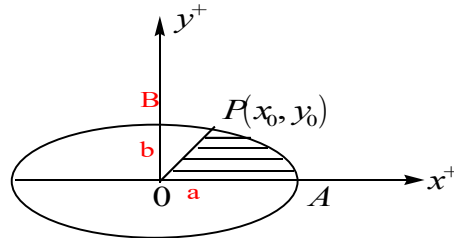
A) 2

B) 1

C) $\ln \frac{3}{2}$

D) $\ln \frac{13}{11}$

16. An ellipse shown here has equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b$. If $P(x_0, y_0)$ is any point on it and area of curvilinear part OPA is denoted by Δ_0 . Then which of the following statements can be wrong?



- A) $P = \left(a \cos \left(\frac{2\Delta_0}{ab} \right), b \sin \left(\frac{2\Delta_0}{ab} \right) \right)$ B) $P = \left(a \cos \left(\frac{2\sqrt{\Delta_0}}{ab} \right), b \sin \left(\frac{2\sqrt{\Delta_0}}{ab} \right) \right)$
 C) $P = \left(a \cos \left(\frac{ab}{2\Delta_0} \right), b \sin \left(\frac{ab}{2\Delta_0} \right) \right)$ D) $P = \left(a \cos \left(\frac{\sqrt{ab}}{2\Delta_0} \right), b \sin \left(\frac{\sqrt{ab}}{2\Delta_0} \right) \right)$

INTEGER ANSWER TYPE

17. The area of the figure bounded by $x = -1, x = 2$ and $y = \begin{cases} -x^2 + 2; & x \leq 1 \\ 2x - 1; & x > 1 \end{cases}$ is λ then

$[\lambda] = \underline{\hspace{2cm}}$ (Where $[\cdot]$ is G.I.F)

18. Consider a rectangle ABCD formed by the points A(0,0) B(6,0) C(6,4) D(0,4).

P is a moving interior point of the rectangle, moving in such a way $d(P, AB) \leq \min \{d(P, BC), d(P, CD), d(P, AD)\}$ and here $d(P, AB), d(P, BC), d(P, CD), d(P, AD)$ represents that the distances of the point P from the sides AB, BC, CD and AD respectively. Area of the region representing all possible positions of the point P is equal to.....

19. Consider $f(x) = \begin{cases} \cos x, & 0 \leq x < \frac{\pi}{2} \\ \left(\frac{\pi}{2} - x\right)^2, & \frac{\pi}{2} \leq x < \pi \end{cases}$ such that $f(x)$ is periodic with period π then the

area bounded by $y = f(x)$ and the x-axis from $x = -n\pi$ to $x = n\pi$ is $an \left(1 + \frac{\pi^3}{b} \right)$, $n \in N$ for a given n, n is a natural number then $b/(2a) = \dots\dots\dots$

20. Area bounded by $2 \geq \max. \{|x - y|, |x + y|\}$ is k sq. units then k =

21. The area bounded by the curves $y = \ln x, y = \ln |x|, y = |\ln x|, y = |\ln |x||$ is

22. If $f(x) = 2\sin x$ and $g(x) = 2\cos x$ and the area between $p(x) = \max\{f(x), g(x)\}$ and $q(x) = \min\{f(x), g(x)\}$ from $x = 0$ to $x = \frac{\pi}{2}$ is A then the value of $\sqrt{2}(A + 4)$ is
23. If the area bounded by the curve $(y - \arcsin x)^2 = x - x^2$, is 'A', then the value of $[4A]$ is equal to ([.] denotes G.I.F)
24. Let $f(x) = x^2 + 6x + 1$ and S denotes the set of points (x, y) in the coordinate plane such that $f(x) + f(y) \leq 0$ and $f(x) - f(y) \leq 0$. The area of S is $k\pi$ then $k = \dots$
25. The area of the region bounded by the curves $y = \frac{4 - x^2}{4 + x^2}$, $25y^2 = 9x$ and $y = \frac{3}{5}|x| - \frac{6}{5}$ which contains $(1, 0)$ point in its interior, is $\pi - a \tan^{-1} \frac{1}{2} + \frac{1}{b}$ where $b - 2a$ is equal to $(a, b \text{ are positive integers})$
26. The minimum area bounded by the function $y = f(x)$ and $y = \alpha x + 9$ ($\alpha \in R$) where f satisfies the relation $f(x + y) = f(x) + f(y) + y\sqrt{f(x)} \quad \forall x, y \in R$ and $f'(0) = 0$ is $9A$, value of 'A' is
27. If the area bounded by the curves $y = -x^2 + 6x - 5$, $y = -x^2 + 4x - 3$ and the line $y = 3x - 15$ is $\frac{73}{\lambda}$, where $x > 1$ then the value of λ is
28. Let the straight line $y = k$ divide the area enclosed by $x = (1 - y)^2$, $x = 0$ and $y = 0$ into two parts of areas, A_1 ($0 \leq y \leq k$) and A_2 ($k \leq y \leq 1$) such that $\frac{A_1}{A_2} = 7$. Then $\frac{1}{k}$ is equal to
29. Let $f(x) = \int_x^{x^2 + \frac{\pi}{3}} 4\sin^2 t \, dt$ for all $x \in R$ and $g: \left[0, \frac{1}{2}\right] \rightarrow [0, \infty]$ be a continuous function. For $\alpha \in \left[0, \frac{1}{2}\right]$, if $f^1(\alpha) + 2$ is the area of the region bounded by
30. If A be the area of the region represented by $\{(x, y) \in R^2 : y \geq \sqrt{|x + 3|}, 5y \leq x + 9 \leq 15\}$, Then the value of $4A$ is equal to

30. If A be the area of the region represented by

$\{(x, y) \in R^2 : y \geq \sqrt{|x + 3|}, 5y \leq x + 9 \leq 15\}$, Then the value of $4A$ is equal to

KEY

1	D	2	A	3	A	4	C	5	ABC	6	ABCD	7	ABD	8	CD
9	AC	10	AD	11	ACD	12	D	13	AD	14	ABCD	15	BCD	16	BCD
17	5	18	8	19	6	20	8	21	4	22	8	23	3	24	8
25	2	26	8	27	6	28	2	29	6	30	6				



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BASARA GNANA SARASWATHI CAMPUS KAKATIYA HILLS

Sec: Star 5C

Test on Areas

Date: 17-09-2020

SINGLE ANSWER TYPE

- For the curve $f(x) = \frac{1}{1+x^2}$, let two points on it be $A(\alpha, f(\alpha))$, $B\left(\frac{-1}{\alpha}, f\left(\frac{-1}{\alpha}\right)\right)$, $(\alpha > 0)$.
The minimum area bounded by the line segments OA, OB and $f(x)$, where 'O' is the origin, is
A) $\frac{\pi-1}{2}$ B) $\frac{\pi}{2}$ C) $\frac{\pi}{2}-1$ D) $\frac{\pi+1}{2}$
- If $|z - (4 + 4i)| = 4$, then the area of the region bounded by the loci of $z, iz, -z$ and $-iz$ in complex plane, is
A) $4(4-\pi)$ B) $16(4-\pi)$ C) $16(\pi-1)$ D) $4(\pi-1)$
- The area bounded by the curves $y = \sin^{-1}|\sin x|$ and $y = (\sin^{-1}|\sin x|)^2$, where $0 \leq x \leq 2\pi$, is
A) $\frac{1}{3} + \frac{\pi^2}{4}$ sq.units B) $\frac{1}{6} + \frac{\pi^3}{8}$ sq.units
C) 2 sq.units D) none of these
- If $y = f(x)$ makes positive intercepts of 2 and 1 units respectively on x and y-axes and enclose an area of $\frac{3}{4}$ sq. units with the axes then $\int_0^2 x f'(x) dx = \dots\dots\dots$
A) $\frac{3}{2}$ B) $\frac{-3}{4}$ C) $\frac{5}{4}$ D) 1
- The area bounded by $e^{\ln(x+1)} \geq |y|, |x| \leq 1$ is
a) 1 b) 2 c) 3 d) 4
- Let $f(x) = \text{maximum} \{x^2, (1-x)^2, 2x(1-x)\}$, where $0 \leq x \leq 1$. Determine the area of the region bounded by the curves $y = f(x)$, x-axis, $x = 0$ and $x = 1$.
a) $\frac{21}{27}$ sq units b) $\frac{23}{27}$ sq units c) $\frac{25}{27}$ sq units d) $\frac{17}{27}$ sq units
- Compute the area bounded by the curve $y = x^4 - 2x^3 + x^2 + 3$, the axis of abscissa and two ordinates corresponding to the points of minima of the function $y(x)$
a) $\frac{91}{30}$ square units b) $\frac{93}{30}$ square units
c) $\frac{1}{30}$ square units d) $\frac{1}{40}$ square units

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8. Let $A = \int_1^{e^2} \frac{\ln x}{\sqrt{x}} dx$. Identify which of the following statement(s) is (are) correct?
 A) $A < 2\left(e - \frac{1}{e}\right)$ B) $A < (e-1)\left(2 + \frac{1}{\sqrt{e}}\right)$ C) $A = \int_0^2 t\sqrt{e^t} dt$ D) $3 < A < 5$
9. The ratio of the area in which the curve $y = \left[\frac{x^3}{100} + \frac{x}{35}\right]$ divided the circle $x^2 + y^2 - 4x + 2y + 1 = 0$, where $[.]$ denotes the greatest integer function is
 a) $4\pi - 3\sqrt{3} : 8\pi + 3\sqrt{3}$ b) $4\pi + 3\sqrt{3} : 8\pi + 3\sqrt{3}$
 c) $4\pi\sqrt{3} - 9 : 8\pi\sqrt{3} + 9$ d) $4\pi - 3\sqrt{3} : 8\pi - 2\sqrt{3}$
10. If the area enclosed by $y^2 = 4ax$ and line $y = ax$ is $\frac{1}{3}$ sq.units, then the roots of the equation $x^2 + 2x = a$ are
 A) -4 B) 2 C) -2 D) 8
11. Let $f(x) = \frac{x}{1+x^2}$ and $g(x) = \frac{1}{x(1+x^2)}$. If A_1 be the area of region bounded between $y = f(x)$, x -axis, $x = \frac{1}{e}$ and $x = \pi$, and A_2 is the area of region bounded between $y = g(x)$, x -axis and between $x = \frac{1}{\pi}$ and $x = \frac{1}{e}$. Then $|A_1 - A_2|$ is greater than or equal to
 A) 2 B) 1 C) $\ln \frac{3}{2}$ D) $\ln \frac{13}{11}$
12. If $f(x)$ and $g(x)$ be the two continuous functions on $x \in [a, b]$ and $F(x) = \max\{f(x), g(x)\}$ and $G(x) = \min\{f(x), g(x)\}$ the area bounded by $F(x)$ and $G(x)$ between the lines $x = a$, $x = b$, and x -axis, is
 (A) $\int_a^b (f(x) - g(x)) dx$ (B) $\int_a^b |f(x) - g(x)| dx$ (C) $\int_a^b |F(x) - G(x)| dx$ (D) none of these
13. The area bounded by curves $y = x^3$; $y = (x)^{1/3}$
 a) some part lies in 1st quadrant b) some part lies is IIIrd quadrant
 c) is equal to 1 unit² d) is equal to 4 unit²
14. The function $f(\theta) = \frac{d}{d\theta} \int_0^\theta \frac{dx}{(1 - \cos 2\theta)(1 + \cos 2x)}$ satisfies the differential equation
 A) $\frac{df}{d\theta} + 4f(\theta) \cot 2\theta = 0$ B) $\frac{df}{d\theta} - 2f(\theta) \tan 2\theta = 4 \operatorname{cosec}^3 2\theta$
 C) $f(\theta) = \operatorname{cosec}^2 2\theta$ D) $f(\theta) = -(\operatorname{cosec} 2\theta)(\cot 2\theta)$

PASSAGE

Let $f(x) = \sin^4 \pi x$ and $g(x) = \ln x$. Let $x_0, x_1, x_2, \dots, x_{n+1}$

be the roots of $f(x) = g(x)$ in increasing order. Then answer the following

15. The value of n is less than or equal to

- A) 2 B) 3 C) 4 D) 5

16. The whole area (approximately) bounded by $y = f(x)$, $y = g(x)$ and $x = 0$ is less than or equal to

- A) $\frac{17}{8}$ B) $\frac{15}{8}$ C) $\frac{13}{8}$ D) $\frac{11}{8}$

INTEGER TYPE

17. Let $f(x) = \int_x^{x^2 + \frac{\pi}{3}} 4 \sin^2 t \, dt$ for all $x \in \mathbb{R}$ and $g : \left[0, \frac{1}{2}\right] \rightarrow [0, \infty]$ be a continuous

function. For $\alpha \in \left[0, \frac{1}{2}\right]$, if $f^1(\alpha) + 2$ is the area of the region bounded by

$x = 0, y = 0, y = g(x)$ and $x = \alpha$, then the value of $g(0)$ is equal to

18. The area bounded by $y = 2 - |2 - x|$; $y = \frac{3}{|x|}$ is $\frac{k - 3 \ln 3}{2}$, then k is equal to

19. If the area bounded by the region $|y| + 1 = 2e^{-|x|}$ is $k(1 - \log_e 2)$ then the numerical value of 'k' is

PASSAGE-1

If $f : [a, b] \rightarrow \mathbb{R}, g : [a, b] \rightarrow \mathbb{R}$ are continuous and $0 \leq f(x) \leq g(x) \forall x \in [a, b]$, then the area of the region bounded by the curves $y = f(x), y = g(x)$ and the lines $x = a, x = b$ is given

by $\int_a^b [g(x) - f(x)] dx$. Using this information, answer the following questions.

20. The area of the region bounded between the parabola $y = x^2 - x - 6$ and the line $y = -4$ is

- A) $\frac{7}{2}$ B) $\frac{9}{2}$ C) $\frac{5}{2}$ D) 4

21. The area of the region enclosed by the curve $5x^2 + 6xy + 2y^2 + 7x + 6y + 6 = 0$ is

- A) $\frac{\pi}{4}$ sq. unit B) $\frac{\pi}{2}$ sq. unit C) π sq. unit D) $\frac{\pi}{3}$ sq. unit

PASSAGE-2

Consider two curves $x^2y = 1$, $xy - y = \frac{1}{4}$. Then answer the following

22. The value of a ($a > 2$) for which the reciprocal of area of figure bounded by curves and lines $x = 2$, $x = a$ is equal to 'a' itself is
- A) $e + 1$ B) $e^2 + 1$ C) $3 - \sqrt{e}$ D) $\sqrt{2} + e$
23. The value of b ($1 < b < 2$) such that area bounded by these curves lines $x = b$ and $x = 2$ is equal to $1 - \frac{1}{b}$ is
- A) $1 + \frac{1}{e^2}$ B) $1 + \frac{1}{\sqrt{e}}$ C) $2 - \sqrt[3]{e}$ D) $1 + \frac{2}{e}$

PASSAGE-3

The boundaries of quadrilateral Q are given by $1 + x^2y^2 - x^2 - y^2 = 0$. R is a region belonging to Q such that every point 'P' of 'R' is nearer to the origin O(0, 0) than to any edge of 'Q'. Now answer the following questions :

24. Area of the region R is sq. units
- A) $\frac{1+6\sqrt{2}}{5}$ B) $\frac{3\sqrt{2}-1}{3}$ C) $\frac{16\sqrt{2}-20}{3}$ D) $\frac{20\sqrt{2}-16}{5}$
25. Maximum and least values of distance AB where A and B points on boundaries of R and Q respectively are M and L respectively then $\sqrt{\frac{M}{L}} = \dots\dots$
- A) 2 B) $\frac{4}{\sqrt{3}}$ C) $\frac{2}{\sqrt{3}}$ D) $\frac{\sqrt{5}}{3}$

PASSAGE-4

A continuous function $f(x)$ satisfying $x^4 - 4x^2 \leq f(x) \leq 2x^2 - x^3$ for all $x \in [0, 2]$. Such that the area bounded by $y = f(x)$, $y = x^4 - 4x^2$, y-axis and the line $x = t$ ($0 \leq t \leq 2$) is k times the area bounded by

$y = f(x)$, $y = 2x^2 - x^3$, y-axis and the line $x = t$ ($0 \leq t \leq 2$). Answer the following questions

26. If $k = 0$, then $f(x)$ attains local maximum at
- a) 0 b) $\sqrt{2}$ c) $-\sqrt{2}$ d) $\sqrt{3}$
27. The value of $\int_{-1}^1 f(x) dx$ is
- a) $\frac{2}{15(k+1)}(23-10k)$ b) $\frac{2}{15(k+1)}(23+10k)$ c) $\frac{2}{15(k+1)}(10k-17)$ d) $\frac{2}{15(k+1)}(10k+17)$

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KEY

1	A	2	B	3	D	4	B	5	D	6	D	7	A	8	ABCD
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9	AC	10	AB	11	BCD	12	BC	13	ABC	14	BD	15	ABCD	16	ABCD
17	6	18	4	19	4	20	B	21	B	22	B	23	A	24	C
25	A	26	A	27	D										

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BASARA GNANA SARASWATHI CAMPUS KAKATIYA HILLS

Sec: Star SC

Test on Areas

Date: 18-09-2020

SINGLE ANSWER TYPE

1. Let λ be a positive real constant. Consider the curves $S_1 : y = e^x$ and $S_2 : y = e^{\lambda-x}$.

Let A be the area of region surrounded by S_1 , S_2 and y-axis. Then $\lim_{\lambda \rightarrow 0} \frac{A}{\lambda^2} = \dots$

- A) 4 B) $\frac{1}{2}$ C) 0 D) $\frac{1}{4}$

2. Let $f(x) = \min\{x+1, \sqrt{1-x}\}$ for all $x \leq 1$. Then the area bounded by $y = f(x)$ and x-axis is

- A) $\frac{1}{6}$ sq.units B) $\frac{5}{6}$ sq.units C) $\frac{7}{6}$ sq.units D) $\frac{11}{6}$ sq.units

3. If $y = f(x)$ makes positive intercepts of 2 and 1 units respectively on x and y-axes and

enclose an area of $\frac{3}{4}$ sq. units with the axes then $\int_0^2 x f'(x) dx = \dots\dots\dots$

- A) $\frac{3}{2}$ B) $\frac{-3}{4}$ C) $\frac{5}{4}$ D) 1

4. Area enclosed by the curve, $y = f(x)$, that is being defined, parametrically as

$x = \frac{1-t^2}{1+t^2}$ and $y = \frac{2t}{1+t^2}$ is equal to

- a) π Square units b) $\frac{\pi}{2}$ square units c) $\frac{3\pi}{4}$ square units d) $\frac{3\pi}{2}$ square units

5. $d_1((x_1, y_1), (x_2, y_2)) = |x_1 - x_2| + |y_1 - y_2|$ And $d_2((x_1, y_1), (x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ d_1, d_2 denotes the distance between the points (x_1, y_1) and (x_2, y_2) on the coordinates plane, then area of the region enclosed by the set of points (x, y) satisfying

$d_1((x, y), (0, 0)) \geq 1$ and $d_2((x, y), (0, 0)) \leq 1$ is (in sq. units)

- a) $\pi + 4$ b) $\pi + 2$ c) $\pi - 2$ d) $\pi - 4$

6. Area enclosed by the figure described by the equation $x^4 + 1 = 2x^2 + y^2$ is (in sq. units)

- a) 3 b) $\frac{8}{3}$ c) $\frac{7}{3}$ d) $\frac{10}{3}$

7. Let the straight line $x = b$ divide the are enclosed by $y = (1-x)^2$, $y = 0$ and $x = 0$ into two parts $R_1 (0 \leq x \leq b)$ and $R_2 (b \leq x \leq 1)$ such that $R_1 - R_2 = \frac{1}{4}$. Then, b equals

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a) $\frac{3}{4}$

b) $\frac{1}{2}$

c) $\frac{1}{3}$

d) $\frac{1}{4}$

8. Let $f : [-1, 2] \rightarrow [0, \infty)$ be a continuous function such that $f(x) = f(1-x)$ for all $x \in [-1, 2]$. Let $R_1 = \int_{-1}^2 xf(x)dx$ and R_2 be the area of the region bounded by $y = f(x)$, $x = -1$, $x = 2$ and the x-axis. Then,

a) $R_1 = 2R_2$

b) $R_1 = 3R_2$

c) $2R_1 = R_2$

d) $3R_1 = R_2$

9. The area of the region between the curves $y = \sqrt{\frac{1+\sin x}{\cos x}}$ and $y = \sqrt{\frac{1-\sin x}{\cos x}}$ and bounded by the lines $x = 0$ and $x = \frac{\pi}{4}$ is

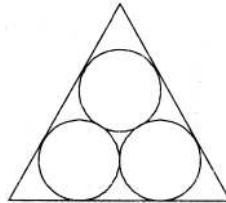
a) $\int_0^{\sqrt{2}-1} \frac{t}{(1+t^2)\sqrt{1-t^2}} dt$

b) $\int_0^{\sqrt{2}-1} \frac{4t}{(1+t^2)\sqrt{1-t^2}} dt$

c) $\int_0^{\sqrt{2}+1} \frac{4t}{(1+t^2)\sqrt{1-t^2}} dt$

d) $\int_0^{\sqrt{2}+1} \frac{t}{(1+t^2)\sqrt{1-t^2}} dt$

10. The area of the equilateral triangle, in which three coins of radius 1 cm are placed, as shown in the figure, is



a) $(6 + 4\sqrt{3})$ sq cm b) $(4\sqrt{3} - 6)$ sq cm c) $(7 + 4\sqrt{3})$ sq cm d) $4\sqrt{3}$ sq cm

11. Let $A(x) = k \sin x \int_0^x \cos t dt + 2 \int_0^x t dt + k \cos^2 x - x^2$. If $x^2 - 2x + 3 \geq A(x)$, $\forall x \in R$

then the greatest area bounded by $y = xA(x)$, $y = 0$ and $x = 5$ is equal to

A) 16 sq. units

B) 15 sq. units

C) 8 sq. units

D) None of these

12. Let $f(x)$ be a real valued continuous function defined on R as $f(x) = x^2 e^{-|x|}$; then area bounded by curve $y = f(x)$, x-axis & ordinates at points of local maximum is

A) $2 \left(\frac{e^2 - 5}{e^2} \right)$

B) $4 \left(\frac{e^2 - 5}{e^2} \right)$

C) $2 \left(\frac{e^2 + 5}{e^2} \right)$

D) $4 \left(\frac{e^2 + 5}{e^2} \right)$

MULTIPLE ANSWER TYPE

13. Area bounded by curves $y = \frac{x^2}{4a}$ and $y = \frac{8a^3}{x^2 + 4a^2}$ is

a) $\frac{1}{3}(6\pi - 4)$ if $a = 1$ b) $\frac{1}{3}(4\pi + 3)$ if $a = 1$ c) $\frac{4}{3}(6\pi - 4)$ if $a = 2$ d) $\frac{1}{3}(2\pi + 3)$ if $a = 1$

14. Let $f : R \rightarrow R$ is $f(x) = x^3 + 3x^2 + 26x + 24$; then

A) If $x \in [-2, 2]$; then range of $f(x)$ is $[-24, 96]$

B) If $x \in [-2, 2]$; then range of $f(x)$ is $[-48, 48]$

C) Area bounded by $y = f^{-1}(x)$ with x-axis & lines $x = -24$ & $x = 96$ is 128 units

D) Area bounded by $y = f^{-1}(x)$ with x-axis & lines $x = -24$ & $x = 96$ is 116 units

15. Consider $f : (R - \{-2\}) \rightarrow (R - \{1\})$ as $f(x) = \frac{x+1}{x+2}$. Let $g(x)$ be the inverse function of $f(x)$; then

A) Number of real solutions of $f(x) = g(x)$ is 2.

B) Area bounded by asymptotes of $y = f(x)$ and $y = g(x)$ is 9.

C) Area bounded by $y = f(x)$ with asymptotes of $y = g(x)$ is greater than area of Δ formed by their points of intersection.

D) Area bounded by $y = g(x)$ with asymptotes of $y = f(x)$ is less than area of Δ formed by their points of intersection.

16. Consider $y = f(x)$ defined on R , such that $\forall \alpha \in R$, tangent line to $y = f(x)$ at point with abscissa $x = \alpha$ has x-intercept equal to $\alpha - 2$. If $f(0) = 1$ and A is area bounded by $y = f(x)$ with axes & line $x = 2$; then __ (Note: [.] represents greatest integer function.)

A) $\lim_{x \rightarrow 0^+} \left[\frac{f(x) - 1}{\sin x} \right] = 0$

B) $\lim_{x \rightarrow 0} \frac{f(x) - 1}{\sin x} = 1$

C) $A > (1 + \sqrt{e})$

D) $A < e + \sqrt{e}$

17. Let $y = f(x)$ be a differentiable function defined on $R \rightarrow R$ such that $f''(x) = f(x)$. If $f(0) = 2$; $f'(0) = 3$; then __ (Note: [.] represents greatest integer function.)

A) Range of $f(x)$ is R

B) If A is the area bounded by $y = f(x)$ with axes then $[A] = 0$

C) Area bounded by curve $y = f(-|x|)$ with x-axis is $6 - 2\sqrt{5}$

D) Area bounded by curve $y = f(x)$, lines $x = \pm \frac{1}{2} \ln 5$ & $y = f(c)$ (where $c \in \left(-\frac{1}{2} \ln 5, \frac{1}{2} \ln 5 \right)$) is minimum when $c = \lambda$, then $[\lambda] = 1$

18. Consider the curve $x^2 + y = 1$ and tangent T to the curve at point P having x-coordinate $= \alpha$; $0 < \alpha < 1$; then

A) Area bounded by curve with x-axis is $2/3$

B) Area bounded by tangent with axes will be minimum when $\alpha = \frac{1}{\sqrt{3}}$

C) Minimum value of the area of triangle bounded by tangent with axes is $\frac{2}{3\sqrt{3}}$

D) Ratio of minimum value of area of triangle bounded by tangent with axes to area bounded by curve with axes ($0 \leq x \leq 1$) is $\frac{2}{\sqrt{3}}$

INTEGER TYPE

19. Let $g(x) = f(x) + 1$ is a differentiable function, where

$$f(x) + \int_0^x f(x-t) \tan(x-t) dt = - \int_0^x \tan(x-t) dt \text{ for } x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right); \text{ then area enclosed}$$

by $g(x)$ with x-axis & ordinates $x = \pm \frac{\pi}{4}$ is λ where λ^4 is

PASSAGE-1

Consider two curves $x^2y = 1$, $xy - y = \frac{1}{4}$. Then answer the following

20. The value of a ($a > 2$) for which the reciprocal of area of figure bounded by curves and lines $x = 2$, $x = a$ is equal to 'a' itself is

- A) $e + 1$ B) $e^2 + 1$ C) $3 - \sqrt{e}$ D) $\sqrt{2} + e$

21. The value of b ($1 < b < 2$) such that area bounded by these curves lines $x = b$ and $x = 2$ is equal to $1 - \frac{1}{b}$ is

- A) $1 + \frac{1}{e^2}$ B) $1 + \frac{1}{\sqrt{e}}$ C) $2 - \sqrt[3]{e}$ D) $1 + \frac{2}{e}$

PASSAGE-2

A continuous function $f(x)$ satisfying $x^4 - 4x^2 \leq f(x) \leq 2x^2 - x^3$ for all $x \in [0, 2]$, such that the area bounded by $y = f(x)$, $y = x^4 - 4x^2$, the y-axis and the line $x = t$ ($0 \leq t \leq 2$) is k times the area bounded by $y = f(x)$, $y = 2x^2 - x^3$, y-axis and the line $x = t$ ($0 \leq t \leq 2$).

Answer the following questions.

22. If $k = 2$, then $f(x)$ attains point of inflection at

- a) 0 b) $\sqrt{2}$ c) $-\sqrt{2}$ d) none of these

23. If $k = 0$, then $f(x)$ attains local maximum at

- a) 0 b) $\sqrt{2}$ c) $-\sqrt{2}$ d) none of these

24. The value of $\int_{-1}^1 f(x) dx$, is

- a) $\frac{2}{15(k+1)}(23-10k)$ b) $\frac{2}{15(k+1)}(23+10k)$ c) $\frac{2}{15(k+1)}(10k-17)$ d) $\frac{2}{15(k+1)}(10k+17)$

PASSAGE-3

Let $A_r (r \in N)$ be the area, bounded by $(6\pi^3 ry^2 - x)(6e^2 y - x) = 0$ then

25. $\sqrt{A_1}, \sqrt{A_2}, \sqrt{A_3}, \dots$ are in
a) AP b) GP c) HP d) none
26. $\lim_{n \rightarrow \infty} \{ \sqrt{A_1 A_2 A_3} + \sqrt{A_2 A_3 A_4} + \sqrt{A_3 A_4 A_5} + \dots + \text{up to } n \text{ terms} \}$ is.....
a) $\frac{1}{2} \left(\frac{e}{\pi} \right)^9$ b) $\frac{1}{4} \left(\frac{e}{\pi} \right)^9$ c) $\frac{1}{3} \left(\frac{e}{\pi} \right)^9$ d) $\frac{2}{3} \left(\frac{e}{\pi} \right)^9$
27. $\lim_{n \rightarrow \infty} \left\{ e^{\frac{-1}{\sqrt{A_1}}} + e^{\frac{-1}{\sqrt{A_2}}} + e^{\frac{-1}{\sqrt{A_3}}} + \dots + \text{up to } n \text{ terms} \right\}$
a) $\left(e^{\frac{\pi^3}{e^3}} - 1 \right)^{-1}$ b) $\left(e^{\frac{\pi}{e}} - 1 \right)^{-1}$ c) $2 \left(e^{\frac{\pi^3}{e^3}} - 1 \right)^{-1}$ d) $\left(e^{\frac{\pi^2}{e^2}} - 1 \right)^{-1}$

27. $\lim_{n \rightarrow \infty} \left\{ e^{\frac{-1}{\sqrt{A_1}}} + e^{\frac{-1}{\sqrt{A_2}}} + e^{\frac{-1}{\sqrt{A_3}}} + \dots + \text{up to } n \text{ terms} \right\}$

a) $\left(e^{\frac{\pi^3}{e^3}} - 1 \right)^{-1}$ b) $\left(e^{\frac{\pi}{e}} - 1 \right)^{-1}$ c) $2 \left(e^{\frac{\pi^3}{e^3}} - 1 \right)^{-1}$ d) $\left(e^{\frac{\pi^2}{e^2}} - 1 \right)^{-1}$

KEY

1	D	2	C	3	B	4	A	5	C	6	B	7	B	8	C
9	B	10	A	11	B	12	B	13	AC	14	AC	15	ABC	16	ACD
17	ABC	18	BD	19	4	20	B	21	A	22	A	23	A	24	C
25	C	26	B	27	A										



BASARA GNANA SARASWATHI CAMPUS KAKATIYA HILLS

Sec: Star 5C

Differential equations

Date: 25-09-2020

MULTIPLE ANSWER TYPE

- Q.61** The solution of the differential equation $\left(\frac{dy}{dx}\right)^2 - \frac{dy}{dx}(e^x + e^{-x}) + 1 = 0$ is/are-
- (A) $y + e^{-x} = c$ (B) $y - e^{-x} = c$ (C) $y + e^x = c$ (D) $y - e^x = c$
- Q.62** Let $y = (A + Bx)e^{3x}$ be a solution of the differential equation $\frac{d^2y}{dx^2} + m\frac{dy}{dx} + ny = 0$, $m, n \in I$, then
- (A) $m + n = 3$ (B) $n^2 - m^2 = 64$ (C) $m = -6$ (D) $n = 9$
- Q.63** If $y = f(x)$ passes through $(1, 0)$ and satisfying $x - 1 = xy \frac{dy}{dx} + \frac{x^2 y^2}{2!} \left(\frac{dy}{dx}\right)^2 + \frac{x^3 y^3}{3!} \left(\frac{dy}{dx}\right)^3 + \dots$, then-
- (A) $f(x) = \ln x$ (B) $f(x) = \ln(1/x)$ (C) $f(x) = (\ln x)^2$ (D) $f(x) = \frac{1}{x} \ln x$
- Q.64** If function $y = f(x)$, satisfies the differential equation $\frac{d}{dx}(xy) = 1$ and $f(1) = 2$, then
- (A) 'f' is monotonic (B) 'f' is bounded
(C) 'f' has a horizontal and vertical asymptote (D) Range of 'f' is $R - \{1\}$
- Q.65** If $y = y(x)$ is solution of differential equation $\frac{dy}{dx} = \sin x$; $y(0) = 0$, then -
- (A) $y > 0$ if $x \in \left(-\frac{\pi}{2}, 0\right)$
(B) $y < 0$ if $x \in \left(-\frac{\pi}{2}, 0\right)$
(C) Number of inflection points of $y = y(x)$ $\left(0, \frac{\pi}{2}\right)$ is 0.
(D) $\frac{d^2y}{dx^2} + y = -2 \sin x$
- Q.66** Let $f(x) = \tan^{-1}(e^x)$ and $g(x) = f'(x)$, then which of the following statements is(are) correct?
- (A) $g(x)$ is an even function (B) $f(x)$ is derivable for all real values of x .
(C) Range of $g(x)$ is $(0, 1]$ (D) $f'(0) + 2g(0) = \frac{3}{2}$
- Q.67** If $f(x)$ be twice differentiable function such that $f''(x)$ is positive in $[0, 4]$, then which of the following is/are correct
- (A) $f(0) + f(4) = 2f(c)$ for atleast one $c \in (0, 4)$ (B) $f(0) + f(4) < 2f(2)$
(C) $f(0) + f(4) > 2f(2)$ (D) $2f(0) + f(4) > 3f\left(\frac{4}{3}\right)$

- Q.68** The differential equation $\frac{d^2y}{dx^2} + y + \cot^2 x = 0$ must be satisfied by
- (A) $y = 2 + c_1 \cos x + \sqrt{c_2} \sin x$ (B) $y = \cos x \cdot \ln\left(\tan \frac{x}{2}\right) + 2$
- (C) $y = 2 + c_1 \cos x + c_2 \sin x + \cos x \log\left(\tan \frac{x}{2}\right)$ (D) All the above
- Q.69** The normal at a general point (a, b) on a curve makes an angle θ with x-axis which satisfies $b(a^2 \tan \theta - \cot \theta) = a(b^2 + 1)$. The equation of the curve can be
- (A) $y = e^{x^2/2} + c$ (B) $\log(ky^2) = x^2$ (C) $y = ke^{x^2/2}$ (D) $x^2 - y^2 = k$
- Q.70** The equation of the curve such that the distance between the origin & the tangent line at an arbitrary point is equal to the distance between the origin & the normal at the same point is :
- (A) $x^2 + y^2 = c \cdot e^{\arctan(y/x)}$ (B) $\sqrt{x^2 + y^2} = c \cdot e^{-\arctan(y/x)}$
- (C) $\sqrt{x^2 + y^2} \cdot c \cdot e^{\arctan(y/x)}$ (D) $x^2 + y^2 = c \cdot e^{\arctan(y/x)}$
- where c is an arbitrary constant.
- Q.71** Identify the statement(s) which is/are True.
- (A) $f(x, y) = e^{y/x} + \tan \frac{y}{x}$ is a homogeneous function of degree zero
- (B) $x \cdot \ln \frac{y}{x} dx + \frac{y^2}{x} \sin^{-1} \frac{y}{x} dy = 0$ is a homogeneous differential equation of degree one
- (C) $f(x, y) = x^2 + \sin x \cdot \cos y$ is not homogeneous function
- (D) $(x^2 + y^2)dx - (xy^2 - y^3)dy = 0$ is a homogenous differential equation
- Q.72** A function $y = f(x)$ satisfying the differential equation $\frac{dy}{dx} \cdot \sin x - y \cos x + \frac{\sin^2 x}{x^2} = 0$ is such that, $y \rightarrow 0$ as $x \rightarrow \infty$ then the statement which is correct is
- (A) $\lim_{x \rightarrow 0} f(x) = 1$ (B) $\int_0^{\pi/2} f(x) dx$ is less than $\frac{\pi}{2}$
- (C) $\int_0^{\pi/2} f(x) dx$ is greater than unity (D) $f(x)$ is an odd function
- Q.73** Let function $y = f(x)$ satisfies the differential equation $x^2 \frac{dy}{dx} = y^2 e^{\frac{1}{x}}$ ($x \neq 0$) and $\lim_{x \rightarrow 0^+} f(x) = 1$. Identify the correct statement(s) ?
- (A) Range of $f(x)$ is $(0, 1) - \left\{\frac{1}{2}\right\}$ (B) $f(x)$ is bounded
- (C) $\lim_{x \rightarrow 0^+} f(x) = 1$ (D) $\int_0^e f(x) dx > \int_0^1 f(x) dx$
- Q.74** Let $\frac{dy}{dx} + y = f(x)$ where y is a continuous function of x with $y(0) = 1$ and $f(x) = \begin{cases} e^{-x} & \text{if } 0 \leq x \leq 2 \\ e^{-2} & \text{if } x > 2 \end{cases}$.
- Which of the following hold (s) good ?
- (A) $y(1) = 2e^{-1}$ (B) $y'(1) = -e^{-1}$ (C) $y(3) = -2e^{-3}$ (D) $y'(3) = -2e^{-3}$
- Q.75** The tangent at any point P on a curve $f(x, y) = 0$ cuts the y-axis at T. If the distance of the point T from P equals the distance of T from the origin then the curve with this property represents a family of circles. Which of the following is/are **CORRECT** ?
- (A) Any arbitrary line $y = mx$ cuts every member of this family at the point where the slopes of these members are equal.
- (B) $f(x, y) = 0$ is orthogonal to the family of circles $x^2 + y^2 - ky = 0 \forall k \in \mathbb{R}$
- (C) If $f(x, y) = 0$ passes through (2, 2) then the intercept made by its director circle on the y-axis is equal to 8.
- (D) If $f(x, y) = 0$ passes through (-1, 1) then image of its centre in the line $y = x$, is (1, 0)

- Q.76** Consider the functions $f(x)$ and $g(x)$, both defined from $\mathbb{R} \rightarrow \mathbb{R}$ and are defined as $f(x) = 2x - x^2$ and $g(x) = x^n$ where $n \in \mathbb{N}$. If the area between $f(x)$ and $g(x)$ in first quadrant is $\frac{1}{2}$ then n is a divisor of
- (A) 12 (B) 15 (C) 20 (D) 30
- Q.77** Let C be the family of curves $f(x, y, c) = 0$ (no member of C is x -axis) such that length of subnormal at any point $P(x, y)$ on the curve C is equal to four times that of the length of subtangent at the same point. Which of the following statements(s) is (are) correct ?
- (A) Equation of the line with positive y -intercept passing through $(4, 2)$ and perpendicular to the curve C is $x + 2y = 8$.
- (B) Orthogonal trajectory of C is family of parallel lines having gradient ± 2 .
- (C) Order and degree of the differential equation of family of curves C are 1 and 2 respectively.
- (D) Differential equation of family of curves is $2y' \pm x = 0$.

PASSAGE BASED QUESTIONS

Passage # 1 (Q.78 & 79)

Let the solution of the differential equation $y'' = -\frac{y - y'(x+1)}{y^2}$, where $y'(0, 1) = -1$ be a curve which passes through origin. M and m be the maximum and minimum distance of the points on the curve from $(2, 0)$ respectively.

On the basis of above information answer the following:

- Q.78** Equation of the curve is
- (A) $y^2 = 2x$ (B) $y^2 + 3x + y = 0$ (C) $x^2 + y^2 + 2x = 0$ (D) None of these
- Q.79** $M + m$ is equal to
- (A) 4 (B) 6 (C) 8 (D) 12

Passage # 2 (Q.80 & 81)

A rumour spreads through a population of 5000 people at a rate proportional to the product of the number of people who have heard it and the number who have not. Suppose that 100 people initiate the rumour and that a total of 500 people know the rumour after two days.

Let $y(t)$ denotes the number of people who know the rumour at time t . Take $\frac{\log 9}{\log 49} = \frac{129}{229}$

On the basis of above information answer the following.

- Q.80** The time required for half the population to hear rumour is-
- (A) 4.22 days (B) 6.24 days (C) 4.58 days (D) 3.32 days
- Q.81** The function $y(t)$ is given by-
- (A) $\frac{5000}{1 + 50e^{-5000kt}}$ (B) $\frac{5000}{1 + 49e^{-5000kt}}$ (C) $\frac{5000}{1 - 50e^{-5000kt}}$ (D) $\frac{500}{1 + 49e^{-5000kt}}$

Passage # 3 (Q.82 & 83)

A tangent drawn to the curve $y = f(x)$ at $P(x, y)$ cuts the x -axis and y -axis at A and B respectively such that $BP : AP = 2 : 1$, given that $f(1) = 1$.

- Q.82** Equation of curve is-
- (A) $y = \frac{1}{x}$ (B) $y = \frac{1}{x^2}$ (C) $y = \frac{1}{x^3}$ (D) None of these

- Q.83** Equation of normal to curve at (1, 1) is
(A) $x - 4y + 3 = 0$ (B) $x - 3y + 2 = 0$ (C) $x - 2y + 1 = 0$ (D) None of these

Passage # 4 (Q 84 to 86)

Let the function f satisfies $f(x) \cdot f'(-x) = f(-x) \cdot f'(x)$ for all x and $f(0) = 3$.

- Q.84** The value of $f(x) \cdot f(-x)$ for all x , is
(A) 4 (B) 9 (C) 12 (D) 16
- Q.85** $\int_{-51}^{51} \frac{dx}{3+f(x)}$ has the value equal to
(A) 17 (B) 34 (C) 102 (D) 0
- Q.86** Number of roots of $f(x) = 0$ in $[-2, 2]$ is
(A) 0 (B) 1 (C) 2 (D) 4

Passage # 5 (Q. 87 to 91)

Let $f(x)$ and $g(x)$ be two real valued functions satisfying $f(x+y) = f(x) + f(y) - 8xy$ for $\forall x, y \in \mathbb{R}$ and $g(x+y) = g(x) + g(y) + 3xy(x+y)$ for $\forall x, y \in \mathbb{R}$, also $f'(0) = 8$ and $g'(0) = -4$.

- Q.87** Equation $f(x) = k$ has real roots for
(A) $k \geq -4$ (B) $k \leq 4$ (C) $k \geq 4$ (D) $k \leq -4$
- Q.88** If $k \in [-3, 3]$ then the equation $g(x) = k$ has exactly
(A) two roots (B) three roots (C) one root (D) no root
- Q.89** The graph $y = g(x)$ is symmetric w.r.t.
(A) y axis (B) x axis (C) origin (D) $y = x$ line
- Q.90** The area of the figure bounded by the curves $y = f(x)$ and $y = |g(x)|$ is
(A) $\frac{4}{3}$ (B) $\frac{4}{7}$ (C) $\frac{28}{3}$ (D) $\frac{16}{3}$
- Q.91** The area of the figure bounded by $y = f(x)$ and $y = g(x)$ for $x \geq 0$ is divided by x -axis in the ratio
(A) $\frac{4}{3}$ (B) $\frac{4}{7}$ (C) $\frac{28}{3}$ (D) $\frac{16}{3}$

Passage # 6 (Q.92 to 94)

A curve $y = f(x)$ satisfies the differential equation $(1+x^2) \frac{dy}{dx} + 2yx = 4x^2$ and passes through the origin.

- Q.92** The function $y = f(x)$
(A) is strictly increasing $\forall x \in \mathbb{R}$
(B) is such that it has a minima but no maxima
(C) is such that it has a maxima but no minima
(D) has no inflection point
- Q.93** The area enclosed by $y = f^{-1}(x)$, the x -axis and the ordinate at $x = 2/3$ is
(A) $2 \ln 2$ (B) $\frac{4}{3} \ln 2$ (C) $\frac{2}{3} \ln 2$ (D) $\frac{1}{3} \ln 2$

Q.94 For the function $y = f(x)$ which one of the following does not hold good?

- (A) $f(x)$ is a rational function
- (B) $f(x)$ has the same domain and same range
- (C) $f(x)$ is a transcendental function
- (D) $y = f(x)$ is a bijective mapping

Passage # 7 (Q.95 & 96)

Let f be a real valued differentiable function satisfying $f(x + y) = f(x) + f(y) - xy - 1$, $\forall x, y \in \mathbb{R}$ and $f(1) = 1$

Q.95 Equation of the normal to the graph of $f(x)$ at the point where $y = f(x)$ cuts the y -axis, is

- (A) $2x - y + 1 = 0$
- (B) $2x + y - 1 = 0$
- (C) $x - 2y + 2 = 0$
- (D) $x + 2y - 2 = 0$

Q.96 Area enclosed by the curve $y = f(x)$ in the first quadrant is

- (A) $\frac{5}{3}$
- (B) $\frac{4}{3}$
- (C) $\frac{5}{4}$
- (D) $\frac{9}{4}$

COLUMN MATCHING TYPE QUESTIONS

Q.97 Match the following:

Column I

- (A) Solution of differential equation $[3x^2y + 2xy - e^x(1+x)] dx + (x^3 + x^2) dy = 0$
- (B) Solution of differential equation $y dx - x dy - 3xy^2 e^{x^2} dx = 0$
- (C) Solution of differential equation $\frac{dy}{dx} = xy(x^2y^2 - 1)$
- (D) Solution of differential equation $\frac{dy}{dx}(x^2y^3 + xy) = 1$

Column II

- (P) $y^2(x^2 + 1 + ce^{x^2}) = 1$
- (Q) $(x^2 + x^3)y - xe^x = c$
- (R) $\frac{x}{y} - \frac{3}{2}e^{x^2} = c$
- (S) $\frac{1}{x} = 2 - y^2 + ce^{-y^2/2}$
- (T) $\frac{2}{x} = 1 - y^2 + ce^{-y/2}$

Q.98 Match the following :

Column -I

- (A) The number of ways in which three numbers can be selected from the set $\{2, 2^2, 2^3, \dots, 2^{11}\}$ so that they form G.P. are
- (B) Let z_1 and z_2 be the non real roots of the equation $z^2 + 6z + b = 0$. If the origin and the points represented by z_1 and z_2 in argand plane forms an equilateral triangle then value of 'b' is
- (C) In the expansion of $(1 + x)^{36}$, 7th and 8th terms are equal and the value of $\left(\frac{7}{x} + 6\right)^2$ is λ , then the value of $\lambda^{1/4}$ is
- (D) The order of the differential equation whose solution is $y = \frac{C_1}{C_2} \cos(x + C_3) - C_4 e^{x+C_5}$ is

Column-II

- (P) 12
- (Q) 6
- (R) 3
- (S) 25
- (T) 24

Q.99 Match the following :

Column -I

Column -II

(A) $x dy = y (dx + y dy)$, $y > 0$, $y(1) = 1$ and $y(x_0) = -3$, then $x_0 = \lambda$ is less than

(P) $\frac{1}{4}$

(B) If $y(t)$ is solution of $(t+1) \frac{dy}{dt} - ty = 1$, $y(0) = -1$ and $y(1) = \lambda$, then λ is greater than

(Q) -15

(C) $(x^2 + y^2) dx = xy dx$ and $y(1) = 1$ and $y(x_0) = \sqrt{e}$, then $x_0 = \lambda$, then λ is greater than

(R) $-\frac{1}{2}$

(D) $\frac{dy}{dx} + \frac{2y}{x} = 0$, $y(1)$, then $y(2)$ is equal to

(S) $\sqrt{3} e$

NUMERIC RESPONSE TYPE QUESTIONS

Q.100 If $f''(x) + f'(x) + f^2(x) = x^2$ be the differential equation of a curve and let P be the point of maxima then number of tangents which can be drawn from P to $x^2 - y^2 = a^2$ is/are

Q.101 The differential equation of $(y\sqrt{x^2 + y^2} - x)dx + (x\sqrt{x^2 + y^2} - y)dy = 0$ represents a curve given by $f(x, y) = 0$. The point $(0, 2)$ satisfies the equation of the curve and the value of x when $y = 1$ is $-\frac{h}{4}$. Then h is equal to

Q.102 Let $y = f(x)$ be a curve C_1 passing through $(2, 2)$ and $(8, \frac{1}{2})$ and satisfying a differential equation $y \left(\frac{d^2y}{dx^2} \right) = 2 \left(\frac{dy}{dx} \right)^2$. Curve C_2 is the director circle of the circle $x^2 + y^2 = 2$. If the shortest distance between the curves C_1 and C_2 is $(\sqrt{p} - q)$ where $p, q \in \mathbb{N}$, then find the value of $(p^2 - q)$.

KEY

Multiple Correct Option type Questions

61. (A,D) 62. (A,C,D) 63. (A,B) 64. (A,C,D) 65. (A,C,D) 66. (A,B,D) 67. (C,D)
68. (B,C) 69. (C,D) 70. (B,C) 71. (A,B,C) 72. (A,B,C) 73. (A,B,D) 74. (A,B,D)
75. (A,B) 76. (B,C,D) 77. (A,C)

Passage Based Questions

78. (C) 79. (B) 80. (C) 81. (B) 82. (B) 83. (C) 84. (B)
85. (A) 86. (A) 87. (B) 88. (B) 89. (C) 90. (A) 91. (A)
92. (A) 93. (C) 94. (C) 95. (B) 96. (A)

Column Matching Type Questions

97. $A \rightarrow Q; B \rightarrow R; C \rightarrow P; D \rightarrow S$ 98. $A \rightarrow S; B \rightarrow P; C \rightarrow Q; D \rightarrow R$
99. $A \rightarrow P,R,S; B \rightarrow Q; C \rightarrow P,Q,R; D \rightarrow P$

Numeric Response Type Questions

100. 2 101. 3 102. 62



NARAYANA
IIT ACADEMY
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Sec: Star 5C

Test on Differential equations

Date: 22-09-2020

SINGLE ANSWER TYPE

1. Solution of the differential equation $\frac{x+y \frac{dy}{dx}}{y-x \frac{dy}{dx}} = \frac{x \sin^2(x^2+y^2)}{y^3}$ is
- A) $-\cot(x^2+y^2) = \left(\frac{x}{y}\right)^2 + c$ B) $\frac{y^2}{x^2+y^2 c} = -\tan^2(x^2+y^2)$
- C) $-\cot(x^2+y^2) = \left(\frac{y}{x}\right)^2 + c$ D) $\frac{y^2+x^2 c}{x^2} = -\tan^2(x^2+y^2)$
2. The solution of the differential equation $\frac{\sqrt{x}dx + \sqrt{y}dy}{\sqrt{x}dx - \sqrt{y}dy} = \sqrt{\frac{y^3}{x^3}}$ is given by
- a) $\frac{3}{2} \log\left(\frac{y}{x}\right) + \log\left|\frac{x^{3/2} + y^{3/2}}{x^{3/2}}\right| + \tan^{-1}\left(\frac{y}{x}\right)^{3/2} + c = 0$
- b) $\frac{2}{3} \log\left(\frac{y}{x}\right) + \log\left|\frac{x^{3/2} + y^{3/2}}{x^{3/2}}\right| + \tan^{-1} \frac{y}{x} + c = 0$
- c) $\frac{2}{3} \log\left(\frac{y}{x}\right) + \log\left(\frac{x+y}{x}\right) + \tan^{-1}\left(\frac{y^{3/2}}{x^{3/2}}\right) + c = 0$ d) None of these
3. The differential equation representing the family of curves $y^2 = 2c(x + \sqrt{c})$, where $c > 0$ is a parameter, is of order and degree as follows :
- a) order 1, degree 3 b) order 2, degree 3
- c) order 1, degree 2 d) order 2, degree 1
4. Water is initially 12.5 meters deep in a conical tank of radius 6 meters and depth 15 meters. It evaporates at a rate proportional to the exposed surface area. If the depth of the water in the tank is 10 meters after 7 days, when will all the water in the tank evaporate?
- A) 30 days B) 33 days C) 35 days D) 28 days
5. Curve which satisfies the differential equation $(x^3 + 3xy^2) \frac{dy}{dx} = y^3 + 3x^2y$, passing through $(1, \sqrt{2})$ lies is
- A) Second & fourth quadrants only B) first & third quadrants only
- C) First & fourth quadrant only D) second & fourth quadrants only
6. If in a culture of yeast, the active ferment doubles itself in three hours. By what ratio will it increase in 15 hours on the assumption that the quantity increases at a rate proportional to itself?
- A) Multiplies itself 16 times B) Multiplies itself 32 times

- C) Multiplies itself 64 times D) Multiplies itself 10 times
7. If the differential equation of the family of curve given by $y = Ax + Be^{2x}$ where A,B are arbitrary constants is of, the form $(1-2x)\frac{d}{dx}\left(\frac{dy}{dx} + \ell y\right) + k\left(\frac{dy}{dx} + \ell y\right) = 0$
- Then the ordered pair (k, ℓ) is---
- A) $(2, -2)$ B) $(-2, 2)$ C) $(2, 2)$ D) $(-2, -2)$
8. Solution of the differential equation $\left\{\frac{1}{x} - \frac{y^2}{(x-y)^2}\right\}dx + \left\{\frac{x^2}{(x-y)^2} - \frac{1}{y}\right\}dy = 0$
- a) $\ln\left|\frac{x}{y}\right| + \frac{xy}{x-y} = C$ b) $\frac{xy}{x-y} = Ce^{x/y}$ c) $\ln|xy| = C + \frac{xy}{x-y}$ d) $\ln|xy| = C + \frac{2xy}{x-y}$
9. Solve the differential equation $\frac{dy}{dx} + \frac{\cos x(3\cos y - 7\sin x - 3)}{\sin y(3\sin x - 7\cos y + 7)} = 0$
- a) $(\cos y - \sin x + 1)^5 (\cos y - \sin x - 1)^2 = C$ b) $(\cos y - \sin x + 1)^5 (\cos y + \sin x - 1)^2 = C$
- c) $(\cos y - \sin x - 1)^5 (\cos y + \sin x - 1)^2 = C$ d) $(\cos y + \sin x - 1)^5 (\cos y - \sin x - 1)^2 = C$
10. Tangent is drawn at any point P of a curve which passes through $(1, 1)$ cutting x-axis and y-axis at A and B respectively. If $BP : PA = 3 : 1$ then
- a) Differential equation of the curve is $x\frac{dy}{dx} + 3y = 0$ b) Curve is passing through $\left(\frac{1}{8}, 2\right)$
- b) Differential equation of the curve is $3x\frac{dy}{dx} + y = 0$ d) Normal at $(1, 1)$ is $x + 3y = 4$
11. Solution of $y^2 dx + (x^2 - xy + y^2) dy = 0$ is (c is arbitrary constant)
- A) $\tan^{-1}\left(\frac{y}{x}\right) - \log y + c = 0$ B) $2 \tan^{-1}\left(\frac{y}{x}\right) - \log x + c = 0$
- C) $\log\left(y + \sqrt{x^2 + y^2}\right) + \log y + c = 0$ D) $\log\left(y - \sqrt{x^2 + y^2}\right) - \log y + c = 0$
12. Let $f : [0, 2] \rightarrow R$ be a function which is continuous on $[0, 2]$ and is differentiable on $(0, 2)$ with $f(0) = 1$. Let $F(x) = \int_0^x f(\sqrt{t}) dt$ for $x \in [0, 2]$. If $F'(x) = f'(x)$, for all $x \in (0, 2)$, then $F(2)$ equals
- A) $e^2 - 1$ B) $e^4 - 1$ C) $e - 1$ D) e^4
13. A function $y = f(x)$ satisfies the differential equation $\frac{dy}{dx} - y = \cos x - \sin x$ with initial condition that y is bounded when $x \rightarrow \infty$. The area enclosed by $y = f(x)$, $y = \cos x$ and y-axis is first quadrant is sq. units
- A) $\sqrt{2} - 1$ B) $\sqrt{2}$ C) 1 D) $\frac{1}{\sqrt{2}}$
14. At present, a firm is manufacturing 2000 items. It is estimated that the rate of change of production p with respect to additional number of workers 'x' is given by $\frac{dp}{dx} = 100 - 12\sqrt{x}$. If the firm employs 25 more workers then the new level of production of items is
- A) 2500 B) 3000 C) 3500 D) 4500

MULTIPLE ANSWER TYPE

15. The differential equation representing the family of curves $y^2 = 2c(x + \sqrt{c})$, where c is a positive parameter, is of
 A) order 1 B) order 2 C) degree 3 D) degree 4
16. The solution of the differential equation $(y^2 dx - 2xy dy) = x^3 y^3 dy + x^2 y^4 dx$ is
 A) $\log\left(\frac{x}{y^2}\right) = \frac{(xy)^2}{2} + c$ B) $\frac{x}{cy^2} = e^{\frac{(xy)^2}{2}}$ C) $\left(\frac{x}{y^2}\right) = 2\log(xy) + c$ D) $\frac{x}{y^2} = e^{2(xy)} + c$

INTEGER ANSWER TYPE

17. If the solution of differential equation $\frac{dy}{dx} = y + \int_0^1 y dx$, is $y = \frac{1}{3-e}(Ae^x + Be + 1)$; also given that $y = 1$, when $x = 0$. the value of $A + B$ is
18. Let $f(x)$ be a real valued differentiable function on set of real numbers such that $f(1) = 1$. If the y -intercept of tangent at any point $P(x, y)$ on the curve $y = f(x)$ is equal to the cube of abscissa of P then $f(-3) = \dots\dots\dots$
19. Let $h=h(t)$ be a solution to the differential equation $\frac{dh}{dt} + 2ht = t^2$, then $16 \lim_{t \rightarrow \infty} \left(\frac{h}{t}\right)$ is equal to
20. A spherical ball loses volume by evaporation at a rate proportional to its instantaneous area. If the diameter of the ball decreases from 4cm to 1 cm in 3 days, how many more day(s) will it take for the ball to be gone?
21. The rate at which a metal cools in moving air is proportional to the difference between the temperature of the substance and that of the air. If the temperature of the air is 290K and the substance cools from 370K to 330K in 10min, then time required further to reach 295K is 10α min, where $\alpha =$

PASSAGE-1

Consider the differential equation $e^x(ydx - dy) = e^{-x}(ydx + dy)$. Let $y = f(x)$ be a particular solution to this differential equation which passes through the point $(0, 2)$.

Let $C \equiv y = \log_{1/4}\left(x - \frac{1}{4}\right) + \frac{1}{2}\log_4(16x^2 - 8x + 1)$ be another curves

22. The range of the function $g(x) = \log_2(f(x))$ is
 a) $[1, \infty)$ b) $[2, \infty)$ c) $[0, \infty)$ d) $[3, \infty)$
23. If the area bounded by the curve $y = f(x)$, curve C , ordinate $x = \frac{1}{4}$ and the ordinate $x = a$ is $4 - \ln 4 + \frac{1}{e^{1/4}} - e^{1/4}$, then value of 'a' is
 a) $\ln 6$ b) $\ln 4$ c) 4 d) $\ln 12$

MATRIX MATCHING TYPE

24. Match the following:

The differential equation of family of

Column -I

- A) Circles passing through the origin and having their centers on the x axis

Column -II

(P) $xy \frac{d^2 y}{dx^2} + x \left(\frac{dy}{dx}\right)^2 - y \frac{dy}{dx} = 0$

- B) Parabolas with foci at the origin and axis along the x axis (Q) $y^2 = x^2 + 2xy \frac{dy}{dx}$
- C) Parabolas each of which has a latus-rectum $4a$ and whose axes are parallel to x – axis (R) $y \left(\frac{dy}{dx} \right)^2 + 2x \left(\frac{dy}{dx} \right) - y = 0$
- D) Ellipse centered at the origin with an axis along x – axis (S) $2a \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^3 = 0$

- D) Ellipse centered at the origin with an axis along x – axis (S) $2a \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^3 = 0$

KEY

1	A	2	D	3	A	4	C	5	B	6	B	7	A	8	A
9	D	10	A	11	A	12	B	13	A	14	C	15	AC	16	AB
17	1	18	9	19	8	20	1	21	3	22	A	23	B		

24. A-Q; B-RC-S; D-P



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BASARA GNANA SARASWATHI CAMPUS KAKATIYA HILLS

Sec: Star 5C

Test on Differential equations

Date: 23-09-2020

SINGLE ANSWER TYPE

1. A continuous function $f: R \rightarrow R$ satisfying the differential equation

$$f(x) = (1+x^2) \left(1 + \int_0^x \frac{f^2(t)}{1+t^2} dt \right) \text{ then } f(-2) = \underline{\hspace{2cm}}$$

- A) 0 B) $\frac{17}{15}$ C) $-\frac{17}{15}$ D) $\frac{15}{17}$

2. The solution of the differential equation $\left(\frac{1}{x} - \frac{y^2}{(x-y)^2} \right) dx + \left(\frac{x^2}{(x-y)^2} - \frac{1}{y} \right) dy = 0$ is

- A) $\left(\log_e \left(\frac{x}{y} \right) \right) + \frac{xy}{(x-y)} = c$ B) $\log_e(xy) + \frac{xy}{x-y} = C$
C) $\log_e \left(\frac{x}{y} \right) - \frac{xy}{x-y} = c$ D) $\log_e \left(\frac{x}{y} \right) + \frac{1}{x-y} = xy + c$

3. $x \int_0^x y(t) dt = (x+1) \int_1^x t y(t) dt, x > 0$ then $y(x) = \underline{\hspace{2cm}}$

- A) $\frac{ce^{-x}}{x^2}$ B) $\frac{ce^{-1/x}}{x^3}$ C) $\frac{ce^{-x}}{x^3}$ D) $\frac{e^{-x}}{4x^3} + c$

MULTIPLE ANSWER TYPE

4. A function $y = f(x)$ satisfies the following conditions

1) $f(1) = 1$

2) The tangent at any point of the graph of $f(x)$ cuts off on the y-axis an intercept equal to the square of the abscissa of point of tangency.

Which of the following are correct.

A) The graph of $y = f(x)$ is passing through $(0, 0)$

B) $y = f(x)$ represents a conic whose focus is $\left(1, \frac{3}{4}\right)$

C) The area enclosed by $y = f(x)$ and the line $y+3=0$ is $\frac{32}{3}$

D) $y = f(x)$ has no maxima

5. Which of the following statements are correct

A) The solution of the differential equation $y(x^2y + e^x)dx - e^x dy = 0$ is $x^3y + 3e^x = 3cy$ (where c is constant)

B) The solution of $\frac{xdx + ydy}{xdy - ydx} = \sqrt{\frac{a^2 - x^2 - y^2}{x^2 + y^2}}$ is $\tan^{-1}\left(\frac{y}{x}\right) + \sin^{-1}\left(\frac{\sqrt{x^2 + y^2}}{a}\right) = c$

C) $xdy = (y + x^2 + 9y^2)dx$ then its solution is $\tan^{-1}\left(\frac{3y}{x}\right) = 3x + c$

D) The solution of the differential equation $\frac{dy}{dx} - y \cdot \frac{\phi'(x)}{\phi(x)} = \frac{-y^2}{\phi(x)}$ is $\phi(x) = y(x + c)$

6. If $x^m + y^m = cx^n$ is the solution of $(x^3 - 2y^3)dx + 3xy^2dy = 0$ then

- a) $m - n = 1$ b) $m + n = 5$ c) $m = 3, n = 2$ d) $2m = 3n$

7. Let $f: \mathbb{R} \rightarrow \mathbb{R}$, such that $f''(x) - 2f'(x) + f(x) = 2e^x$ and $f'(x) > 0, \forall x$. then which of the following is/are true?

- A) $|f(x)| = f(x), \forall x \in \mathbb{R}$ B) $f(x) = -5$ for some $x > 0$
C) $|f(x)| = -f(x), \forall x \in \mathbb{R}$ D) $f(x)$ can be 5 for some $x > 0$

8. Let $f: \mathbb{R}^+ \rightarrow \mathbb{R}$ be a strictly increasing function such that $f(x) > \frac{-1}{x}, \forall x > 0$ and

$f(x) \cdot f\left(f(x) + \frac{1}{x}\right) = 1, \forall x > 0$ then which of the following is/are false?

- A) maximum value of $f(x)$ in $[1, 2]$ is $\frac{1 + \sqrt{5}}{4}$
B) minimum value of $f(x)$ in $[1, 2]$ is $\frac{1 - \sqrt{5}}{4}$
C) Maximum value of $f(x)$ in $[1, 2]$ is $\frac{1 - \sqrt{5}}{4}$
D) Minimum value of $f(x)$ in $[1, 2]$ is $\frac{1 - \sqrt{5}}{2}$

9. A function $y = f(x)$ satisfying the differential equation $\sin x \frac{dy}{dx} - y \cos x + \frac{\sin^2 x}{x^2} = 0$ is such that $y \rightarrow 0$ as $x \rightarrow \infty$ then which of the following is/are true?

- A) $\lim_{x \rightarrow 0} f(x) = 1$ B) $1 < \int_0^{\pi/2} f(x) dx < \pi/2$
C) $\int_0^{\pi/2} f(x) dx > \pi/2$ D) $f(x)$ is odd function

10. The orthogonal trajectories of the system of curves $\left(\frac{dy}{dx}\right)^2 = \frac{a}{x}$ are

- a) $9a(y + c)^2 = 4x^3$ b) $y + c = \frac{-2}{3\sqrt{a}}x^{3/2}$

c) $y + c = \frac{2}{3\sqrt{a}} x^{3/2}$

d) none of these

11. A solutions of the differential equation $\left(\frac{dy}{dx}\right)^2 - \frac{dy}{dx}(e^x + e^{-x}) + 1 = 0$ are given by

A) $y + e^{-x} = c$

B) $y - e^{-x} = c$

C) $y + e^x = c$

D) $y - e^x = c$

12. The solution of the differential equation $(y^2 dx - 2xy dy) = x^3 y^3 dy + x^2 y^4 dx$ is

A) $\log\left(\frac{x}{y^2}\right) = \frac{(xy)^2}{2} + c$

B) $\frac{x}{cy^2} = e^{\frac{(xy)^2}{2}}$

C) $\left(\frac{x}{y^2}\right) = 2 \log(xy) + c$

D) $\frac{x}{y^2} = e^{2(xy)} + c$

13. If $\left(\frac{x+y-1}{x+y-2}\right) \frac{dy}{dx} = \left(\frac{x+y+1}{x+y+2}\right)$, given that $y = 1$ when $x = 1$, then which of the following is/are **not** a solution of the given differential equation?

A) $\ln\left|\frac{(x-y)^2 - 2}{2}\right| = 2(x+y)$

B) $\ln\left|\frac{(x+y)^2 - 2}{2}\right| = 2(x-y)$

C) $\ln\left|\frac{(x-y)^2 + 2}{2}\right| = 2(x+y)$

D) $\ln\left|\frac{(x+y)^2 + 2}{2}\right| = 2(x-y)$

14. The gradient of curve passing through $(4, 0)$ is given by $\frac{dy}{dx} - \frac{y}{x} + \frac{5x}{(x+2)(x-3)} = 0$.

If the point $(5, a)$ lies on the curve then

A) Domain of $y(x)$ is $(-2, 3)$

B) Domain of $y(x)$ is $(-\infty, -2) \cup (3, \infty)$

C) $a = 5 \ln\left(\frac{12}{7}\right)$

D) $a = -5 \ln\left(\frac{12}{7}\right)$

PASSAGE-1

Let $y = f(x)$ satisfy the differential equation

$$f(x) = (e^x + e^{-x}) \cos x - 2x - \int_0^x (x-t) f'(t) dt$$

15. If y satisfies the differential equation $\frac{dy}{dx} + y = e^x g(x) + e^{-x} g'(x)$ then $g(x)$ is ...

A) $\sin x + \cos x$

B) $\sin x - \cos x$

C) $\cos x - \sin x$

D) $-\sin x - \cos x$

16. The value of $f'(\pi) + f''(\pi) = \dots\dots$

A) -1

B) 0

C) 1

D) 2

INTEGER TYPE

17. A curve passing through $(2, 3)$ and satisfying $\int_0^x t y(t) dt = x^2 y(x)$; $x > 0$ is $xy = \lambda$ then

$$\lambda = \underline{\hspace{2cm}}$$

18. If $f(x) = e + (1-x) \log_e\left(\frac{x}{e}\right) + \int_1^x f(t) dt$ then $f(e) - e^e =$

19. If $x \frac{dy}{dx} + 2y = \log_e x$ then $\left[\frac{e^2 y(e) - y(1)}{e^2 + 1} \right] = \text{---}$ (Where $[.]$ is G.I.F)
20. Let $y = f(x)$ be a curve passing through (e, e^e) which satisfy the differential equation $(2ny + xy \log_e x)dx - x \log_e x dy = 0, x > 0, y > 0$. If $g(x) = \lim_{n \rightarrow \infty} f(x)$, then $\int_{1/e}^e g(x) dx =$
21. A function $y = f(x)$ satisfies the differential equation $f(x) \sin 2x - \cos x + (1 + \sin^2 x) f'(x) = 0$ with initial condition $y(0) = 0$ then the value of $(10 f(\frac{\pi}{6}))$ is equal to
22. Let $y = f(x)$ be a curve passing through (e, e^e) which satisfy the differential equation $(2ny + xy \log_e x)dx - x \log_e x dy = 0, x > 0, y > 0$. If $g(x) = \lim_{n \rightarrow \infty} f(x)$, then $\int_{1/e}^e g(x) dx =$
23. If $y = f(x)$ be a curve passing through $(1, 3)$ satisfying the differential equation $(x^2 + xy + 4x + 2y + 4) \frac{dy}{dx} = y^2 (x > 0)$, then the number of solutions of the equation $f(x) = x + 2$ is _____

MATCHING TYPE

24. Match the following:

The differential equation of family of

Column -I

Column -II

(A) Circles passing through the

origin and having their centers on the x axis

(p) $xy \frac{d^2 y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 - y \frac{dy}{dx} = 0$

(B) Parabolas with foci at the origin

and axis along the x axis

(q) $y^2 = x^2 + 2xy \frac{dy}{dx}$

(C) Parabolas each of which has a

latus-rectum $4a$ and whose axes are parallel to

x - axis

(r) $y \left(\frac{dy}{dx} \right)^2 + 2x \left(\frac{dy}{dx} \right) - y = 0$

(D) Ellipse centered at the origin with an

axis along x - axis

(s) $2a \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx} \right)^3 = 0$

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axis along x - axis

$$(s) \quad 2a \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx} \right)^3 = 0$$

+

KEY

1	D	2	A	3	B	4	ABC	5	ACD	6	ABCD	7	AD	8	AB
---	---	---	---	---	---	---	-----	---	-----	---	------	---	----	---	----

9	AD	10	ABC	11	AD	12	AB	13	ACD	14	BD	15	C	16	B
17	6	18	1	19	0	20	0	21	4	22	0	23	1		

24. A-Q;B-R;C-S;D-P

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Sec: Star 5C

Test on Differential equations

Date: 24-09-2020

SINGLE ANSWER TYPE

1. Let $f:[0,2] \rightarrow R$ be a function which is continuous on $[0,2]$ and is differentiable on $(0,2)$ with $f(0)=1$. Let $F(x) = \int_0^{x^2} f(\sqrt{t}) dt$ for $x \in [0,2]$. If $F'(x) = f'(x)$, for all $x \in (0,2)$, then $F(2)$ equals
- A) $e^2 - 1$ B) $e^4 - 1$ C) $e - 1$ D) e^4
2. The solution of the differential equation $y_1 y_3 = 3y_2^2$ is
- a) $x = A_1 y^2 + A_2 y + A_3$ b) $x = A_1 y + A_2$
c) $x = A_1 y^2 + A_2 y$ d) none of these
3. The orthogonal trajectories of the family of curves $a^{n-1}y = x^n$ are given by
- a) $x^n + n^2 y = \text{constant}$ b) $ny^2 + x^2 = \text{constant}$
c) $n^2 x + y^n = \text{constant}$ d) $n^2 x - y^n = \text{constant}$
4. A curve is such that the mid point of the portion of the tangent intercepted between the point where the tangent is drawn and the point where the tangent meets y-axis, lies on the line $y = x$. If the curve passes through $(1, 0)$, then the curve is
- (A) $2y = x^2 - x$ (B) $y = x^2 - x$ (C) $y = x - x^2$ (D) $y = 2(x - x^2)$
5. A function $y = f(x)$ has a second order derivative $f'' = 6(x-1)$. If its graph passes through the point $(2,1)$ and at that point the tangent to the graph is $y = 3x - 5$, then the function is
- a) $(x-1)^2$ b) $(x-1)^3$ c) $(x+1)^2$ d) $(x+1)^3$
6. The normal to a curve at $P(x,y)$ meets the x -axis at G. If the distance of G from the origin is twice the abscissa of P, then the curve is
- a) a parabola b) a circle c) a hyperbola d) an ellipse
7. A conic C satisfies the differential equation $(1+y^2)dx - xydy = 0$ and passes through the point $(1,0)$. An ellipse E which is confocal with C has its eccentricity $\sqrt{2/3}$. The angle of intersection of the curves C and E is
- a) $\frac{\pi}{6}$ b) $\frac{\pi}{4}$ c) $\frac{\pi}{3}$ d) $\frac{\pi}{2}$
8. The solution of the differential equation $2x^3 y dy + (1-y^2)(x^2 y^2 + y^2 - 1) dx = 0$
[Where c is a constant]

(A) $x^2y^2 = (cx+1)(1-y^2)$

(B) $x^2y^2 = (cx+1)(1+y^2)$

(C) $x^2y^2 = (cx-1)(1-y^2)$

(D) none of these

9. The differential equation of all circles in a plane must be

a) $y_3(1+y_1^2) - 3y_1y_2^2 = 0$

b) of order 3 and degree 3

c) of order 3 and degree 2

d) $y_3(1-y_1^2) - 3y_1y_2^2 = 0$

10. If the population of a country doubles in 50 years in how many years will it become thrice the original, assume the rate of increase is proportional to the number of inhabitants

a) 75

b) $50 \log_2 3$ c) $50 \log_3 2$

d) 100

11. The function $y=f(x)$ is the solution of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2-1} = \frac{x^4+2x}{\sqrt{1-x^2}}$ in
 $(-1,1)$ satisfying $f(0)=0$. Then $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)dx$ is

(a) $\frac{\pi}{3} - \frac{\sqrt{3}}{2}$

(b) $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$

(c) $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$

(d) $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

MULTIPLE ANSWER TYPE12. Let $y(x)$ be a solution of the differential equation $(1+e^x)y' + ye^x = 1$. If $y(0) = 2$, then which of the following statements is (are) true?

a) $y(-4) = 0$

b) $y(-2) = 0$

c) $y(x)$ has a critical point in the interval $(-1,0)$ d) $y(x)$ has no critical point in the interval $(-1,0)$ 13. Consider the family of all circles whose centers lie on the straight line $y = x$. If this family of circles is represented by the differential equation $Py'' + Qy' + 1 = 0$, where P, Q are functions of x, y and y' (here $y' = \frac{dy}{dx}$, $y'' = \frac{d^2y}{dx^2}$), then which of the following statements is (are) true?

a) $P = y + x$

b) $P = y - x$

c) $P + Q = 1 - x + y + y' + (y')^2$

d) $P - Q = x + y - y' - (y')^2$

14. The solution of $\frac{dy}{dx} = \frac{x^2+y^2}{2xy}$ satisfying $y(1)=1$ is given by $(x, y \neq 0)$

A) a pair of straight lines

B) a system of circles

C) $y^2 = x^2$

D) $(x-2)^2 + (y-3)^2 = 5$

15. Solution of the differential equation :

$(3 \tan x + 4 \cot y - 7) \sin^2 y \, dx - (4 \tan x + 7 \cot y - 5) \cos^2 x \, dy = 0$ is

(A) $\frac{3}{2} \cot^2 x - 7 \cot x + \frac{7}{2} \tan^2 y - 5 \tan y + 4 \cot x \cdot \tan y = c$

(B) $\frac{3}{2} \tan^2 x - 7 \tan x + \frac{7}{2} \cot^2 y - 5 \cot y + 4 \tan x \cdot \cot y = c$

(C) $3 \tan^2 y - 14 \cot x \cdot \tan^2 y + 7 \cot^2 x - 10 \tan y \cot^2 x + 8 \cot x \cdot \tan y + 2 \cot^2 x \tan^2 y = 0$.

(D) $3 \cot^2 y - 14 \cot x \cdot \cot^2 y + 7 \cot^2 x + 10 \cot y \tan^2 x + 8 \tan x \cdot \cot y = 0$.

16. The coordinates of a point $P(x, y)$ are functions of time t and satisfy the relations

$$\frac{dx}{dt} + \frac{dy}{dt} = t \text{ and } \frac{dx}{dt} - 2 \frac{dy}{dt} = t^2 \text{ at any instant of time } t. \text{ The locus of point } P(x, y) \text{ is a curve}$$

given by (assume $x(0) = y(0) = 0$)

a) $(x + y)^3 = (x - 2y)^2$

b) $x = \frac{t^2}{3} + \frac{t^3}{9}, y = \frac{t^2}{6} - \frac{t^3}{9}$

c) $9(x + y)^3 = 8(x - 2y)^2$

d) $8(x + y)^3 = 9(x - 2y)^2$

17. The solution of the differential equation $(y^2 dx - 2xy dy) = x^3 y^3 dy + x^2 y^4 dx$ is

A) $\log\left(\frac{x}{y^2}\right) = \frac{(xy)^2}{2} + c$

B) $\frac{x}{cy^2} = e^{\frac{(xy)^2}{2}}$

C) $\left(\frac{x}{y^2}\right) = 2 \log(xy) + c$

D) $\frac{x}{y^2} = e^{2(xy)} + c$

18. Let $y = (A + Bx)e^{3x}$ is a solution of the differential equation $\frac{d^2 y}{dx^2} + m \frac{dy}{dx} + ny = 0$, $m, n \in I$, then

a) $m = -6$

b) $n = -6$

c) $m = 9$

d) $n = 9$

19. If $y(x)$ satisfies the differential equation $y' - y \tan x = 2x \sec x$ and $y(0) = 0$, then

a) $y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{8\sqrt{2}}$

b) $y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{18}$

c) $y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{9}$

d) $y\left(\frac{\pi}{3}\right) = \frac{4\pi}{3} + \frac{2\pi^2}{3\sqrt{3}}$

20. If $x^m + y^m = cx^n$ is the solution of $(x^3 - 2y^3)dx + 3xy^2 dy = 0$ then

a) $m - n = 1$

b) $m + n = 5$

c) $m = 3, n = 2$

d) $2m = 3n$

INTEGER TYPE

21. Let $h = h(t)$ be a solution to the differential equation $\frac{dh}{dt} + 2ht = t^2$, then $16 \lim_{t \rightarrow \infty} \left(\frac{h}{t}\right)$ is equal to

22. A spherical ball loses volume by evaporation at a rate proportional to its instantaneous area. If the diameter of the ball decreases from 4cm to 1 cm in 3 days, how many more day(s) will it take for the ball to be gone?

23. The rate at which a metal cools in moving air is proportional to the difference between the temperature of the substance and that of the air. If the temperature of the air is 290K and the substance cools from 370K to 330K in 10min, then time required further to reach 295K is 10α min, where $\alpha =$

24. Differential equation, having $y = (\sin^{-1} x)^2 + A(\cos^{-1} x) + B$ where A and B are arbitrary constants is $(p - x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} = q$ then $p + q =$

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KEY															
1	B	2	A	3	B	4	C	5	B	6	C	7	D	8	C
9	A	10	B	11	B	12	AC	13	BC	14	AC	15	BC	16	BD
17	AB	18	AD	19	AD	20	ABCD	21	8	22	1	23	3	24	3

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BASARA GNANA SARASWATHI CAMPUS KAKATIYA HILLS

Sec: Star 5C

AOD

Date: 02-10-2020

SINGLE ANSWER TYPE

- Let $f(x)$ and $g(x)$ defined from $[0, \infty)$ to $[0, \infty)$. If $f(x)$ be an increasing function and $g(x)$ be a decreasing function. Also, $h(x) = f(g(x))$, $h(0) = 0$ then $h(x) - h(1)$ is
 A) Always zero
 B) Always negative
 C) Always positive
 D) Strictly increasing
- A person wishes to lay a straight fence across a triangular field ABC, with $\angle A < \angle B < \angle C$ so as to divide it into two equal areas. The length of the fence with minimum expense, is
 A) $\sqrt{2\Delta \cot \frac{B}{2}}$
 B) $\sqrt{2\Delta \tan \frac{C}{3}}$
 C) $\sqrt{\tan \frac{A}{2} \tan \frac{B}{2} \tan \frac{C}{2}}$
 D) $\sqrt{2\Delta \tan \frac{A}{2}}$
- Let a, b, c, d, e, f, g, h be distinct elements in the set $\{-7, -5, -3, -2, 2, 4, 6, 13\}$. The minimum value of $(a + b + c + d)^2 + (e + f + g + h)^2$ is
 A) 30
 B) 32
 C) 34
 D) 40
- Let $\phi(x) = \frac{(x-b)(x-c)}{(a-b)(a-c)}f(a) + \frac{(x-c)(x-a)}{(b-c)(b-a)}f(b) + \frac{(x-a)(x-b)}{(c-a)(c-b)}f(c) - f(x)$ Where $a < c < b$ and $f^{(1)}(x)$ exists at all points in (a, b) . Then, there exists a real number $\mu, a < \mu < b$ such that $\frac{f(a)}{(a-b)(a-c)} + \frac{f(b)}{(b-c)(b-a)} + \frac{f(c)}{(c-a)(c-b)} =$
 a) $f^{(1)}(\mu)$
 b) $2f^{(1)}(\mu)$
 c) $\frac{1}{2}f^{(1)}(\mu)$
 d) $\frac{1}{3}f^{(1)}(\mu)$
- The least value of 'a' for which the equation $\frac{4}{\sin x} + \frac{1}{1 - \sin x} = a$ for atleast one solution on the interval $\left(0, \frac{\pi}{2}\right)$ is,
 a) 1
 b) 4
 c) 8
 d) 9
- Water is dripping out from a conical funnel, at the uniform rate of $2\text{ cm}^3/\text{sec}$ through a tiny hole at the vertex at the bottom. When the slant height of the water is 4 cm, find the rate of decrease of the slant height of the water given that the vertical angle of the funnel is 120° .
 A) $3\pi\text{ cm/sec}$
 B) $\frac{1}{3\pi}\text{ cm/sec}$
 C) $4\pi\text{ cm/sec}$
 D) $\frac{1}{4\pi}\text{ cm/sec}$
- Let A be the point where the curve $5x^2x^3 + 10ax^2 + x + 2y - 4 = 0$ ($a \in \mathbb{R} - \{0\}$) meets y-axis. Then the equation of tangent to the curve at the point where the normal at A meets the curve again, is
 A) $x - ay + 2a = 0$
 B) $ax + y - 2 = 0$
 C) $2x - y + 2 = 0$
 D) $x + 2y - 4 = 0$

8. Two runners A and B start at the origin and run along positive x-axis, with B running three times as fast as A. An observer, standing one unit above the origin, keeps A and B in view. Then the maximum angle of sight ' θ ' between the observer's view of A and B is
- A) $\pi/8$ B) $\pi/6$ C) $\pi/3$ D) $\pi/4$
9. The tangent to the curve $y = e^x$ drawn at the point (c, e^c) intersects the line joining the points $(c-1, e^{c-1})$ and $(c+1, e^{c+1})$
- A) on the left of $x = c$ B) on the right of $x = c$
C) at no point D) at all points
10. Let $f(x) = (1-x)^2 \sin^2 x + x^2$ for all $x \in \mathbb{R}$. Consider the statements :
- P: There exists some $x \in \mathbb{R}$ such that $f(x) + 2x = 2(1+x^2)$
Q: There exists some $x \in \mathbb{R}$ such that $2f(x) + 1 = 2x(1+x)$
- Then
- A) Both P and Q are true B) P is true and Q is false
C) P is false and Q is true D) Both P and Q are false
11. The number of values of x at which the function, $f(x) = (x-1)x^{2/3}$ has extreme values, is
- A) 4 B) 3 C) 2 D) 1
12. Tangents are drawn from origin to the curve $y = \sin x + \cos x$. Then their points of contact lie on the curve
- A) $\frac{1}{x^2} + \frac{2}{y^2} = 1$ B) $\frac{2}{x^2} - \frac{1}{y^2} = 1$ C) $\frac{2}{x^2} + \frac{1}{y^2} = 1$ D) $\frac{2}{y^2} - \frac{1}{x^2} = 1$
13. A rectangle is inscribed in an equilateral Δ of side length 2a units. Maximum area of this rectangle is
- A) $\sqrt{3}a^2$ B) $\frac{\sqrt{3}a^2}{4}$ C) a^2 D) $\frac{\sqrt{3}a^2}{2}$
14. A person wishes to lay a straight fence across a triangular field ABC, with $\angle A < \angle B < \angle C$ so as to divide it into two equal areas. The length of the fence with minimum expense, is
- A) $\sqrt{2\Delta \cot \frac{B}{2}}$ B) $\sqrt{2\Delta \tan \frac{C}{3}}$ C) $\sqrt{\tan \frac{A}{2} \tan \frac{B}{2} \tan \frac{C}{2}}$ D) $\sqrt{2\Delta \tan \frac{A}{2}}$

(where 'Δ' represents, area of triangle ABC)

15. A variable line through A(6,8) meets the curve $x^2 + y^2 = 2$ at B and C. P is a point on BC such that AB, AP, AC are in HP. The minimum distance of the origin from the locus of P is

A) 1 B) $\frac{1}{2}$ C) $\frac{1}{3}$ D) $\frac{1}{5}$

MULTIPLE ANSWER TYPE

16. Tangent at a point P_1 [other than (0, 0)] on the curve $y = x^3$ meets the curve again at P_2 . The tangent at P_2 meets the curve at P_3 and so on. Then

A) abscissae of $P_1, P_2, P_3, \dots, P_n$ are in A. P.

B) abscissae of $P_1, P_2, P_3, \dots, P_n$ form a G. P.

C) area $(\Delta P_2 P_3 P_4) = 16$ area $(\Delta P_1 P_2 P_3)$ D) area $(\Delta P_1 P_2 P_3) = 16$ area $(\Delta P_2 P_3 P_4)$

17. Let $g'(x) > 0$ and $f'(x) < 0 \forall x \in \mathbb{R}$, then

A) $g(f(x+1)) > g(f(x-1))$

B) $g(f(x+1)) < g(f(x-1))$

C) $f(g(x-1)) > f(g(x+1))$

D) $f(g(x-1)) < f(g(x+1))$

18. If $|f(x)| = |ax^2 + bx + c| \leq 1$ for $|x| \leq 1$ $(a, b, c) \in \mathbb{R}$ and $\frac{8}{3}a^2 + 2b^2$ is maximum, then

which of the following is/are **true**?

A) $b = 0$

B) $|a + c| = 1$

C) $ac = 1$

D) $\left| \frac{c}{a} \right| = \frac{1}{2}$

19. Let $f(x)$ be a differentiable function upto any order such that

$f(x) + |f''(x)| \leq 0, \forall x \in \mathbb{R}$. If α and β ($\alpha < \beta$) be the two consecutive real roots of $f(x) = 0$, then which of the following is/are **true**?

A) $f''(x)$ must not be equal to zero for any $x \in (\alpha, \beta)$

B) $f'''(x)$ must be equal to zero for atleast one $x \in (\alpha, \beta)$

C) $f'(x) \neq 0 \forall x \in (\alpha, \beta)$

D) $f'(x)$ must be equal to zero for atleast one $x \in (\alpha, \beta)$

INTEGER ANSWER TYPE

20. The minimum value of, $\frac{\sec^4 \alpha}{\tan^2 \beta} + \frac{\sec^4 \beta}{\tan^2 \alpha}$, $\alpha, \beta \neq \frac{K\pi}{2}$, $K \in \mathbb{I}$, is
21. $f(x)$ is a twice differentiable function in x such that $f(0.9) = 0$, $f(1.9) = 3$, and $f(2.9) = -4$, $f(3.9) = 6$, $f(4.9) = -2$ and $f(5.9) = 2$ then the minimum number of zeros of $g(x) = (f(x))^2 + f(x)f'(x)$ in the interval $[0.9, 5.9]$ is
22. $f(x) = \lim_{n \rightarrow \infty} \left(\sin^4 x + \frac{1}{4} \sin^4 2x + \dots + \frac{1}{4^n} \sin^4 (2^n x) \right)$ and $g(x)$ is a differentiable function satisfying $g(x) + f(x) = 1$, the maximum value of $(\sqrt{f(x)} + \sqrt{g(x)})^4$ is _____
23. Let f be a cubic polynomial function having relative extremum at $x = -1$ and $x = \frac{1}{3}$. Such that $\int_1^{-1} f(t) dt = \frac{14}{3}$. If $f(-2) = 0$, $|f(f(f(0)) - 1)| = \underline{\hspace{2cm}}$
24. Tangent to a curve $x^4 y^3 = C^7$ at point P in I^{st} quadrant intersect x, y axis at A, B respectively. If P divides AB in the ratio $m:n$ (in lowest form) then $m = \underline{\hspace{2cm}}$
25. Number of values of x for which the function $f(x) = (x - \alpha_1)^2 (x - \alpha_2)^3 (x - \alpha_3)^2 (x - \alpha_4)^3 (x - \alpha_5)^2$ attains a local maximum or local minimum is _____
($\alpha_1, \alpha_2, \dots, \alpha_5$ are distinct real numbers)
26. f, g are continuous, differentiable and monotonically increasing functions $\forall x \in \mathbb{R}$ such that $g(x) = f(x) \sqrt{1 - 2(f(x))^2}$. If $6|f(x)| \leq k \forall x$ the minimum value of k is _____

PASSAGE-1

For a polynomial function $y = f(x)$

Points of extrema are obtained at points where

$f'(x) = 0$ and

$f''(x_1) > 0 \Rightarrow x_1$ is a point of minima

$f''(x_1) < 0 \Rightarrow x_1$ is a point of maxima

Let $f(x) = x^3 - 3(7 - a)x^2 - 3(9 - a^2)x + 2$

27. The values of parameter 'a' if $f(x)$ has a negative point of local minimum are
A) ϕ B) $\left(-\infty, \frac{58}{14}\right)$ C) $(-3, 3)$ D) $(-\infty, -3) \cup \left(\frac{58}{14}, \infty\right)$
28. The values of parameter 'a' if $f(x)$ has a positive point of local maxima are

- A) ϕ B) $(-\infty, -3) \cup \left(\frac{58}{14}, \infty\right)$ C) $\left(-\infty, \frac{58}{14}\right)$ D) $(-\infty, -3) \cup \left(3, \frac{58}{14}\right)$

29. The values of parameter 'a' if f(x) has points of extrema which are opposite in sign are

- A) ϕ B) $(-3, 3)$ C) $\left(-\infty, \frac{58}{14}\right)$ D) $(-\infty, -3) \cup \left(\frac{58}{14}, \infty\right)$

PASSAGE-2

$$f(x) = \sin 2\pi x + \{x\} : x \in [0, 10].$$

30. Number of points where f achieves local maximum is

- A) 10 B) 11 C) 20 D) 19

31. Number of roots of $f(x) = 0$ in $(0, 10)$ is

- A) 20 B) 30 C) 31 D) 29

PASSAGE-3

$$\text{Consider } f(x) = x + \cos x - \alpha$$

32. Which of the following holds good for the above $f(x)$

- A) $\alpha > 1$ for which $f(x)$ has exactly one positive root
 B) $\alpha > 2$ for which $f(x)$ has exactly one -ve root
 C) $1 < \alpha < 2$ $f(x)$ will have an imaginary set of roots
 D) None of these

33. Which of the following is true for $f(x)$

- A) For $0 < \alpha < \beta < \frac{\pi}{2}$ $\sin \alpha > \frac{\cos \alpha - \cos \beta}{\beta - \alpha} > \cos \beta$
 B) For $0 < \alpha < \beta < \frac{\pi}{2}$ $\sin \alpha < \frac{\cos \alpha - \cos \beta}{\beta - \alpha} < \sin \beta$
 C) For $0 < \alpha < \beta < \frac{\pi}{2}$ $\tan \alpha < \frac{\cos \alpha - \cos \beta}{\beta - \alpha} < \tan \beta$
 D) None of these

KEY

1	A	2	D	3	C	4	C	5	D
6	B	7	C	8	B	9	A	10	C
11	C	12	D	13	D	14	D	15	D
16	BC	17	BC	18	ABD	19	AC	20	8
21	3,9	22	4	23	3	24	3	25	7
26	3	27	A	28	D	29	B	30	A
31	D	32	A	33	B				


BASARA GNANA SARASWATHI CAMPUS KAKATIYA HILLS
Sec: Star 5C
TEST ON AOD
Date: 01-10-2020
SINGLE ANSWER TYPE

- If the equation $|x^2 - 5x + 6| - \lambda x + 7\lambda = 0$ has exactly 3 solutions, then λ is equal to
 A) $-7 + \sqrt{23}$ B) $-9 + 4\sqrt{5}$ C) $-7 - \sqrt{23}$ D) $-9 - 4\sqrt{5}$
- Two runners A and B start at the origin and run along positive x-axis, with B running three times as fast as A. An observer, standing one unit above the origin, keeps A and B in view. Then the maximum angle of sight ' θ ' between the observer's view of A and B is
 A) $\pi/8$ B) $\pi/6$ C) $\pi/3$ D) $\pi/4$
- Let $f(x) = (1-x)^2 \sin^2 x + x^2$ for all $x \in \mathbb{R}$. Consider the statements :
 P: There exists some $x \in \mathbb{R}$ such that $f(x) + 2x = 2(1+x^2)$
 Q: There exists some $x \in \mathbb{R}$ such that $2f(x) + 1 = 2x(1+x)$
 Then
 A) Both P and Q are true B) P is true and Q is false
 C) P is false and Q is true D) Both P and Q are false
- If the function $f(x) = ax^3 + bx^2 + 11x - 6$ satisfies conditions of Rolle's theorem in $[1, 3]$ and $f'\left(2 + \frac{1}{\sqrt{3}}\right) = 0$, then value of a and b are respectively
 A) 1, -6 B) -1, 6 C) -2, 1 D) -1, 1/2
- If the petrol burnt per hour in driving a motor boat varies as the cube of its velocity. When going against a current of 'C' kmph, the most economical speed is (in kmph)
 A) $\frac{C}{2}$ B) $\frac{3C}{2}$ C) $\frac{\sqrt{3}C}{2}$ D) C
- Let the smallest positive value of x for which the function $f(x) = \sin \frac{x}{3} + \sin \frac{x}{11}$, ($x \in \mathbb{R}$) achieves its maximum value be x_0 . Express x_0 in degrees i.e, $x_0 = \alpha^\circ$. Then the sum of the digits in α is
 A) 15 B) 17 C) 16 D) 18
- Tangents are drawn from origin to the curve $y = \sin x + \cos x$. Then their points of contact lie on the curve
 A) $\frac{1}{x^2} + \frac{2}{y^2} = 1$ B) $\frac{2}{x^2} - \frac{1}{y^2} = 1$ C) $\frac{2}{x^2} + \frac{1}{y^2} = 1$ D) $\frac{2}{y^2} - \frac{1}{x^2} = 1$
- Let $f(x) = x^2 - bx + c$, b is a odd positive integer, $f(x) = 0$ have two prime numbers as roots and $b + c = 35$. Then the global minimum value of $f(x)$ is
 A) $-\frac{183}{4}$ B) $\frac{173}{16}$ C) $-\frac{81}{4}$ D) data not sufficient

MULTIPLE ANSWER TYPE

9. Let $f(x) = \sin x + ax + b$. Then $f(x) = 0$ has:
- Only one real root which is positive if $a > 1, b < 0$
 - Only one real root which is negative if $a > 1, b > 0$
 - Only one real root which is negative if $a < -1, b < 0$
 - Only one real root which is positive if $a < -1, b < 0$
10. Let $f(x) = \frac{e^x}{1+x^2}$ and $g(x) = f'(x)$, then
- $g(x)$ has two local maxima and two local minima points
 - $g(x)$ has exactly one local maxima and one local minima point
 - $x = 1$ is a point of local maxima for $g(x)$
 - there is a point of local maxima for $g(x)$ in the interval $(-1, 0)$
11. Let f be a real-valued function defined on the interval $(0, \infty)$ by
- $$f(x) = \ln x + \int_0^x \sqrt{1 + \sin t} \, dt.$$
- Then which of the following statement(s) is (are) true?
- $f''(x)$ exists for all $x \in (0, \infty)$
 - $f'(x)$ exists for all $x \in (0, \infty)$ and f' is continuous on $(0, \infty)$, but not differentiable on $(0, \infty)$
 - There exists $\alpha > 1$ such that $|f'(x)| < |f(x)|$ for all $x \in (\alpha, \infty)$
 - There exists $\beta > 0$ such that $|f(x)| + |f'(x)| \leq \beta$ for all $x \in (0, \infty)$
12. The parabola $y = x^2 + px + q$ cuts the straight line $y = 2x - 3$ at a point with abscissa 1. If the distance between the vertex of the parabola and the x-axis is the least, then
- $p = 0$ and $q = -2$
 - $p = -2$ and $q = 0$
 - The least distance between the parabola and the x-axis is 2
 - the least distance between the parabola and the x-axis is 1
13. If the equation $x^5 - 10a^3x^2 + b^4x + c^5 = 0$ has three equal roots, then
- $2b^2 - 10a^3b^2 + c^5 = 0$
 - $6a^5 + c^5 = 0$
 - $2c^5 - 10a^3b^2 + b^4c^5 = 0$
 - $b^4 = 15a^4$
14. $g(x) = f(x^2) + f(1-x^2)$ and $f''(x) < 0$ for $0 \leq x \leq 1$, then
- $g(x)$ decreases in $\left(0, \frac{1}{\sqrt{2}}\right)$
 - $g(x)$ increases in $\left(0, \frac{1}{\sqrt{2}}\right)$
 - $g(x)$ decreases in $\left(\frac{1}{\sqrt{2}}, 1\right)$
 - $g(x)$ increases in $\left(\frac{1}{\sqrt{2}}, 1\right)$

INTEGER ANSWER TYPE

15. Let α and β respectively be the number of solutions of $e^x = x^2$ and $e^x = x^3$. Then, the numerical value of $2\alpha + 3\beta$, is
16. If $f(x) = 4x^3 - x^2 - 2x + 1$ and $g(x) = \begin{cases} \min\{f(t) : 0 \leq t \leq x; & 0 \leq x \leq 1 \\ 3 - x & 1 < x \leq 2 \end{cases}$
- with $g\left(\frac{1}{4}\right) + g\left(\frac{3}{4}\right) + g\left(\frac{5}{4}\right) = \frac{k}{2}$ then the value of $k =$ _____

17. If $a, b, c \in \mathbb{N}$, and if $\frac{ax^4 - bx^3 + cx^2 - bx + a}{(x^2 + 1)^2}$ attains minimum value at $x = 2$ or $1/2$ then the A.M of the least possible values of a, b and c is _____
18. In the coordinate plane, the region M consists of all points (x, y) satisfying the inequalities $y \geq 0, y \leq x$, and $y \leq 2 - x$ simultaneously. The region N which varies with parameter t , consists of all the points (x, y) satisfying the inequalities $t \leq x \leq t + 1$ and $0 \leq t \leq 1$ simultaneously. If the area of the region $M \cap N$ is a function of t , i.e., $M \cap N = f(t)$ and if α is the value of t for which this area is maximum, then the numerical value of 2α is _____
19. Number of solution of the equation $3 \tan x + x^3 = 2$ in $\left(0, \frac{\pi}{4}\right)$ is _____
20. Find the number of real values of x satisfy the equation $x^2 = x \sin x + \cos x$.
21. The least value of a for which the equation $\frac{4}{\sin x} + \frac{1}{1 - \sin x} = a$ has atleast one solution in $\left(0, \frac{\pi}{2}\right)$ is _____
22. Let C be the curve $y = x^3$ (where x takes all real values). The tangent at A meets the curve again at B . If the gradient of the curve at B is K times the gradient at A then K is equal to _____
23. The number of positive integral values of a for which $y = a^x$ intersects the line $y = x$ is _____

PASSAGE-1

Let $f(x) = 2 + |x - 1|$ and $g(x) = \min(f(t))$ where $x \leq t \leq x^2 + x + 1$, then

24. $g\left(-\frac{1}{2}\right)$ is equal to _____
 A) 2 B) $\frac{1}{2}$ C) $\frac{9}{4}$ D) $\frac{4}{9}$
25. Which of the following is incorrect?
 A) $g(0) = g(1)$ B) $g(2) = g(3)$ C) $g(-3) = g(-7)$ D) $g\left(\frac{1}{2}\right) + g\left(\frac{1}{3}\right) = 4$

PASSAGE-2

We are given the curves $y = \int_{-\infty}^x f(t) dt$ through the point $\left(0, \frac{1}{2}\right)$ and $y = f(x)$, where

$f(x) > 0$ and $f(x)$ is differentiable, $\forall x \in \mathbb{R}$ through $(0, 1)$. If tangents drawn to both the curves at the points with equal abscissa intersect on the same point on the x -axis, then

26. $\lim_{x \rightarrow \infty} (f(x))^{f(-x)}$ is _____
 A) 3 B) 6 C) 1 D) 2
27. The function $f(x)$ is _____
 A) Increasing for all x B) Non – monotonic
 C) Decreasing for all x D) Decreasing exactly some finite set of values of x

KEY

1	B	2	B	3	C	4	A	5	B
6	D	7	D	8	C	9	ABC	10	BD
11	BC	12	BD	13	BD	14	BC	15	8
16	5	17	4	18	1	19	1	20	2
21	9	22	4	23	1	24	C	25	B
26	C	27	A						



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Sec: Star SC

TEST ON AOD

Date: 29-09-2020

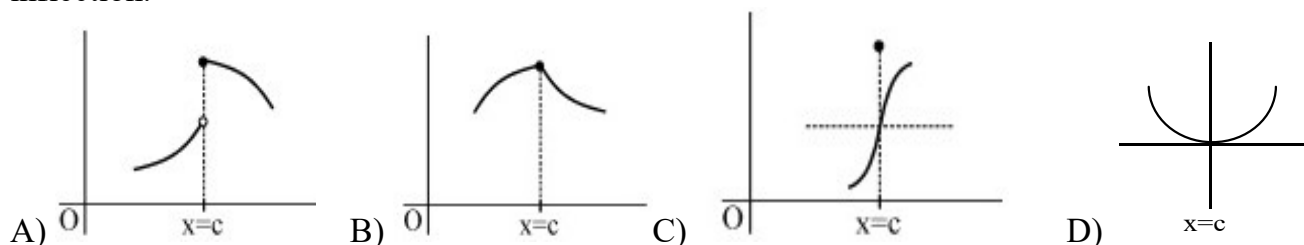
SINGLE ANSWER TYPE

- Let $F(x) = \int_{\sin x}^{\cos x} e^{(1+\sin^{-1}t)^2} dt$ on $\left[0, \frac{\pi}{2}\right]$ then which of the following is correct?
 A) $F''(c) = 0$ for all $c \in \left(0, \frac{\pi}{2}\right)$ B) $F''(c) = 0$ for some $c \in \left(0, \frac{\pi}{2}\right)$
 C) $F'(c) = 0$ for some $c \in \left(0, \frac{\pi}{2}\right)$ D) $F(c) \neq 0$ for all $c \in \left(0, \frac{\pi}{2}\right)$
- Let $f(x)$ and $g(x)$ be two differentiable function in R and $f(2) = 8$, $g(2) = 0$, $f(4) = 10$ and $g(4) = 8$ then which of the following is correct?
 A) $g'(x) > 4f'(x) \forall x \in (2, 4)$ B) $3g'(x) = 4f'(x)$ for at least one $x \in (2, 4)$
 C) $g(x) > f(x) \forall x \in (2, 4)$ D) $g'(x) = 4f'(x)$ for at least one $x \in (2, 4)$
- The set of values of p for which the equation $|\ln x| - px = 0$ possess three distinct roots is
 A) $\left(0, \frac{1}{e}\right)$ B) $(0, 1)$ C) $(1, e)$ D) $(0, e)$
- Let $f(x) = 1 + x^m(x-1)^n$ where $m, n \in N$. Then in $(0, 1)$ the equation $f'(x) = 0$ has
 A) no root B) at least one root
 C) at most one root D) exactly one root
- For a steamer the consumption of petrol (per hour) varies as the cube of its speed (in km). If the speed of the current is steady at C km/hr then the most economical speed of the steamer going against the current will be
 A) $1.25C$ B) $1.5C$ C) $1.75C$ D) $2C$
- If $f(x) = 4x^3 - x^2 - 2x + 1$ and $g(x) = \begin{cases} \text{Min } \{f(t) : 0 \leq t \leq x\} & ; 0 \leq x \leq 1 \\ 3 - x & ; 1 < x \leq 2 \end{cases}$ then $g\left(\frac{1}{4}\right) + g\left(\frac{3}{4}\right) + g\left(\frac{5}{4}\right)$ has the value equal to
 A) $\frac{7}{4}$ B) $\frac{9}{4}$ C) $\frac{13}{4}$ D) $\frac{5}{2}$
- Let f be a continuous function on $[a, b]$. If $F(x) = \left(\int_a^x f(t) dt - \int_x^b f(t) dt \right) (2x - (a+b))$ then there exist some $c \in (a, b)$ such that
 A) $\int_a^c f(t) dt = \int_c^b f(t) dt$ B) $\int_a^c f(t) dt - \int_c^b f(t) dt = f(c)(a+b-2c)$
 C) $\int_a^c f(t) dt - \int_c^b f(t) dt = f(c)(2c - (a+b))$ D) $\int_a^c f(t) dt + \int_c^b f(t) dt = f(c)(2c - (a+b))$
- Consider $f(x) = \int_1^x \left(t + \frac{1}{t}\right) dt$ and $g(x) = f'(x)$, for $x \in \left[\frac{1}{2}, 3\right]$ if P is a point on the curve $y = g(x)$ such that the tangent to this curve at P is parallel to a chord joining the points

$\left(\frac{1}{2}, g\left(\frac{1}{2}\right)\right)$ and $(3, g(3))$ of the curve, then the coordinates of the point P

- A) Cant be found out B) $\left(\frac{7}{4}, \frac{65}{28}\right)$ C) (1,2) D) $\left(\sqrt{\frac{3}{2}}, \frac{5}{\sqrt{6}}\right)$

9. Number of solutions satisfying the equation, $3x^2 - 2x^3 = \log_2(x^2 + 1) - \log_2 x$ is:
 A) 1 B) 2 C) 3 D) None
10. The tangent at a variable point P of the curve $y=x^2-x^3$ meets it again at Q. Find the locus of the middle point of PQ is
 A) $y=1-9x+28x^2-28x^3$ B) $y=1-2x+28x^2-28x^3$
 C) $y=1-3x+28x^2-28x^3$ D) $y=1-4x+28x^2-28x^3$
11. $f: R \rightarrow R$ be a twice differentiable function $\forall x \in R$. If tangent drawn to the curve at any point $x \in (a, b)$ always lie below the curve, then which of the following is always correct?
 A) $f'(x)$ is increasing and $f''(x) < 0 \forall x \in (a, b)$
 B) $f'(x)$ is decreasing and $f''(x) < 0 \forall x \in (a, b)$
 C) $f'(x)$ is increasing and $f''(x) > 0 \forall x \in (a, b)$
 D) $f'(x)$ is decreasing and $f''(x) > 0 \forall x \in (a, b)$
12. Read the following mathematical statements carefully:
 I. A differentiable function 'f' with maximum at $x = c \Rightarrow f''(c) < 0$.
 II. Antiderivative of a periodic function is also a periodic function.
 III. If f is a periodic function with period T, then for any $a \in R$, $\int_0^T f(x) dx = \int_0^T f(x+a) dx$
 IV. If f(x) has a maxima at $x = c$, then 'f' is increasing in $(c-h, c)$ and decreasing in $(c, c+h)$ as $h \rightarrow 0$ for $h > 0$.
 Now indicate the correct alternative.
 A) exactly one statement is correct. B) exactly two statements are correct.
 C) exactly three statements are correct. D) All the four statements are correct.
13. Let h be a twice continuously differentiable positive function on an open interval J and $g(x) = \log_e(h(x))$ such that $(h'(x))^2 > h''(x)h(x)$ for each $x \in J$. Then which of the following is/are correct?
 A) g is increasing on J B) g is decreasing on J
 C) g is concave up on J D) g is concave down on J
14. In the following graphs, in which option(s), at the point $x=c$ always represents point of inflection.



MULTIPLE ANSWER TYPE

15. Suppose that f is a polynomial of degree 3 and that $f''(x) \neq 0$ at any of the stationary point. Then which of the following is/are true?
 A) f has exactly one such stationary point. B) f must have no such stationary point.
 C) f can have two such stationary points. D) f can have zero such stationary points.
16. $f(x) = \begin{cases} e^x & 0 \leq x \leq 1 \\ 2 - e^{x-1} & 1 < x \leq 2 \\ x - e & 2 < x \leq 3 \end{cases}$ and $g(x) = \int_0^x f(t) dt, x \in [1, 3]$ then g(x) has

- A) local maxima at $x = 1 + \ln 2$ and local minima at $x = e$
 B) local minima at $x = 1$ and local maxima at $x = 3$
 C) no local maxima D) no local minima
17. Assume that inverse of a differentiable function f is denoted by g . Then which of the following statement hold good?
 A) If f is increasing then g is also increasing.
 B) If f is decreasing then g is increasing.
 C) The function f is injective. D) The function g is onto.
18. Suppose $f: \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function satisfying $f(x+y) = f(x) + f(y) + xy(x+y)$ for every $x, y \in \mathbb{R}$. If $f'(0) = 0$ then which of the following hold(s) good?
 A) f is an odd function B) f is a bijective mapping
 C) f has a minima but no maxima D) f has an inflection point
19. The extremum value(s) of the function $f(x) = \frac{1}{\sin x + 4} - \frac{1}{\cos x - 4}$, where $x \in \mathbb{R}$ is:
 A) $\frac{4}{8 - \sqrt{2}}$ B) $\frac{2\sqrt{2}}{8 - \sqrt{2}}$ C) $\frac{2\sqrt{2}}{4\sqrt{2} + 1}$ D) $\frac{4}{8 + \sqrt{2}}$
20. If $f(x) = a^{\{a^{|x|} \operatorname{sgn} x\}}$; $g(x) = a^{\lfloor a^{|x|} \operatorname{sgn} x \rfloor}$ for $a > 0$, $a \neq 1$ and $x \in \mathbb{R}$, where $\{\}$ & $\lfloor \rfloor$ denote the fractional part and integral part functions respectively, then which of the following statements can hold good for the function $h(x)$, where $(\ln a) h(x) = (\ln f(x) + \ln g(x))$.
 A) 'h' is even and increasing B) 'h' is odd and decreasing
 C) 'h' is even and decreasing D) 'h' is odd and increasing.
21. If $p'(x) > p(x)$ for all $x \geq 1$ and $p(1) = 0$ then which of the following is/are true?
 A) $e^{-x} p(x)$ is an increasing function for all $x \geq 1$
 B) $e^x p(x)$ is a decreasing function for all $x \geq 1$
 C) $p(x) > 0$ for all x in $[1, \infty)$ D) $p(x) < 0$ for all x in $[1, \infty)$
22. Find the equations of the tangents drawn to the curve $y^2 - 2x^3 - 4y + 8 = 0$ from the point $(1, 2)$.
 A) $2\sqrt{3}x - y = 2(\sqrt{3} - 1)$ B) $2\sqrt{3}x + y = 2(\sqrt{3} + 1)$
 C) $\sqrt{3}x + y = 2(\sqrt{3} + 1)$ D) $\sqrt{3}x - y = 2(\sqrt{3} - 1)$
23. The equation of the normal to the curve $\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2$ ($n \in \mathbb{N}$) at the point with abscissa equal to 'a' can be:
 A) $ax + by = a^2 - b^2$ B) $ax + by = a^2 + b^2$ C) $ax - by = a^2 - b^2$ D) $bx - ay = a^2 - b^2$
24. Let $f(x) = \frac{x-1}{x^2}$ then which of the following is/are correct?
 A) $f(x)$ has minima but no maxima.
 B) $f(x)$ increases in the interval $(0, 2)$ and decreases in the interval $(-\infty, 0) \cup (2, \infty)$.
 C) $f(x)$ is concave down in $(-\infty, 0) \cup (0, 3)$. D) $x = 3$ is the point of inflection.
25. Which of the following options are correct
 A) $x \cdot \sin x - (1/2) \sin^2 x < (1/2)(\pi - 1)$ for $0 < x < \pi/2$
 B) $\tan^2 x + 6 \ln \sec x + 2 \cos x + 4 > 6 \sec x$ for $x \in \left(\frac{3\pi}{2}, 2\pi\right)$
 C) $2x \ln x > 4(x-1) - 2 \ln x$ for $x > 1$ D) $e^\pi > \pi^e$

26. A function f is defined by $f(x) = \int_0^x \cos t \cos(x-t) dt$, $0 \leq x \leq 2\pi$ then which of the following hold(s) good?
- A) $f(x)$ is continuous but not differentiable in $(0, 2\pi)$ B) Maximum value of f is π
- C) There exists atleast one $c \in (0, 2\pi)$ such that $f'(c) = 0$. D) Minimum value of f is $-\frac{\pi}{2}$.
27. If $\phi(x) = f(x) + f(2a - x)$ and $f''(x) > 0$, $a > 0$, $0 \leq x \leq 2a$ then which of the following is/are correct?
- A) $\phi(x)$ increases in $(a, 2a)$ B) $\phi(x)$ increases in $(0, a)$
- C) $\phi(x)$ decreases in $(0, a)$ D) $\phi(x)$ decreases in $(a, 2a)$

KEY

1	B	2	D	3	A	4	D	5	B
6	D	7	B	8	D	9	A	10	A
11	C	12	A	13	D	14	C	15	CD
16	AB	17	ACD	18	ABD	19	ACD	20	BD
21	AC	22	AB	23	AC	24	BCD	25	ABCD
26	CD	27	AC						



NARAYANA
IIT ACADEMY
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BASARA GNANA SARASWATHI CAMPUS KAKATIYA HILLS

Sec: Star 5C

AOD

Date: 02-10-2020

SINGLE ANSWER TYPE

- Let $f(x)$ and $g(x)$ defined from $[0, \infty)$ to $[0, \infty)$. If $f(x)$ be an increasing function and $g(x)$ be a decreasing function. Also, $h(x) = f(g(x))$, $h(0) = 0$ then $h(x) - h(1)$ is
 A) Always zero
 B) Always negative
 C) Always positive
 D) Strictly increasing
- A person wishes to lay a straight fence across a triangular field ABC, with $|A| < |B| < |C|$ so as to divide it into two equal areas. The length of the fence with minimum expense, is
 A) $\sqrt{2\Delta \cot \frac{B}{2}}$
 B) $\sqrt{2\Delta \tan \frac{C}{3}}$
 C) $\sqrt{\tan \frac{A}{2} \tan \frac{B}{2} \tan \frac{C}{2}}$
 D) $\sqrt{2\Delta \tan \frac{A}{2}}$
- Let a, b, c, d, e, f, g, h be distinct elements in the set $\{-7, -5, -3, -2, 2, 4, 6, 13\}$. The minimum value of $(a + b + c + d)^2 + (e + f + g + h)^2$ is
 A) 30
 B) 32
 C) 34
 D) 40
- Let $\phi(x) = \frac{(x-b)(x-c)}{(a-b)(a-c)}f(a) + \frac{(x-c)(x-a)}{(b-c)(b-a)}f(b) + \frac{(x-a)(x-b)}{(c-a)(c-b)}f(c) - f(x)$ Where $a < c < b$ and $f^{(1)}(x)$ exists at all points in (a, b) . Then, there exists a real number $\mu, a < \mu < b$ such that $\frac{f(a)}{(a-b)(a-c)} + \frac{f(b)}{(b-c)(b-a)} + \frac{f(c)}{(c-a)(c-b)} =$
 a) $f^{(1)}(\mu)$
 b) $2f^{(1)}(\mu)$
 c) $\frac{1}{2}f^{(1)}(\mu)$
 d) $\frac{1}{3}f^{(1)}(\mu)$
- The least value of 'a' for which the equation $\frac{4}{\sin x} + \frac{1}{1 - \sin x} = a$ for atleast one solution on the interval $\left(0, \frac{\pi}{2}\right)$ is,
 a) 1
 b) 4
 c) 8
 d) 9
- Water is dripping out from a conical funnel, at the uniform rate of $2\text{cm}^3/\text{sec}$ through a tiny hole at the vertex at the bottom. When the slant height of the water is 4 cm, find the rate of decrease of the slant height of the water given that the vertical angle of the funnel is 120° .
 A) $3\pi\text{cm}/\text{sec}$
 B) $\frac{1}{3\pi}\text{cm}/\text{sec}$
 C) $4\pi\text{cm}/\text{sec}$
 D) $\frac{1}{4\pi}\text{cm}/\text{sec}$
- Let A be the point where the curve $5\alpha^2x^3 + 10\alpha x^2 + x + 2y - 4 = 0$ ($\alpha \in \mathbb{R} - \{0\}$) meets y-axis. Then the equation of tangent to the curve at the point where the normal at A meets the curve again, is
 A) $x - \alpha y + 2\alpha = 0$
 B) $\alpha x + y - 2 = 0$
 C) $2x - y + 2 = 0$
 D) $x + 2y - 4 = 0$

8. Two runners A and B start at the origin and run along positive x-axis, with B running three times as fast as A. An observer, standing one unit above the origin, keeps A and B in view. Then the maximum angle of sight ' θ ' between the observer's view of A and B is
- A) $\pi/8$ B) $\pi/6$ C) $\pi/3$ D) $\pi/4$
9. The tangent to the curve $y = e^x$ drawn at the point (c, e^c) intersects the line joining the points $(c-1, e^{c-1})$ and $(c+1, e^{c+1})$
- A) on the left of $x = c$ B) on the right of $x = c$
C) at no point D) at all points
10. Let $f(x) = (1-x)^2 \sin^2 x + x^2$ for all $x \in \mathbb{R}$. Consider the statements :
- P: There exists some $x \in \mathbb{R}$ such that $f(x) + 2x = 2(1+x^2)$
Q: There exists some $x \in \mathbb{R}$ such that $2f(x) + 1 = 2x(1+x)$
- Then
- A) Both P and Q are true B) P is true and Q is false
C) P is false and Q is true D) Both P and Q are false
11. The number of values of x at which the function, $f(x) = (x-1)x^{2/3}$ has extreme values, is
- A) 4 B) 3 C) 2 D) 1
12. Tangents are drawn from origin to the curve $y = \sin x + \cos x$. Then their points of contact lie on the curve
- A) $\frac{1}{x^2} + \frac{2}{y^2} = 1$ B) $\frac{2}{x^2} - \frac{1}{y^2} = 1$ C) $\frac{2}{x^2} + \frac{1}{y^2} = 1$ D) $\frac{2}{y^2} - \frac{1}{x^2} = 1$
13. A rectangle is inscribed in an equilateral Δ of side length 2a units. Maximum area of this rectangle is
- A) $\sqrt{3}a^2$ B) $\frac{\sqrt{3}a^2}{4}$ C) a^2 D) $\frac{\sqrt{3}a^2}{2}$
14. A person wishes to lay a straight fence across a triangular field ABC, with $\angle A < \angle B < \angle C$ so as to divide it into two equal areas. The length of the fence with minimum expense, is
- A) $\sqrt{2\Delta \cot \frac{B}{2}}$ B) $\sqrt{2\Delta \tan \frac{C}{3}}$ C) $\sqrt{\tan \frac{A}{2} \tan \frac{B}{2} \tan \frac{C}{2}}$ D) $\sqrt{2\Delta \tan \frac{A}{2}}$

(where 'Δ' represents, area of triangle ABC)

15. A variable line through A(6,8) meets the curve $x^2 + y^2 = 2$ at B and C. P is a point on BC such that AB, AP, AC are in HP. The minimum distance of the origin from the locus of P is

A) 1 B) $\frac{1}{2}$ C) $\frac{1}{3}$ D) $\frac{1}{5}$

MULTIPLE ANSWER TYPE

16. Tangent at a point P_1 [other than (0, 0)] on the curve $y = x^3$ meets the curve again at P_2 . The tangent at P_2 meets the curve at P_3 and so on. Then

A) abscissae of $P_1, P_2, P_3, \dots, P_n$ are in A. P.

B) abscissae of $P_1, P_2, P_3, \dots, P_n$ form a G. P.

C) area $(\Delta P_2 P_3 P_4) = 16$ area $(\Delta P_1 P_2 P_3)$ D) area $(\Delta P_1 P_2 P_3) = 16$ area $(\Delta P_2 P_3 P_4)$

17. Let $g'(x) > 0$ and $f'(x) < 0 \forall x \in R$, then

A) $g(f(x+1)) > g(f(x-1))$

B) $g(f(x+1)) < g(f(x-1))$

C) $f(g(x-1)) > f(g(x+1))$

D) $f(g(x-1)) < f(g(x+1))$

18. If $|f(x)| = |ax^2 + bx + c| \leq 1$ for $|x| \leq 1$ $(a, b, c) \in R$ and $\frac{8}{3}a^2 + 2b^2$ is maximum, then

which of the following is/are **true**?

A) $b = 0$

B) $|a + c| = 1$

C) $ac = 1$

D) $\left| \frac{c}{a} \right| = \frac{1}{2}$

19. Let $f(x)$ be a differentiable function upto any order such that

$f(x) + |f''(x)| \leq 0, \forall x \in R$. If α and β ($\alpha < \beta$) be the two consecutive real roots of $f(x) = 0$, then which of the following is/are **true**?

A) $f''(x)$ must not be equal to zero for any $x \in (\alpha, \beta)$

B) $f'''(x)$ must be equal to zero for atleast one $x \in (\alpha, \beta)$

C) $f'(x) \neq 0 \forall x \in (\alpha, \beta)$

D) $f'(x)$ must be equal to zero for atleast one $x \in (\alpha, \beta)$

INTEGER ANSWER TYPE

20. The minimum value of, $\frac{\sec^4 \alpha}{\tan^2 \beta} + \frac{\sec^4 \beta}{\tan^2 \alpha}$, $\alpha, \beta \neq \frac{K\pi}{2}$, $K \in I$, is
21. $f(x)$ is a twice differentiable function in x such that $f(0.9) = 0$, $f(1.9) = 3$, and $f(2.9) = -4$, $f(3.9) = 6$, $f(4.9) = -2$ and $f(5.9) = 2$ then the minimum number of zeros of $g(x) = (f(x))^2 + f(x)f'(x)$ in the interval $[0.9, 5.9]$ is
22. $f(x) = \lim_{n \rightarrow \infty} \left(\sin^4 x + \frac{1}{4} \sin^4 2x + \dots + \frac{1}{4^n} \sin^4 (2^n x) \right)$ and $g(x)$ is a differentiable function satisfying $g(x) + f(x) = 1$, the maximum value of $(\sqrt{f(x)} + \sqrt{g(x)})^4$ is _____
23. Let f be a cubic polynomial function having relative extremum at $x = -1$ and $x = \frac{1}{3}$. Such that $\int_1^{-1} f(t) dt = \frac{14}{3}$. If $f(-2) = 0$, $|f(f(f(0)) - 1)| = \underline{\hspace{2cm}}$
24. Tangent to a curve $x^4 y^3 = C^7$ at point P in I^{st} quadrant intersect x, y axis at A, B respectively. If P divides AB in the ratio $m:n$ (in lowest form) then $m = \underline{\hspace{2cm}}$
25. Number of values of x for which the function $f(x) = (x - \alpha_1)^2 (x - \alpha_2)^3 (x - \alpha_3)^2 (x - \alpha_4)^3 (x - \alpha_5)^2$ attains a local maximum or local minimum is _____
($\alpha_1, \alpha_2, \dots, \alpha_5$ are distinct real numbers)
26. f, g are continuous, differentiable and monotonically increasing functions $\forall x \in \mathbb{R}$ such that $g(x) = f(x) \sqrt{1 - 2(f(x))^2}$. If $6|f(x)| \leq k \forall x$ the minimum value of k is _____

PASSAGE-1

For a polynomial function $y = f(x)$

Points of extrema are obtained at points where

$f'(x) = 0$ and

$f''(x_1) > 0 \Rightarrow x_1$ is a point of minima

$f''(x_1) < 0 \Rightarrow x_1$ is a point of maxima

Let $f(x) = x^3 - 3(7 - a)x^2 - 3(9 - a^2)x + 2$

27. The values of parameter 'a' if $f(x)$ has a negative point of local minimum are
A) ϕ B) $\left(-\infty, \frac{58}{14}\right)$ C) $(-3, 3)$ D) $(-\infty, -3) \cup \left(\frac{58}{14}, \infty\right)$
28. The values of parameter 'a' if $f(x)$ has a positive point of local maxima are

- A) ϕ B) $(-\infty, -3) \cup \left(\frac{58}{14}, \infty\right)$ C) $\left(-\infty, \frac{58}{14}\right)$ D) $(-\infty, -3) \cup \left(3, \frac{58}{14}\right)$

29. The values of parameter 'a' if $f(x)$ has points of extrema which are opposite in sign are

- A) ϕ B) $(-3, 3)$ C) $\left(-\infty, \frac{58}{14}\right)$ D) $(-\infty, -3) \cup \left(\frac{58}{14}, \infty\right)$

PASSAGE-2

$$f(x) = \sin 2\pi x + \{x\} : x \in [0, 10].$$

30. Number of points where f achieves local maximum is

- A) 10 B) 11 C) 20 D) 19

31. Number of roots of $f(x) = 0$ in $(0, 10)$ is

- A) 20 B) 30 C) 31 D) 29

PASSAGE-3

$$\text{Consider } f(x) = x + \cos x - \alpha$$

32. Which of the following holds good for the above $f(x)$

- A) $\alpha > 1$ for which $f(x)$ has exactly one positive root
 B) $\alpha > 2$ for which $f(x)$ has exactly one -ve root
 C) $1 < \alpha < 2$ $f(x)$ will have an imaginary set of roots
 D) None of these

33. Which of the following is true for $f(x)$

- A) For $0 < \alpha < \beta < \frac{\pi}{2}$ $\sin \alpha > \frac{\cos \alpha - \cos \beta}{\beta - \alpha} > \cos \beta$
 B) For $0 < \alpha < \beta < \frac{\pi}{2}$ $\sin \alpha < \frac{\cos \alpha - \cos \beta}{\beta - \alpha} < \sin \beta$
 C) For $0 < \alpha < \beta < \frac{\pi}{2}$ $\tan \alpha < \frac{\cos \alpha - \cos \beta}{\beta - \alpha} < \tan \beta$
 D) None of these

KEY

1	A	2	D	3	C	4	C	5	D
6	B	7	C	8	B	9	A	10	C
11	C	12	D	13	D	14	D	15	D
16	BC	17	BC	18	ABD	19	AC	20	8
21	3,9	22	4	23	3	24	3	25	7
26	3	27	A	28	D	29	B	30	A
31	D	32	A	33	B				


BASARA GNANA SARASWATHI CAMPUS KAKATIYA HILLS
Sec: Star 5C
TEST ON AOD
Date: 01-10-2020
SINGLE ANSWER TYPE

- If the equation $|x^2 - 5x + 6| - \lambda x + 7\lambda = 0$ has exactly 3 solutions, then λ is equal to
 A) $-7 + \sqrt{23}$ B) $-9 + 4\sqrt{5}$ C) $-7 - \sqrt{23}$ D) $-9 - 4\sqrt{5}$
- Two runners A and B start at the origin and run along positive x-axis, with B running three times as fast as A. An observer, standing one unit above the origin, keeps A and B in view. Then the maximum angle of sight ' θ ' between the observer's view of A and B is
 A) $\pi/8$ B) $\pi/6$ C) $\pi/3$ D) $\pi/4$
- Let $f(x) = (1-x)^2 \sin^2 x + x^2$ for all $x \in \mathbb{R}$. Consider the statements :
 P: There exists some $x \in \mathbb{R}$ such that $f(x) + 2x = 2(1+x^2)$
 Q: There exists some $x \in \mathbb{R}$ such that $2f(x) + 1 = 2x(1+x)$
 Then
 A) Both P and Q are true B) P is true and Q is false
 C) P is false and Q is true D) Both P and Q are false
- If the function $f(x) = ax^3 + bx^2 + 11x - 6$ satisfies conditions of Rolle's theorem in $[1, 3]$ and $f'\left(2 + \frac{1}{\sqrt{3}}\right) = 0$, then value of a and b are respectively
 A) 1, -6 B) -1, 6 C) -2, 1 D) -1, 1/2
- If the petrol burnt per hour in driving a motor boat varies as the cube of its velocity. When going against a current of 'C' kmph, the most economical speed is (in kmph)
 A) $\frac{C}{2}$ B) $\frac{3C}{2}$ C) $\frac{\sqrt{3}C}{2}$ D) C
- Let the smallest positive value of x for which the function $f(x) = \sin \frac{x}{3} + \sin \frac{x}{11}$, ($x \in \mathbb{R}$) achieves its maximum value be x_0 . Express x_0 in degrees i.e., $x_0 = \alpha^\circ$. Then the sum of the digits in α is
 A) 15 B) 17 C) 16 D) 18
- Tangents are drawn from origin to the curve $y = \sin x + \cos x$. Then their points of contact lie on the curve
 A) $\frac{1}{x^2} + \frac{2}{y^2} = 1$ B) $\frac{2}{x^2} - \frac{1}{y^2} = 1$ C) $\frac{2}{x^2} + \frac{1}{y^2} = 1$ D) $\frac{2}{y^2} - \frac{1}{x^2} = 1$
- Let $f(x) = x^2 - bx + c$, b is a odd positive integer, $f(x) = 0$ have two prime numbers as roots and $b + c = 35$. Then the global minimum value of $f(x)$ is
 A) $-\frac{183}{4}$ B) $\frac{173}{16}$ C) $-\frac{81}{4}$ D) data not sufficient

MULTIPLE ANSWER TYPE

9. Let $f(x) = \sin x + ax + b$. Then $f(x) = 0$ has:
- Only one real root which is positive if $a > 1, b < 0$
 - Only one real root which is negative if $a > 1, b > 0$
 - Only one real root which is negative if $a < -1, b < 0$
 - Only one real root which is positive if $a < -1, b < 0$
10. Let $f(x) = \frac{e^x}{1+x^2}$ and $g(x) = f'(x)$, then
- $g(x)$ has two local maxima and two local minima points
 - $g(x)$ has exactly one local maxima and one local minima point
 - $x = 1$ is a point of local maxima for $g(x)$
 - there is a point of local maxima for $g(x)$ in the interval $(-1, 0)$
11. Let f be a real-valued function defined on the interval $(0, \infty)$ by
- $$f(x) = \ln x + \int_0^x \sqrt{1 + \sin t} \, dt.$$
- Then which of the following statement(s) is (are) true?
- $f''(x)$ exists for all $x \in (0, \infty)$
 - $f'(x)$ exists for all $x \in (0, \infty)$ and f' is continuous on $(0, \infty)$, but not differentiable on $(0, \infty)$
 - There exists $\alpha > 1$ such that $|f'(x)| < |f(x)|$ for all $x \in (\alpha, \infty)$
 - There exists $\beta > 0$ such that $|f(x)| + |f'(x)| \leq \beta$ for all $x \in (0, \infty)$
12. The parabola $y = x^2 + px + q$ cuts the straight line $y = 2x - 3$ at a point with abscissa 1. If the distance between the vertex of the parabola and the x-axis is the least, then
- $p = 0$ and $q = -2$
 - $p = -2$ and $q = 0$
 - The least distance between the parabola and the x-axis is 2
 - the least distance between the parabola and the x-axis is 1
13. If the equation $x^5 - 10a^3x^2 + b^4x + c^5 = 0$ has three equal roots, then
- $2b^2 - 10a^3b^2 + c^5 = 0$
 - $6a^5 + c^5 = 0$
 - $2c^5 - 10a^3b^2 + b^4c^5 = 0$
 - $b^4 = 15a^4$
14. $g(x) = f(x^2) + f(1-x^2)$ and $f''(x) < 0$ for $0 \leq x \leq 1$, then
- $g(x)$ decreases in $\left(0, \frac{1}{\sqrt{2}}\right)$
 - $g(x)$ increases in $\left(0, \frac{1}{\sqrt{2}}\right)$
 - $g(x)$ decreases in $\left(\frac{1}{\sqrt{2}}, 1\right)$
 - $g(x)$ increases in $\left(\frac{1}{\sqrt{2}}, 1\right)$

INTEGER ANSWER TYPE

15. Let α and β respectively be the number of solutions of $e^x = x^2$ and $e^x = x^3$. Then, the numerical value of $2\alpha + 3\beta$, is
16. If $f(x) = 4x^3 - x^2 - 2x + 1$ and $g(x) = \begin{cases} \min\{f(t) : 0 \leq t \leq x; & 0 \leq x \leq 1 \\ 3 - x & 1 < x \leq 2 \end{cases}$
- with $g\left(\frac{1}{4}\right) + g\left(\frac{3}{4}\right) + g\left(\frac{5}{4}\right) = \frac{k}{2}$ then the value of $k =$ _____

17. If $a, b, c \in \mathbb{N}$, and if $\frac{ax^4 - bx^3 + cx^2 - bx + a}{(x^2 + 1)^2}$ attains minimum value at $x = 2$ or $1/2$ then the A.M of the least possible values of a, b and c is _____
18. In the coordinate plane, the region M consists of all points (x, y) satisfying the inequalities $y \geq 0, y \leq x$, and $y \leq 2 - x$ simultaneously. The region N which varies with parameter t , consists of all the points (x, y) satisfying the inequalities $t \leq x \leq t + 1$ and $0 \leq t \leq 1$ simultaneously. If the area of the region $M \cap N$ is a function of t , i.e., $M \cap N = f(t)$ and if α is the value of t for which this area is maximum, then the numerical value of 2α is _____
19. Number of solution of the equation $3 \tan x + x^3 = 2$ in $\left(0, \frac{\pi}{4}\right)$ is _____
20. Find the number of real values of x satisfy the equation $x^2 = x \sin x + \cos x$.
21. The least value of a for which the equation $\frac{4}{\sin x} + \frac{1}{1 - \sin x} = a$ has atleast one solution in $\left(0, \frac{\pi}{2}\right)$ is _____
22. Let C be the curve $y = x^3$ (where x takes all real values). The tangent at A meets the curve again at B . If the gradient of the curve at B is K times the gradient at A then K is equal to _____
23. The number of positive integral values of a for which $y = a^x$ intersects the line $y = x$ is _____

PASSAGE-1

Let $f(x) = 2 + |x - 1|$ and $g(x) = \min(f(t))$ where $x \leq t \leq x^2 + x + 1$, then

24. $g\left(-\frac{1}{2}\right)$ is equal to
 A) 2 B) $\frac{1}{2}$ C) $\frac{9}{4}$ D) $\frac{4}{9}$
25. Which of the following is incorrect?
 A) $g(0) = g(1)$ B) $g(2) = g(3)$ C) $g(-3) = g(-7)$ D) $g\left(\frac{1}{2}\right) + g\left(\frac{1}{3}\right) = 4$

PASSAGE-2

We are given the curves $y = \int_{-\infty}^x f(t) dt$ through the point $\left(0, \frac{1}{2}\right)$ and $y = f(x)$, where

$f(x) > 0$ and $f(x)$ is differentiable, $\forall x \in \mathbb{R}$ through $(0, 1)$. If tangents drawn to both the curves at the points with equal abscissa intersect on the same point on the x -axis, then

26. $\lim_{x \rightarrow \infty} (f(x))^{f(-x)}$ is
 A) 3 B) 6 C) 1 D) 2
27. The function $f(x)$ is
 A) Increasing for all x B) Non – monotonic
 C) Decreasing for all x D) Decreasing exactly some finite set of values of x

KEY

1	B	2	B	3	C	4	A	5	B
6	D	7	D	8	C	9	ABC	10	BD
11	BC	12	BD	13	BD	14	BC	15	8
16	5	17	4	18	1	19	1	20	2
21	9	22	4	23	1	24	C	25	B
26	C	27	A						



BASARA GNANA SARASWATHI CAMPUS KAKATIYA HILLS

Sec: Star SC

TEST ON AOD

Date: 29-09-2020

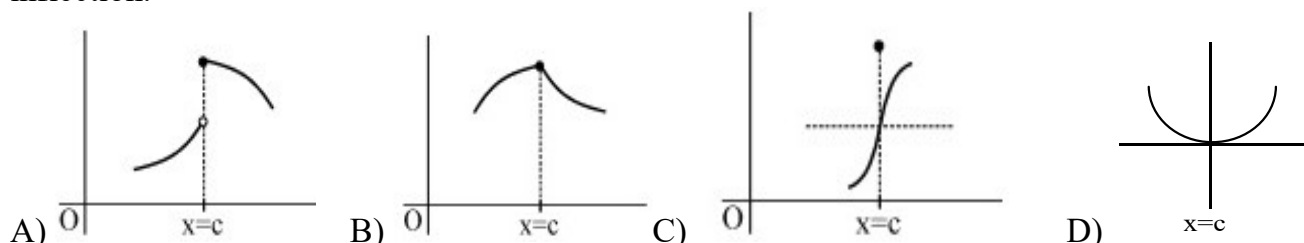
SINGLE ANSWER TYPE

- Let $F(x) = \int_{\sin x}^{\cos x} e^{(1+\sin^{-1}t)^2} dt$ on $\left[0, \frac{\pi}{2}\right]$ then which of the following is correct?
 A) $F''(c) = 0$ for all $c \in \left(0, \frac{\pi}{2}\right)$ B) $F''(c) = 0$ for some $c \in \left(0, \frac{\pi}{2}\right)$
 C) $F'(c) = 0$ for some $c \in \left(0, \frac{\pi}{2}\right)$ D) $F(c) \neq 0$ for all $c \in \left(0, \frac{\pi}{2}\right)$
- Let $f(x)$ and $g(x)$ be two differentiable function in R and $f(2) = 8$, $g(2) = 0$, $f(4) = 10$ and $g(4) = 8$ then which of the following is correct?
 A) $g'(x) > 4f'(x) \forall x \in (2, 4)$ B) $3g'(x) = 4f'(x)$ for at least one $x \in (2, 4)$
 C) $g(x) > f(x) \forall x \in (2, 4)$ D) $g'(x) = 4f'(x)$ for at least one $x \in (2, 4)$
- The set of values of p for which the equation $|\ln x| - px = 0$ possess three distinct roots is
 A) $\left(0, \frac{1}{e}\right)$ B) $(0, 1)$ C) $(1, e)$ D) $(0, e)$
- Let $f(x) = 1 + x^m(x-1)^n$ where $m, n \in N$. Then in $(0, 1)$ the equation $f'(x) = 0$ has
 A) no root B) at least one root
 C) at most one root D) exactly one root
- For a steamer the consumption of petrol (per hour) varies as the cube of its speed (in km). If the speed of the current is steady at C km/hr then the most economical speed of the steamer going against the current will be
 A) $1.25C$ B) $1.5C$ C) $1.75C$ D) $2C$
- If $f(x) = 4x^3 - x^2 - 2x + 1$ and $g(x) = \begin{cases} \text{Min } \{f(t) : 0 \leq t \leq x\} & ; 0 \leq x \leq 1 \\ 3 - x & ; 1 < x \leq 2 \end{cases}$ then $g\left(\frac{1}{4}\right) + g\left(\frac{3}{4}\right) + g\left(\frac{5}{4}\right)$ has the value equal to
 A) $\frac{7}{4}$ B) $\frac{9}{4}$ C) $\frac{13}{4}$ D) $\frac{5}{2}$
- Let f be a continuous function on $[a, b]$. If $F(x) = \left(\int_a^x f(t) dt - \int_x^b f(t) dt \right) (2x - (a+b))$ then there exist some $c \in (a, b)$ such that
 A) $\int_a^c f(t) dt = \int_c^b f(t) dt$ B) $\int_a^c f(t) dt - \int_c^b f(t) dt = f(c)(a+b-2c)$
 C) $\int_a^c f(t) dt - \int_c^b f(t) dt = f(c)(2c - (a+b))$ D) $\int_a^c f(t) dt + \int_c^b f(t) dt = f(c)(2c - (a+b))$
- Consider $f(x) = \int_1^x \left(t + \frac{1}{t}\right) dt$ and $g(x) = f'(x)$, for $x \in \left[\frac{1}{2}, 3\right]$ if P is a point on the curve $y = g(x)$ such that the tangent to this curve at P is parallel to a chord joining the points

$\left(\frac{1}{2}, g\left(\frac{1}{2}\right)\right)$ and $(3, g(3))$ of the curve, then the coordinates of the point P

- A) Cant be found out B) $\left(\frac{7}{4}, \frac{65}{28}\right)$ C) (1,2) D) $\left(\sqrt{\frac{3}{2}}, \frac{5}{\sqrt{6}}\right)$

9. Number of solutions satisfying the equation, $3x^2 - 2x^3 = \log_2(x^2 + 1) - \log_2 x$ is:
 A) 1 B) 2 C) 3 D) None
10. The tangent at a variable point P of the curve $y=x^2-x^3$ meets it again at Q. Find the locus of the middle point of PQ is
 A) $y=1-9x+28x^2-28x^3$ B) $y=1-2x+28x^2-28x^3$
 C) $y=1-3x+28x^2-28x^3$ D) $y=1-4x+28x^2-28x^3$
11. $f: R \rightarrow R$ be a twice differentiable function $\forall x \in R$. If tangent drawn to the curve at any point $x \in (a, b)$ always lie below the curve, then which of the following is always correct?
 A) $f'(x)$ is increasing and $f''(x) < 0 \forall x \in (a, b)$
 B) $f'(x)$ is decreasing and $f''(x) < 0 \forall x \in (a, b)$
 C) $f'(x)$ is increasing and $f''(x) > 0 \forall x \in (a, b)$
 D) $f'(x)$ is decreasing and $f''(x) > 0 \forall x \in (a, b)$
12. Read the following mathematical statements carefully:
 I. A differentiable function 'f' with maximum at $x = c \Rightarrow f''(c) < 0$.
 II. Antiderivative of a periodic function is also a periodic function.
 III. If f is a periodic function with period T, then for any $a \in R$, $\int_0^T f(x) dx = \int_0^T f(x+a) dx$
 IV. If f(x) has a maxima at $x = c$, then 'f' is increasing in $(c-h, c)$ and decreasing in $(c, c+h)$ as $h \rightarrow 0$ for $h > 0$.
 Now indicate the correct alternative.
 A) exactly one statement is correct. B) exactly two statements are correct.
 C) exactly three statements are correct. D) All the four statements are correct.
13. Let h be a twice continuously differentiable positive function on an open interval J and $g(x) = \log_e(h(x))$ such that $(h'(x))^2 > h''(x)h(x)$ for each $x \in J$. Then which of the following is/are correct?
 A) g is increasing on J B) g is decreasing on J
 C) g is concave up on J D) g is concave down on J
14. In the following graphs, in which option(s), at the point $x=c$ always represents point of inflection.



MULTIPLE ANSWER TYPE

15. Suppose that f is a polynomial of degree 3 and that $f''(x) \neq 0$ at any of the stationary point. Then which of the following is/are true?
 A) f has exactly one such stationary point. B) f must have no such stationary point.
 C) f can have two such stationary points. D) f can have zero such stationary points.
16. $f(x) = \begin{cases} e^x & 0 \leq x \leq 1 \\ 2 - e^{x-1} & 1 < x \leq 2 \\ x - e & 2 < x \leq 3 \end{cases}$ and $g(x) = \int_0^x f(t) dt, x \in [1, 3]$ then g(x) has

- A) local maxima at $x = 1 + \ln 2$ and local minima at $x = e$
 B) local minima at $x = 1$ and local maxima at $x = 3$
 C) no local maxima D) no local minima
17. Assume that inverse of a differentiable function f is denoted by g . Then which of the following statement hold good?
 A) If f is increasing then g is also increasing.
 B) If f is decreasing then g is increasing.
 C) The function f is injective. D) The function g is onto.
18. Suppose $f: \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function satisfying $f(x+y) = f(x) + f(y) + xy(x+y)$ for every $x, y \in \mathbb{R}$. If $f'(0) = 0$ then which of the following hold(s) good?
 A) f is an odd function B) f is a bijective mapping
 C) f has a minima but no maxima D) f has an inflection point
19. The extremum value(s) of the function $f(x) = \frac{1}{\sin x + 4} - \frac{1}{\cos x - 4}$, where $x \in \mathbb{R}$ is:
 A) $\frac{4}{8 - \sqrt{2}}$ B) $\frac{2\sqrt{2}}{8 - \sqrt{2}}$ C) $\frac{2\sqrt{2}}{4\sqrt{2} + 1}$ D) $\frac{4}{8 + \sqrt{2}}$
20. If $f(x) = a^{\{a^{|x|} \operatorname{sgn} x\}}$; $g(x) = a^{\lfloor a^{|x|} \operatorname{sgn} x \rfloor}$ for $a > 0$, $a \neq 1$ and $x \in \mathbb{R}$, where $\{\}$ & $\lfloor \rfloor$ denote the fractional part and integral part functions respectively, then which of the following statements can hold good for the function $h(x)$, where $(\ln a) h(x) = (\ln f(x) + \ln g(x))$.
 A) ' h ' is even and increasing B) ' h ' is odd and decreasing
 C) ' h ' is even and decreasing D) ' h ' is odd and increasing.
21. If $p'(x) > p(x)$ for all $x \geq 1$ and $p(1) = 0$ then which of the following is/are true?
 A) $e^{-x} p(x)$ is an increasing function for all $x \geq 1$
 B) $e^x p(x)$ is a decreasing function for all $x \geq 1$
 C) $p(x) > 0$ for all x in $[1, \infty)$ D) $p(x) < 0$ for all x in $[1, \infty)$
22. Find the equations of the tangents drawn to the curve $y^2 - 2x^3 - 4y + 8 = 0$ from the point $(1, 2)$.
 A) $2\sqrt{3}x - y = 2(\sqrt{3} - 1)$ B) $2\sqrt{3}x + y = 2(\sqrt{3} + 1)$
 C) $\sqrt{3}x + y = 2(\sqrt{3} + 1)$ D) $\sqrt{3}x - y = 2(\sqrt{3} - 1)$
23. The equation of the normal to the curve $\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2$ ($n \in \mathbb{N}$) at the point with abscissa equal to ' a ' can be:
 A) $ax + by = a^2 - b^2$ B) $ax + by = a^2 + b^2$ C) $ax - by = a^2 - b^2$ D) $bx - ay = a^2 - b^2$
24. Let $f(x) = \frac{x-1}{x^2}$ then which of the following is/are correct?
 A) $f(x)$ has minima but no maxima.
 B) $f(x)$ increases in the interval $(0, 2)$ and decreases in the interval $(-\infty, 0) \cup (2, \infty)$.
 C) $f(x)$ is concave down in $(-\infty, 0) \cup (0, 3)$. D) $x = 3$ is the point of inflection.
25. Which of the following options are correct
 A) $x \cdot \sin x - (1/2) \sin^2 x < (1/2)(\pi - 1)$ for $0 < x < \pi/2$
 B) $\tan^2 x + 6 \ln \sec x + 2 \cos x + 4 > 6 \sec x$ for $x \in \left(\frac{3\pi}{2}, 2\pi\right)$
 C) $2x \ln x > 4(x-1) - 2 \ln x$ for $x > 1$ D) $e^\pi > \pi^e$

26. A function f is defined by $f(x) = \int_0^x \cos t \cos(x-t) dt$, $0 \leq x \leq 2\pi$ then which of the following hold(s) good?
- A) $f(x)$ is continuous but not differentiable in $(0, 2\pi)$ B) Maximum value of f is π
- C) There exists atleast one $c \in (0, 2\pi)$ such that $f'(c) = 0$. D) Minimum value of f is $-\frac{\pi}{2}$.
27. If $\phi(x) = f(x) + f(2a - x)$ and $f''(x) > 0$, $a > 0$, $0 \leq x \leq 2a$ then which of the following is/are correct?
- A) $\phi(x)$ increases in $(a, 2a)$ B) $\phi(x)$ increases in $(0, a)$
- C) $\phi(x)$ decreases in $(0, a)$ D) $\phi(x)$ decreases in $(a, 2a)$

KEY

1	B	2	D	3	A	4	D	5	B
6	D	7	B	8	D	9	A	10	A
11	C	12	A	13	D	14	C	15	CD
16	AB	17	ACD	18	ABD	19	ACD	20	BD
21	AC	22	AB	23	AC	24	BCD	25	ABCD
26	CD	27	AC						